Optimal Dynamic Selection Under Costly Evaluation: Evidence from Graduate Admissions

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Motivation

- **Economic Problem:** Selecting best options under costly evaluation
  - Determining each option's true quality costs decision-maker time and effort
  - Low-cost screening of all options $\Rightarrow$ high-cost evaluation of surviving options
  - **Examples:** Admissions and hiring, clinical trials, matching platforms
- **Why Graduate Admissions:** Benefits graduate programs and future applicants
  - Graduate programs invest time and money into students' development
  - If programs are undervaluing applicants with certain characteristics, future such applicants may receive formerly-withheld opportunities

Research Questions

1. Are admissions committees optimizing, or are they making mistakes?
    - Goal is to accept applicants with highest success probabilities who will matriculate
    - Per literature, measure success with program completion and job placement ranking
    - Predictors of acceptance only (success only) may be overweighted (underweighted)
2. If they are making mistakes, how can they make better admissions decisions?
    - In turn, by changing decision rules, how much better of a cohort can they admit?

★ This paper's methods generalize to admissions, hiring, and other settings

Methodology and Main Results

- Build structural models of admissions committee's decision problem that support:
  - Statistical tests of whether committee's decision rule aligns with desired objective
  - Counterfactual simulations quantifying gains from deviating to model's decision rule
- Estimate model parameters with admissions data from a major research university
  - For many applicants, estimated success probabilities align with those implied by committee's decisions. But some applicant characteristics may be misweighted
  - Reweighting these characteristics would increase admitted cohort's average success probability without requiring committee to read more application files

Literature Review

- **Graduate Admissions:** Regress acceptance and success on applicant charateristics
  - **Original Literature:** Attiyeh & Attiyeh (1997); Krueger & Wu (2000); Grove & Wu (2007); Athey, Katz, Krueger, Levitt & Poterba (2007)
  - **Study of 2002 Cohort:** Stock, Finegan & Siegfried (2006–2015)
  - **New Predictors/Methods:** Bai, Esche, MacLeod & Shi (2022)
- **Optimal Dynamic Selection Under Costly Evaluation:**
  - **Sequential Search:** McCall (1970); Mortensen (1970); Weitzman (1979)
  - **Rational Inattention:** Sims (2003); Matějka & McKay (2015); Caplin & Dean (2015)

★ **This Paper:** Builds and estimates dynamic selection models of admissions, which test optimality by relating acceptance effects to success effects

Synthetic Data Protocol (IRB-Approved)

1. **Administrative office receives data from university departments**
    - Algorithmically extracts variables from textual files, such as recommendation letters
2. **Synthetic data (Patki et al. 2016) used for model development**
    - Estimate joint distribution of variables, and simulate synthetic dataset from it
    - Rows do not correspond to real subjects, preventing identity disclosure
    - Cells with < 5 comparable observations binned to prevent attribute disclosure
3. **Administrative office executes final estimation code on actual data**
    - Only summary statistics and parameter estimates are returned to researcher

★ **Contribution:** Privacy-protecting framework for researchers studying sensitive data

▸▸ Synthetic Data Example

Outline

Data Description

- **Sample:** Applicants to R1 university's economics PhD program from 2015–2019
- **Dependent Variables:** Measures of admission and success
  - **Admission:** Indicators for all stages of admissions process, including first-round filter
  - **Success:** Completion and placement category/quality for **all** applicants, with academia-equivalent rankings for non-academic placements (Krueger & Wu 2000)
- **Independent Variables:** Objective vs. subjective
  - **Objective:** Demographics and performance measures, such as GRE scores
  - **Subjective:** Textual processing of recommendation letters, application essays, CVs, and supplemental forms using dictionaries to construct variables

▸ Variables ▸ Inference vs. Prediction ▸ Sample Selection

Regression Results

- **Method:** Logistic regressions for acceptance, matriculation, and success
    - **Success:** *Top 100 Placement* $= 1$ for 20% of applicants vs. 34% of matriculants
- **Finding:** Different independent variables best predict each dependent variable
    - **Accept:** *Committee Score* is best predictor. Other predictors include demographics, *GRE Quantitative Percentile*, *Work Experience*, and *CV Coding* (negative)
    - **Matriculate:** Predictors of acceptance often negatively predict matriculation. However, best predictor of matriculation is *Open House* attendance
    - **Success:** *PhD Program Rank* is best predictor. Other predictors include *Age*, *TOEFL/IELTS*, *Work Experience*, and *Essay Positive Terms (%)*

Summary Statistics

Acceptance, Matriculation, and Success AMEs

Dependent Variable	Accept	Matriculate	T100 Placement
Winsorized Age	-0.0001	-0.0223**	0.0053*
Female	0.0163**	-0.0490	-0.0032
U.S.	-0.0013	0.1472*	0.0192
International Asian	-0.0214***	0.1644**	-0.0127
GRE Quantitative Percentile	0.0011**	-0.0147**	0.0006
Undergraduate GPA	0.0195	-0.1026	-0.1096**
TOEFL/IELTS	-0.0022	-0.0631	0.0409*
Work Experience	0.0140**	0.1591**	0.0358**
Committee Score	0.0788***	-0.0371	0.0007
Advanced Math	0.0246	-0.1950**	-0.0287
Letters Terms (%) PC1	0.0055	0.0671**	0.0159
CV Coding	-0.0044*	-0.0414**	0.0125
Essay Economics Terms (%)	0.0001	0.0366**	-0.0104
Essay Positive Terms (%)	-0.0013	-0.0072	0.0123*
Open House		0.2615***	
Program Rank			-0.0011***
Missing Program Rank			-0.3965***

Year Dummies
GRE Verbal Percentile
GRE Analytic Percentile
Elite U.S. University
Elite U.S. Liberal Arts College
Elite Foreign University
Other Foreign University
Graduate Degree
#Professor Recommenders
#Prolific Recommenders
#Math Courses
Letters Length/100
Letters Terms (%) PC2
CV Length/100
CV Publication/Working Paper
Essay Length/100
Essay Specific Professor
Essay Research Terms (%)
Essay Negative Terms (%)
Missing Dummies

Full Logistic Regression AMEs   Lasso Logistic Regression Results

Acceptance vs. Success AMEs

Dependent Variable	Accept	T100 Placement
Winsorized Age	-0.0001	0.0053*
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International Asian	-0.0214***	-0.0127
GRE Quantitative Percentile	0.0011**	0.0006
TOEFL/IELTS	-0.0022	0.0409*
Work Experience	0.0140**	0.0358**
#Professor Recommenders	-0.0033	-0.0069
Missing TOEFL/IELTS	0.0070	-0.0008
Committee Score	0.0788***	0.0007
CV Coding	-0.0044*	0.0125
Essay Economics Terms (%)	0.0001	-0.0104
Essay Positive Terms (%)	-0.0013	0.0123*

Regressions vs. Model

- Admissions literature regresses acceptance and success on variables in applications, and qualitatively compares the effects
- While such comparisons are of interest, they are insufficient to test optimality

Insufficiency of Comparing AMEs

- Cannot test optimality by comparing AMEs across regressions. Suppose:
  - Work experience has large effect on success, graduate degree smaller effect
  - 25% of applicants have both, 25% experience only, 25% degree only, 25% neither
  - Committee can accept half of applicants, and other half attend different university
- Optimal committee accepts half with experience and ignores degree
  - **Acceptance Regression:** $\text{AME}_{\text{Experience}} = 1$, $\text{AME}_{\text{Degree}} = 0$
  - **Success Regression:** $1 > \text{AME}_{\text{Experience}} > \text{AME}_{\text{Degree}} > 0$
  - Despite optimality, acceptance and success AMEs not equal across regressions
- **Upshot:** Relationship between AMEs must be interpreted through a model

Wald Tests

Graduate School Regression Results

Outline

Why Structural Models?

- **Reason 1:** Test whether committee's decision rule aligns with desired objective
  - Per literature, committee should accept applicants with highest success probabilities
  - Committees also accept applicants with higher matriculation probabilities
- **Reason 2:** Find counterfactual decision rule in which committee optimizes
  - Quantify gains from deviating from actual to counterfactual decision rule

Model Framework

- **Portfolio Choice Problem:** Accept subset of applicants with highest sum of success probabilities subject to capacity constraint on number of acceptances

- For each applicant $n \in \{1, \ldots, N\}$, make decision $d_n \in \{\text{Accept}, \text{Reject}\}$ given:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} \underbrace{P(y_n = S|x_n, z_n)}_{\text{Success probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}}, \text{ such that: } \sum_{n=1}^{N} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \leq N_A,$$

where x = objective variables, z = subjective variables, $y \in \{\text{Success}, \text{Failure}\}$, N = number of applicants, and N_A = maximum number of acceptances

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- BEMS (2022) prove that portfolio choice problem reduces to cutoff rule problem
  - Accept applicant if and only if their success probability exceeds cutoff probability
  - Decisions across applicants now independent, so no need to consider all subsets

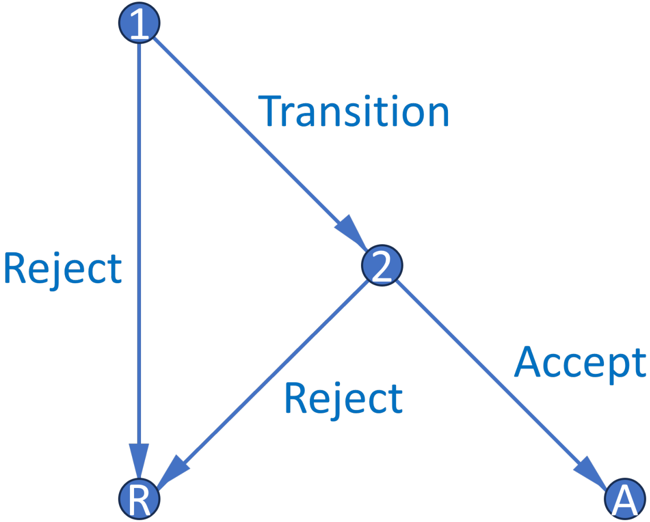
Portfolio Choice to Cutoff Rule

Cutoff Rule Problem

- Start with two-round cutoff rule problem that accounts for costly evaluation:
  - **Round 1:** See only objective variables (e.g. GRE). Can transition application file to Round 2 to see subjective variables (e.g. recommendation letters), or can reject
  - **Round 2:** If chose to transition application file in Round 1, can accept or reject
  - Assume for now committee transitions and accepts optimal number of files
- Along with the baseline dynamic model, there are additional models
  - Static model is simpler and has interpretable cutoff probabilities in both rounds
  - Matriculation models account for both success and matriculation probabilities

Models of Admissions

Decision Graph



Dynamic Model

- **States:** r = round, x = objective variables, z = subjective variables, t = year, $\overline{k}$ = fixed program rank, ϵ_r = T1EV preference shock with scale parameter σ
- **Controls:** $d_1 \in \{\text{Transition}, \text{Reject}\}$, $d_2 \in \{\text{Accept}, \text{Reject}\}$
- **Outcomes:** $y \in \{\text{Success}, \text{Failure}\}$, y_r^* = marginal outcome

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Round 2:

$$v_2(x, z, t, \bar{k}, \epsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x, z, t, \bar{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \bar{k}, d_2) = \underbrace{P(y = S | x, z, t, \bar{k}) \mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S | t) \mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

Round 2 Portfolio Choice

Dynamic Model

Round 1:

$$v_1(x,t,\overline{k},\epsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x,t,\overline{k},d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x,t,\overline{k},d_1) = \underbrace{\mathbb{E}[v_2(x,z,t,\overline{k},\epsilon_2)|x,t,\overline{k}]\mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Social Surplus}]} + \underbrace{L(y_1^* = S|t)\mathbb{1}(d_1 = R)}_{\text{If reject, get } L(\text{Cutoff})}, \text{such that:}$$

$$\mathbb{E}[v_2(x,z,t,\overline{k},\epsilon_2)|x,t,\overline{k}] = \int_{\widetilde{z}} \sigma \log\left(e^{P(y=S|x,\widetilde{z},t,\overline{k})/\sigma} + e^{P(y_2^*=S|t)/\sigma}\right) dF(\widetilde{z}|x,t)$$

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Conditional Choice Probabilities:

$$P(d_r|x, (z), t, \overline{k}) = \frac{\exp(u_r(x, (z), t, \overline{k}, d_r)/\sigma)}{\sum_{\widetilde{d}_r \in D_r} \exp(u_r(x, (z), t, \overline{k}, \widetilde{d}_r)/\sigma)}$$

▸▸ Round 1 Portfolio Choice ▸▸ Dynamic Parameterization

Static Model

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Round 2:

$$v_2(x, z, t, \overline{k}, \epsilon_2) = \max_{d_2 \in \{A, R\}} u_2(x, z, t, \overline{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

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- $P(y=S|x,t,\overline{k})$ estimates inconsistent, but fitted values account for z's effect on y

Round 1 Portfolio Choice

Portfolio Choice with Matriculation

- Accept subset of applicants with highest sum of success times matriculation probabilities subject to capacity constraint on expected number of matriculants

- For each applicant $n \in \{1, \ldots, N\}$, make decision $d_n \in \{\text{Accept}, \text{Reject}\}$ given:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} \underbrace{P(y_n = S|x_n, z_n)}_{\text{Success probability}} \underbrace{P(m_n = M|x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}},$$

$$\text{such that: } \sum_{n=1}^{N} \underbrace{P(m_n = M|x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \leq N_M,$$

where $(x, z) =$ variables, $y \in \{\text{Success}, \text{Failure}\}$, $m \in \{\text{Matriculate}, \text{Decline}\}$, $N =$ number of applicants, and $N_M =$ maximum number of matriculants

Portfolio Choice with Matriculation

- Accept subset of applicants with highest sum of success times matriculation probabilities subject to capacity constraint on expected number of matriculants

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Theorem 1 (Proof)

For large N, this problem has a cutoff rule solution on **success** *probabilities.*

- Likely matriculants more valuable, but also more expensive $\Rightarrow$ effects cancel
- However, matriculation probabilities matter in Round 1 of a two-round model

Cutoff Rule with Matriculation

Matriculation-Dynamic Model

Round 2:

$$v_2(x, z, t, \overline{k}, \epsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x, z, t, \overline{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

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$$\mathbb{E}[v_2(x,z,t,\overline{k},\epsilon_2)|x,t,\overline{k}] = \int_{\widetilde{z}} \sigma \log\left(e^{[P(y=S|x,\widetilde{z},t,\overline{k}) - P(y_2^* = S|t)]/\sigma} + 1\right) P(m = M|x,\widetilde{z},t)\, dF(\widetilde{z}|x,t)$$

Dynamic Portfolio Choice with Matriculation

Outline

Maximum Likelihood Estimation

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \prod_{n=N_T+1}^{N} \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}}$$

$$\prod_{n=1}^{N} \underbrace{\left[P_{y_n}(\theta_{S_1}) P_{y_n}(\theta_{S_2})\right]^{\mathbb{1}(y_n=S)} \left[(1 - P_{y_n}(\theta_{S_1}))(1 - P_{y_n}(\theta_{S_2}))\right]^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- In unrestricted likelihood $\mathcal{L}^U$, parameterize CCPs with flexible logistic specifications
- In restricted likelihood $\mathcal{L}^R$, use model CCPs, which depend on structural parameters
- Do success probabilities implied by outcomes align with those implied by CCPs?

Dynamic Likelihood | Matriculation Likelihood | Empirical Considerations

Cutoff Probability Identification

Theorem 2 (Proof)

The maximum likelihood cutoff probability estimate is such that the number of acceptances N_A equals the expected N_A. Also, for any cutoff probability and N_A, the likelihood is highest when the committee accepts the best N_A applicants.

- Identify each year's cutoffs by how selective committee is that year
- While selectivity identifies cutoffs, sorting improves restricted likelihood
  - If committee engages in sorting, Akaike Information Criterion more favorable
  - AIC $= 2\left[K - \log(\hat{\mathcal{L}}^R)\right]$, such that $K =$ number of parameters

Restricted Static Estimates

	R1 Cutoff Probability	R2 Cutoff Probability	Sigma
2015	29.36%	39.20%	
	(23.20%, 36.38%)	(32.75%, 46.05%)	
2016	29.17%	40.53%	
	(22.21%, 37.27%)	(33.55%, 47.92%)	
2017	37.06%	41.69%	
	(32.10%, 42.29%)	(34.83%, 48.90%)	
2018	38.12%	39.30%	
	(33.98%, 42.44%)	(31.80%, 47.33%)	
2019	34.04%	39.62%	
	(28.67%, 39.85%)	(32.54%, 47.16%)	
Sigma			0.0064
			(0.0035)
Observations = 2329	Parameters = 80	LL = -2992.0	AIC = 6144.1

Full Static Estimates

Restricted Dynamic Estimates

	R1 Cutoff Level	R2 Cutoff Probability	Sigma
2015	0.3731	36.58%	
	(0.3067, 0.4539)	(29.81%, 43.92%)	
2016	0.3795	37.21%	
	(0.3025, 0.4761)	(29.20%, 45.99%)	
2017	0.4126	40.27%	
	(0.3563, 0.4778)	(34.52%, 46.30%)	
2018	0.3836	37.68%	
	(0.3271, 0.4497)	(31.79%, 43.97%)	
2019	0.3887	38.14%	
	(0.3268, 0.4624)	(31.89%, 44.81%)	
Sigma			0.0020
			(0.0013)
Observations = 2329	Parameters = 529	LL = -14131.0	AIC = 29319.9

▸▸ Full Dynamic Estimates ▸▸ Full Myopic Estimates

Restricted Matriculation-Dynamic Estimates

	R1 Cutoff Level	R2 Cutoff Probability	Sigma
2015	0.0071	36.35%	
	(0.0026, 0.0194)	(29.47%, 43.84%)	
2016	0.0071	38.25%	
	(0.0028, 0.0180)	(31.05%, 46.00%)	
2017	0.0105	40.65%	
	(0.0041, 0.0267)	(35.15%, 46.39%)	
2018	0.0095	36.59%	
	(0.0038, 0.0240)	(30.01%, 43.71%)	
2019	0.0095	37.41%	
	(0.0035, 0.0255)	(31.04%, 44.26%)	
Sigma			0.0020**
			(0.0010)
Observations = 2329	Parameters = 568	LL = -14260.9	AIC = 29657.8

Full Matriculation-Dynamic Estimates

Outline

Deviation Gains Test

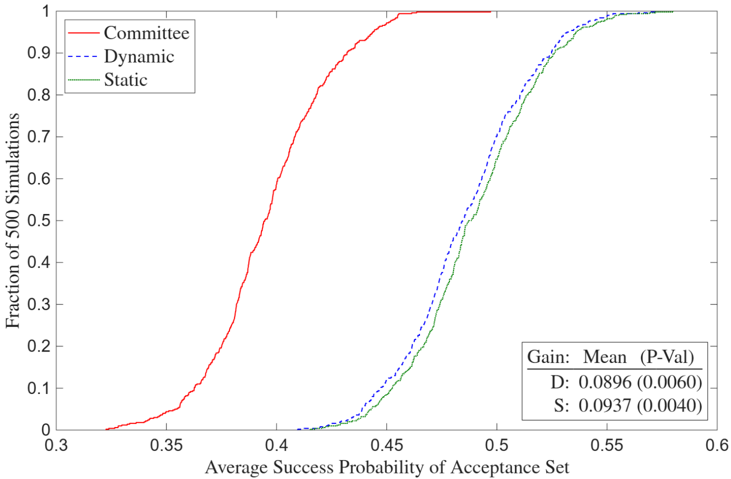
- Suppose committee selected model's chosen set. Would average success probability of acceptance/matriculation set have increased, and if so, by how much?
  - Solve model deterministically, compute average yearly success probability of model's acceptance/matriculation set, and compare to that of actual decision rule's set
  - $\hat{\theta}^R_S$ may be biased to rationalize suboptimal behavior, so plug in $\hat{\theta}^U_S$
- **Inference:** Too few observations for holdout set $\Rightarrow$ start with randomization test
  - Compute average success probabilities with $B = 500$ perturbations $\tilde{\theta}^U_{S,b}$ of $\hat{\theta}^U_S$
  - P-value is fraction of perturbations with model's $\overline{P(\text{Success})} \leq$ committee's

Theorem 3 (Proof)

For large B, the fraction of $\tilde{\theta}^U_{S,b}$ with gain $g(\tilde{\theta}^U_{S,b}) \leq 0$ is a p-value for $H_0 : g_0 \leq 0$.

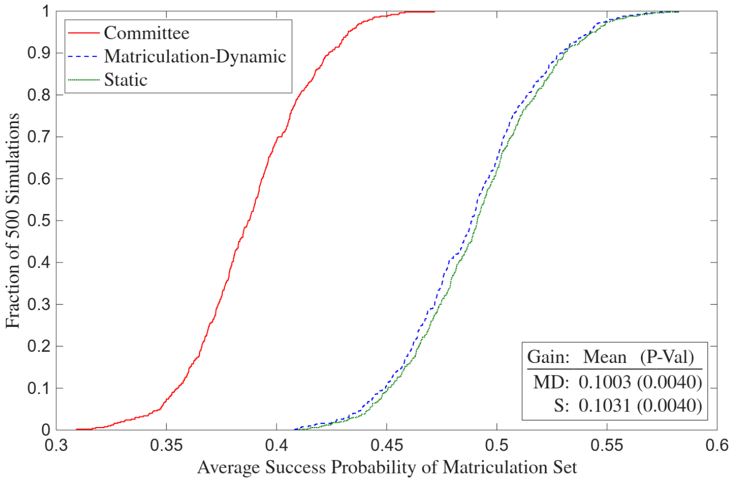
Deviation Gains Algorithm | Randomization Test

Deviation Gains



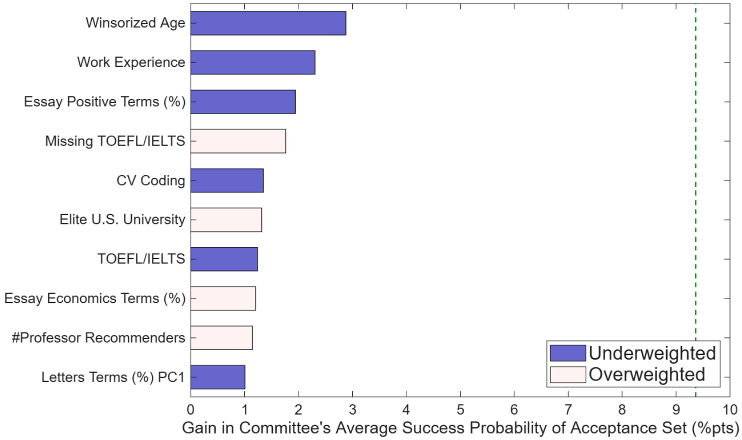
Full Deviation Gains

Deviation Gains (Matriculation)



Full Deviation Gains (Matriculation)

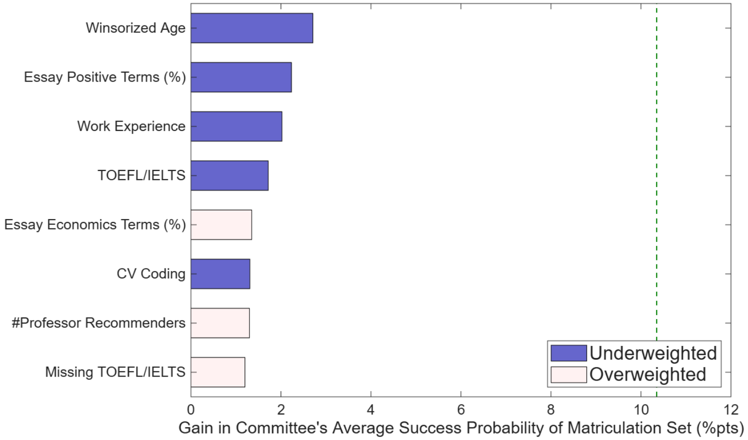
Deviation Gains Attribution



1. Experience
2. English Proficiency
3. Essay/Letters Subtext
4. Industry Background
5. Undergraduate

⏩ Full Deviation Gains Attribution (Matriculation) ⏩ Deviation Gains Attribution Table (Matriculation)

Deviation Gains Attribution (Matriculation)



1. Experience
2. English Proficiency
3. Essay Subtext
4. Industry Background

Full Deviation Gains Attribution (Matriculation)
Deviation Gains Attribution Table (Matriculation)

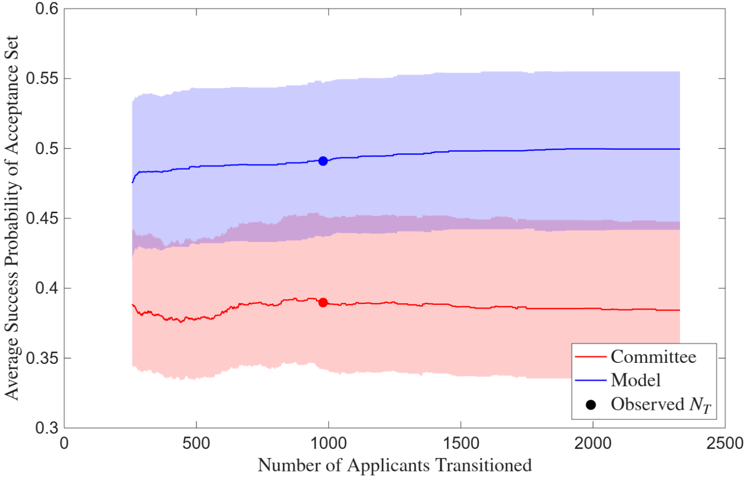
Regressions vs. Model

Dependent Variable	Accept	T100 Placement
Winsorized Age	-0.0001	0.0053*
Female	0.0163**	-0.0032
International Asian	-0.0214***	-0.0127
GRE Quantitative Percentile	0.0011**	0.0006
TOEFL/IELTS	-0.0022	0.0409*
Work Experience	0.0140**	0.0358**
#Professor Recommenders	-0.0033	-0.0069
Missing TOEFL/IELTS	0.0070	-0.0008
Committee Score	0.0788***	0.0007
CV Coding	-0.0044*	0.0125
Essay Economics Terms (%)	0.0001	-0.0104
Essay Positive Terms (%)	-0.0013	0.0123*

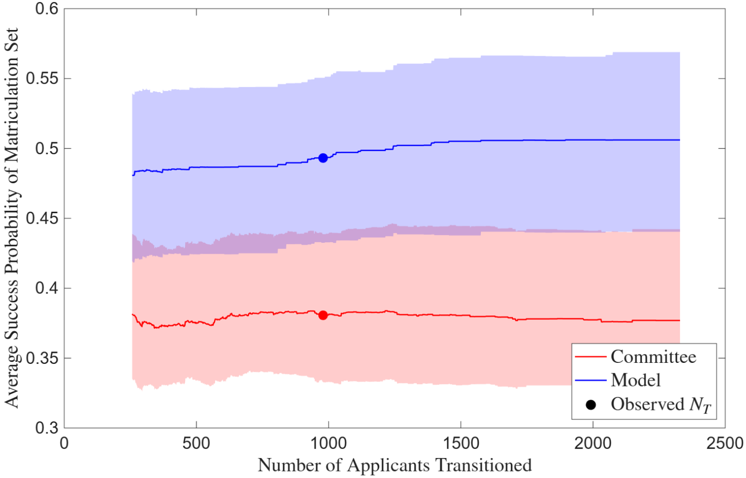
Acceptance vs. Success AMEs

- Blue = underweighted, white = correctly weighted, red = overweighted
- AMEs don't reveal if committee deviates from optimal ranking near the cutoff

Benefit of Reading More Files



Benefit of Reading More Files (Matriculation)



Deviation Gains Interpretation

- Under meaningfully-sized deviation gains, committee may have incorrect beliefs about which types of applicants are most likely to succeed
- Randomization test is not a true out-of-sample test, and it cannot account for changes in the determinants of success over time, such as writing essays with AI
  - See Andrews, Kitagawa & McCloskey (2023)'s on the "winner's curse"
  - Figures should be interpreted judiciously with those caveats in mind
- Committee may use success-predicting variables that I do not observe
  - Because the success probability measure used to evaluate gains cannot detect such variables, any edge the committee's private information provides goes unmeasured
  - Comprehensive data, especially committee score in z, protects against this limitation

Conclusion

- **Economic Problem:** Selecting best options under costly evaluation
- **Application:** Graduate admissions using synthetic data protocol to protect privacy
- **Contribution:** Build models to test optimality and quantify deviation gains
  1. For many applicants, estimated success probabilities align with those implied by committee's decisions. But some applicant characteristics may be misweighted
  2. Reweighting these characteristics would increase admitted cohort's average success probability without requiring committee to read more application files

▸▸ Simulated Results ▸▸ Multiyear Models

Thank You!

A1: Synthetic Rows $\neq$ Actual Applicants

Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
1	1	0	0	0	89	0	0	0	0	5	150
1	0	1	0	1	96	0	1	0	0	6	23
0	0	0	0	0	88	0	1	0	1	.	150
0	0	0	0	0	80	1	1	0	1	.	150
0	0	0	0	1	99	0	1	0	0	.	29
0	0	0	0	0	81	0	0	1	1	.	23
1	0	1	1	1	92	0	1	0	0	3	37
1	1	1	1	1	93	0	0	0	1	3	37
0	0	0	0	0	83	1	0	1	0	.	48
1	1	0	0	0	87	0	1	0	1	5	34

$\Downarrow$

Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
0	0	0	0	0	77	0	0	0	1	.	30
0	0	0	0	1	89	0	0	0	0	.	25
0	0	0	0	0	85	1	0	0	0	.	150
0	0	0	0	0	76	0	1	0	1	.	51
1	1	0	0	0	88	0	1	0	0	8	150
1	0	1	0	0	88	0	0	0	1	5	150
1	1	0	0	0	84	0	0	1	0	2	20
0	0	0	0	0	82	0	1	0	0	.	35
0	0	0	0	0	70	1	0	0	0	.	150
1	1	0	0	1	93	0	1	0	0	5	57

Synthetic Data Protocol

A1: Dependent Variables

Table 1a: Dependent Variables

Admission	Success
$\mathbb{1}$(Transition)	**$\mathbb{1}$(Complete Program)**
$\mathbb{1}$(Accept)	**$\mathbb{1}$(Academic Placement)**
$\mathbb{1}$(Waitlist)	**$\mathbb{1}$(Government Placement)**
$\mathbb{1}$(Open House)	**$\mathbb{1}$(Consulting Placement)**
$\mathbb{1}$(Matriculate)	**$\mathbb{1}$(Tech Placement)**
$\mathbb{1}$(Attend Program)	**$\mathbb{1}$(Other Placement)**
Program Rank	**$\mathbb{1}$(No Placement)**
	Placement Rank
	$\mathbb{1}$(Top 100 Placement)

- Economics-only variables bold, non-economics-only variables italic
- U.S. News PhD program rankings, IDEAS/RePEc placement rankings
  - For non-tenure track, use weighted average of university's and maximum rankings
  - For non-academia, use average of ChatGPT, Claude, and Gemini's rankings

Data Description

A1: Objective Variables

Table 1b: Objective Independent Variables

Demographic	Performance
Year	GRE Verbal Percentile
$\mathbb{1}$*(Covid Year)*	GRE Quantitative Percentile
Winsorized Age	GRE Analytic Percentile
$\mathbb{1}$(Female)	Undergraduate GPA
$\mathbb{1}$**(U.S.)**	TOEFL/IELTS
$\mathbb{1}$*(U.S. African/Hispanic/Native)*	$\mathbb{1}$(Elite U.S. University)
$\mathbb{1}$*(U.S. Asian)*	$\mathbb{1}$(Elite U.S. Liberal Arts College)
$\mathbb{1}$*(U.S. Other)*	$\mathbb{1}$(Other U.S. University/College)
$\mathbb{1}$(International Asian)	$\mathbb{1}$(Elite Foreign University)
$\mathbb{1}$(International Other)	$\mathbb{1}$(Other Foreign University)
	$\mathbb{1}$(Graduate Degree)
	$\mathbb{1}$(Work Experience)
	#Professor Recommenders
	#Prolific Recommenders
	Concentration Dummy Variables

Data Description

A1: Subjective Variables

Table 1c: Subjective Independent Variables

Score/Recommendation Letters	Supplement/CV/Essay
Committee Score	**#Math Courses**
Letters Length	**$\mathbb{1}$(Advanced Math)**
Positive Terms (%) (YS 2012)	**CV Length**
Negative Terms (%) (YS 2012)	**CV Coding**
Standout Terms (%) (BEMS 2022)	**CV $\mathbb{1}$(Publication/Working Paper)**
Ability Terms (%) (BEMS 2022)	**Essay Length**
Research Terms (%) (BEMS 2022)	**Essay $\mathbb{1}$(Specific Professor)**
Grindstone Terms (%) (BEMS 2022)	**Essay Economics Terms (%)**
Teaching Terms (%) (BEMS 2022)	**Essay Research Terms (%)**
Communal Terms (%) (BEMS 2022)	**Essay Positive Terms (%)**
Agentic Terms (%) (MHM 2009)	**Essay Negative Terms (%)**
	Field Dummy Variables

- Dictionaries from Madera, Hebl & Martin (2009), Young & Soroka (2012), and Bai, Esche, MacLeod & Shi (2022)
- *#Math Courses* and *Advanced Math* objective after 2020

Data Description

A1: Academia-Equivalent Rankings (Government)

- **European Institutions:** European Central Bank (40), HM Treasury (80)
- **Foreign Central Banks:** Banco Central do Brasil (93), Banco de México (93), Bank of Canada (80), Bank of England (67), Bank of Israel (93), Bank of Italy (80), Bank of Japan (80), Bank of Spain (80), Banque de France (80), Central Bank of Chile (93), Central Bank of Colombia (93), Central Bank of Korea (93), Central Bank of Turkey (93), Norges Bank (80), Reserve Bank of Australia (80), Reserve Bank of India (93), Reserve Bank of New Zealand (93), Sveriges Riksbank (80), Swiss National Bank (80)
- **Multilaterals/International:** African Development Bank (93), Asian Development Bank (93), Bank for International Settlements (53), European Bank for Reconstruction and Development (93), European Investment Bank (93), Inter-American Development Bank (93), International Monetary Fund (40), Statistics Canada (107), World Bank (40)
- **Policy Research Institutes:** American Enterprise Institute (93), American Institutes for Research (107), Becker Friedman Institute (80), Brookings Institution (80), Center for Global Development (93), Center for Migration Studies in New York (112), Center for Strategic and International Studies (107), Cowles Foundation (80), Harris School Policy Labs (93), Hoover Institution (80), IDinsight (112), Innovations for Poverty Action (93), J-PAL (80), MDRC (107), National Bureau of Economic Research (67), Peterson Institute for International Economics (80), Pew Research Center (107), RAND Corporation (80), Resources for the Future (93), Urban Institute (107)
- **U.S. Executive/Cabinet:** Department of Commerce (107), Department of Justice (80), Department of Labor (107), Department of the Treasury (80)
- **U.S. Executive Policy Offices:** Council of Economic Advisers (80), Office of Management and Budget (93)
- **U.S. Federal Reserve System:** Board of Governors (40), Atlanta (80), Boston (80), Chicago (80), Cleveland (80), Dallas (80), Kansas City (80), Minneapolis (80), New York (67), Philadelphia (80), Richmond (80), San Francisco (80), St. Louis (80)
- **U.S. GSE/Housing Finance:** Fannie Mae (120), Freddie Mac (120)
- **U.S. Independent Agencies:** Commodity Futures Trading Commission (93), Federal Deposit Insurance Corporation (93), Federal Energy Regulatory Commission (93), Federal Housing Finance Agency (93), Federal Trade Commission (80), Office of Financial Research (93), Securities and Exchange Commission (93)
- **U.S. Legislative Committees:** Congressional Budget Office (80), Congressional Research Service (107), Joint Committee on Taxation (93), Joint Economic Committee (107)
- **U.S. Science and Research:** National Science Foundation (107)
- **U.S. Statistical Agencies:** Bureau of Economic Analysis (93), Bureau of Labor Statistics (93), Census Bureau (93)

Data Description

A1: Academia-Equivalent Rankings (Consulting/Tech/Other)

- **Consulting:** Aecon Consulting (130), AlixPartners (125), Analysis Group (120), Bain & Compnay (120), Bates White (120), Berkeley Research Group (120), Boston Consulting Group (120), Brattle Group (120), Charles River Associates (120), Coleman Research (130), Compass Lexecon (120), Cornerstone Research (120), Deloitte Economic and Financial Advisory (125), Econic Partners (130), Economists Incorporated (120), Edgeworth Economics (120), Ernst & Young Economic Advisory (125), Frontier Economics (120), FTI Consulting (120), Guidehouse (130), KPMG Economics & Valuation (125), Matrix Economics (130), McKinsey & Company (120), Mercer (135), NERA Economic Consulting (120), Oliver Wyman (120), PwC Economics (125), Roland Berger (125), Willis Towers Watson (135)
- **Tech (Athey & Luca 2019):** Airbnb (107), Alibaba (120), Amazon (93), AppNexus (135), Apple (107), Block/Square (120), ByteDance/TikTok (120), CoreLogic (135), Coursera (120), DStillery (135), Didichuxing (125), Digonex (135), DoorDash (107), eBay (107), ECONorthwest (125), Expedia (120), Forkcast (135), Glassdoor (120), Google (93), Granular (135), Groupon (125), Houzz (130), Huawei (120), IBM (120), Indeed (120), ING (130), Instacart (120), Intel (120), Kensho (125), Lending Club (135), LinkedIn (107), Lyft (120), Meta/Facebook (93), Microsoft (93), Netflix (93), Nuna (135), Nvidia (120), Pandora (135), PayPal (120), Pinterest (120), Prattle (135), Quantco (130), Quora (130), Redfin (120), Ripple (135), Roblox (120), Rover (135), Salesforce (120), Shopify (120), Spotify (120), Stripe (107), Trulia (125), Uber (107), Upwork (135), Visa (125), Walmart (125), Wealthfront (135), Yahoo (125), Yelp (125), Zillow (120)
- **Tech Labs:** Amazon Science (40), Apple Machine Learning Research (80), DeepMind (80), Google Research (40), Meta Research (80), Microsoft Research (40), Netflix Research (93), OpenAI Research (80), Uber Research (93)
- **Other:** AIG (135), Bank of America (125), Bloomberg (125), Boeing (135) Capital One (125), Exelon (135), ExxonMobil (135), Ford (135), General Electric (135), General Motors (135), Goldman Sachs (125), JPMorgan Chase (125), Liberty Mutual (135), Moody's Analytics (125), Morningstar (125), Nielsen (135), PNC (135), Progressive (135), S&P Global (125), Shell (135), Swiss Re (130), Toyota (130), Other (135), No Placement (150)

Data Description

A1: Dictionaries

- **Recommendation Letters:**
  - **Positive/Negative Words:** Young & Soroka (2012)'s Lexicoder Sentiment Dictionary
  - **Standout, Ability, Research, Grindstone, Communal Words:** Bai, Esche, MacLeod & Shi (2022)
  - **Agentic Words:** assertive, confident, aggressive, ambitious, dominant, forceful, independent, daring, outspoken, intellectual (Madera, Hebl & Martin 2009); **AND** audacious, bold, command, compete, decisive, determine, drive, enterprise, fearless, influential, leader, pioneer, powerful, proactive, resilient, resourceful, self-assured, strategic, tenacious, vision
  - **Teaching Words:** academic, assess, clarity, coach, collaborate, communicate, cultivate, curriculum, demonstrate, develop, educate, encourage, engage, enlighten, evaluate, explain, facilitate, foster, guide, impart, inspire, insight, instruct, knowledge, learn, lesson, mentor, nurture, participate, pedagogy, present, scholar, support, teach, train, tutor, workshop
- **CVs and Essays:**
  - **Coding:** c/c++, eviews, fortran, java, jax, julia, latex, matlab, python, r, sas, stata, webscrape
  - **Publication:** Journal names from IDEAS/RePEc Aggregate Rankings for Journals
  - **Working Paper:** arxiv, center for economic policy research, cepr, econpapers, national bureau of economic research, nber, research papers in economics, repec, social science research network, ssrn, working paper
  - **Economics:** applied, asset, behavior, climate, computation, data, development, dynamic programming, economics, education, empirical, environment, experiment, finance, fiscal, game theory, health, immigration, industrial, inequality, international, io, labor, linear programming, macro, math, metrics, micro, monetary, policy, political, poverty, pricing, probability, public, sports, statistic, theory, trade, urban

Data Description

A1: Inference vs. Prediction

- Challenging to obtain consistent estimates due to omitted variables
  - **Example:** Don't observe effort (e.g. hours worked). Can bias coefficients upward if effort correlated with variables and with success irrespective of those variables
  - Existing literature also suffers from this problem, though BEMS (2022) parse recommendation letters for "grindstone terms" (e.g. depend*, diligen*)
- Research questions emphasize prediction over inference
  - Mullainathan & Spiess (2017) distinguish $\hat{\beta}$ vs. $\hat{y}$ problems
  - If GRE predicts success, committee should use it regardless of why

Data Description

A1: Sample Selection

- Determinants of success conditional on applying vs. matriculating
- BEMS (2022) prove that if $P(\text{Success}|\text{Skill})$ is an S-curve, then $\hat{\beta}$ will be smaller for more selected samples. Consistent with empirical results
- If there aren't some types of outcomes data (e.g. grades, passing comps) for all applicants, or such data are too hard to obtain, use Heckman correction
    - Let $y_n^* = x_n\beta + u_n$, $d_n = \mathbb{1}(z_n\gamma + v_n > 0)$, and $y_n = d_n y_n^*$
    - **Step 1:** Run probit of d on z and compute inverse Mills $\hat{\lambda}_n = \frac{\phi(z_n\hat{\gamma})}{\Phi(z_n\hat{\gamma})}$
    - **Step 2:** Run OLS or heckprobit of y on $(x, \hat{\lambda})$ to get $\hat{\beta}$
    - **Instruments:** Variables that predict matriculation but not the outcome

Data Description

A1: Wald Tests

- Test if acceptance and/or success AMEs are same across PhD programs
  - **Example:** Does elite undergraduate university predict acceptance and/or success more or less for economics program than for other programs?
  - **Sample:** Applicants to university's economics, government, history, linguistics, and psychology PhD programs from 2020–2023 (when multi-program data is available)
- Can I test if acceptance AMEs equal success AMEs within programs?
  - Predictors of acceptance only (success only) may be overweighted (underweighted), provided committee wants to select based on a "success" model
  - **Problem:** Not a test if admissions decisions are optimal
  - Need structural model of admissions to conclude about optimality

Insufficiency of Comparing AMEs

A1: Logistic Wald Test Across Programs

Assume $\left(Y^E \perp\!\!\!\perp Y^{NE}\right)\big|X$. Let $P\left(Y_n^E = 1\big|x_n^E;\theta^E\right) = \frac{\exp(x_{n,0}^E\theta_0^E + x_{n,1}^E\theta_1^E)}{1+\exp(x_{n,0}^E\theta_0^E + x_{n,1}^E\theta_1^E)}$, and $P\left(Y_n^{NE} = 1\big|x_n^{NE};\theta^{NE}\right) = \frac{\exp(x_{n,0}^{NE}\theta_0^{NE} + x_{n,1}^{NE}\theta_1^{NE})}{1+\exp(x_{n,0}^{NE}\theta_0^{NE} + x_{n,1}^{NE}\theta_1^{NE})}$. Finally, let $N = N_E + N_{NE}$. Then:

$$\mathcal{L}_N^U\left(y\big|x;\theta\right) = \prod_{n=1}^{N_E} P\left(Y_n^E = 1\big|x_n^E;\theta^E\right)^{Y_n^E}\left[1 - P\left(Y_n^E = 1\big|x_n^E;\theta^E\right)\right]^{1-Y_n^E}$$
$$\prod_{n=1}^{N_{NE}} P\left(Y_n^{NE} = 1\big|x_n^{NE};\theta^{NE}\right)^{Y_n^{NE}}\left[1 - P\left(Y_n^{NE} = 1\big|x_n^{NE};\theta^{NE}\right)\right]^{1-Y_n^{NE}}.$$

- $Y \in \{\mathbb{1}(\text{Accept}), \mathbb{1}(\text{Attend Program}), \mathbb{1}(\text{Complete Program}), \mathbb{1}(\text{Good Placement})\}$
- **Test:** Let $g(\hat{\theta}) = \frac{1}{N_E}\sum_{n=1}^{N_E}\hat{P}_n^E(1-\hat{P}_n^E)\hat{\theta}_1^E - \frac{1}{N_{NE}}\sum_{n=1}^{N_{NE}}\hat{P}_n^{NE}(1-\hat{P}_n^{NE})\hat{\theta}_1^{NE}$. Then: $W = g(\hat{\theta})'\left[\nabla_\theta g(\hat{\theta})\hat{V}(\hat{\theta})\nabla_\theta g(\hat{\theta})'\right]^{-1} g(\hat{\theta}) \sim \chi_Q^2$. Compare AMEs per Mood (2010)

Insufficiency of Comparing AMEs

A1: Linear Wald Test Across Programs

Assume $\left(Y^E \perp\!\!\!\perp Y^{NE}\right)\middle|X$, i.e. $\left(\text{Economics} \perp\!\!\!\perp \text{Non-Economics}\right)\middle|\text{Predictors}$.

Let $f\left(y_n^E \middle| x_n^E; \beta^E, \sigma_E^2\right) = \frac{1}{\sqrt{2\pi\sigma_E^2}} \exp\left[\frac{-1}{2\sigma_E^2}\left(y_n^E - x_{n,0}^E\beta_0^E - x_{n,1}^E\beta_1^E\right)^2\right]$, and

$f\left(y_n^{NE} \middle| x_n^{NE}; \beta^{NE}, \sigma_S^2\right) = \frac{1}{\sqrt{2\pi\sigma_{NE}^2}} \exp\left[\frac{-1}{2\sigma_{NE}^2}\left(y_n^{NE} - x_{n,0}^{NE}\beta_0^{NE} - x_{n,1}^{NE}\beta_1^{NE}\right)^2\right]$. Then:

$$\mathcal{L}_N^U\left(y \middle| x; \beta, \sigma^2\right) = \prod_{n=1}^{N_E} f\left(y_n^E \middle| x_n^E; \beta^E, \sigma_E^2\right) \prod_{n=1}^{N_{NE}} f\left(y_n^{NE} \middle| x_n^{NE}; \beta^{NE}, \sigma_{NE}^2\right).$$

- $Y \in \{\mathbb{1}(\text{Standardized Committee Score}), \mathbb{1}(\text{Standardized Placement Rank})\}$
- **Test:** Let $g(\hat{\beta}_1) = \hat{\beta}_1^E - \hat{\beta}_1^{NE}$. (Can also do likelihood-ratio test.)
  Then $\text{W} = g(\hat{\beta}_1)'[\nabla_{\beta,\sigma^2} g(\hat{\beta}_1)\hat{V}(\hat{\beta}, \hat{\sigma}^2)\nabla_{\beta,\sigma^2} g(\hat{\beta}_1)']^{-1} g(\hat{\beta}_1) \sim \chi_Q^2$
- **Assumption:** Homoskedasticity (i.e. $\forall n,\ \sigma_n^2 = \sigma^2$)

Insufficiency of Comparing AMEs

A1: Additional Questions

- Are there different determinants of different types of success? (AKKLP 2007)
  - Need different skills at different stages in the program, such as grades vs. placement
  - For some programs, initial placement matters less than for economics
- Different determinants for international applicants? (KW 2000)
  - **Admission:** May have different preparation and/or interests
  - **Completion:** Depending on country, may have fewer outside options
  - **Placement:** May return to native country after graduation
- Different determinants over time? (2015 to 2023)
  - COVID-19 and other macro factors; changes in professions (e.g. more empirics)
  - For economics, increased importance of elite predocs and Fed RAships

Insufficiency of Comparing AMEs

A2: Portfolio Choice to Cutoff Rule (BEMS 2022)

Portfolio Choice Problem:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} P(y_n = S|x_n, z_n)\mathbb{1}(d_n = A), \text{ such that: } \sum_{n=1}^{N} \mathbb{1}(d_n = A) \leq N_A$$

$$\mathcal{L}(d_1,\ldots,d_N;\lambda) = \sum_{n=1}^{N} \left[P(y_n = S|x_n, z_n)\mathbb{1}(d_n = A) + \lambda\mathbb{1}(d_n = R)\right] + \lambda(N_A - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1,\ldots,N\},\ d_n(x_n, z_n|\lambda^*) = \begin{cases} A & \text{if } P(y_n = S|x_n, z_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S|x_n, z_n) = \lambda^* \\ R & \text{if } P(y_n = S|x_n, z_n) < \lambda^* \end{cases}$$

Upshot: λ^* is cutoff probability. Denote it hereafter as $P(y^* = S)$

- $P(y^* = S) \in \left[P(y_{N_A+1} = S|x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S|x_{N_A}, z_{N_A})\right]$

Model Framework

A2.1: Portfolio Choice to Cutoff Rule with Shocks

Portfolio Choice Problem:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} \Big\{ \big[P(y_n = S|x_n, z_n) + \epsilon_n(A)\big] \mathbb{1}(d_n = A) + \epsilon_n(R)\mathbb{1}(d_n = R) \Big\},$$

$$\text{such that: } \sum_{n=1}^{N} \mathbb{1}(d_n = A) \leq N_A. \quad \text{Then: } \mathcal{L}(d_1, \ldots, d_N; \lambda) =$$

$$\sum_{n=1}^{N} \Big\{ \big[P(y_n = S|x_n, z_n) + \epsilon_n(A)\big] \mathbb{1}(d_n = A) + \big[\lambda + \epsilon_n(R)\big] \mathbb{1}(d_n = R) \Big\} + \lambda(N_A - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \ldots, N\}, d_n(x_n, z_n, \epsilon_n|\lambda^*) = \begin{cases} A & \text{if } P(y_n = S|x_n, z_n) + \epsilon_n(A) > \lambda^* + \epsilon_n(R) \\ \{A, R\} & \text{if } P(y_n = S|x_n, z_n) + \epsilon_n(A) = \lambda^* + \epsilon_n(R) \\ R & \text{if } P(y_n = S|x_n, z_n) + \epsilon_n(A) < \lambda^* + \epsilon_n(R) \end{cases}$$

Theorem 4 (Proof)

For large N, there is a cutoff rule solution with logistic CCPs. Model Framework

A2: Models of Admissions

1. **Dynamic Model:** Committee ranks applicants based on conditional expectation in Round 1 about those who survive to Round 2, reads files of those above Cutoff 1, and rejects the rest. Favors those in Round 1 with objective variables positively correlated with subjective variables that predict success. Then re-ranks survivors based on all variables, accepts survivors above Cutoff 2, and rejects the rest
   1. **Static Model:** In Round 1, rank applicants based only on objective variables
   2. **Myopic Model:** Like dynamic, but distribution of z not conditional on x
2. **Matriculation-Dynamic Model:** In Round 1, rank applicants based on combination of success and matriculation probabilities (largely success probabilities)

Cutoff Rule Problem

A2: Description of Models

- **Round 2:** If accept applicant, get success probability $P(y = S|x, z, t, \overline{k})$, where Success $= \mathbb{1}$(Complete Program) or $\mathbb{1}$(Good Placement). If reject applicant, get $P(y_2^* = S|t) = P$(Success) of year t's marginal accepted applicant
- **Round 1:** If reject applicant, get $L/P(y_1^* = S|t) = L/P$(Success) of year t's marginal transitioned applicant. Can be less or greater than $P(y_2^* = S|t)$ depending on value placed on committee's time vs. admitting a successful cohort
  - **Dynamic:** If transition applicant, get $\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}]$ = expected value of Round 2, where z is integrated out conditional on (x, t) based on $F(z|x, t)$
  - **Static:** If transition applicant, get $P(y = S|x, t, \overline{k})$, or P(Success) unconditional on z. It may differ from $P(y = S|x, z, t, \overline{k})$, though it still accounts for z's effect on y
  - **Myopic:** In $\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}]$, replace $F(z|x, t)$ with $F(z|t)$

Cutoff Rule Problem

A2: Portfolio Choice to Cutoff Rule, Round 2

Portfolio Choice Problem:

$$\max_{(d_{2,1},\ldots,d_{2,N_T})\in\{A,R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S|x_n, z_n)\mathbb{1}(d_{2,n} = A), \text{ such that: } \sum_{n=1}^{N_T} \mathbb{1}(d_{2,n} = A) \leq N_A$$

$$\mathcal{L}(d_{2,1}, \ldots, d_{2,N_T}; \lambda_2) = \sum_{n=1}^{N_T} \left[P(y_n = S|x_n, z_n)\mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)\right] + \lambda_2(N_A - N_T)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \ldots, N_T\},\ d_{2,n}(x_n, z_n|\lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S|x_n, z_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S|x_n, z_n) = \lambda_2^* \\ R & \text{if } P(y_n = S|x_n, z_n) < \lambda_2^* \end{cases}$$

Upshot: λ_2^* is cutoff probability. Denote it hereafter as $P(y_2^* = S)$

- $P(y_2^* = S) \in \left[P(y_{N_A+1} = S|x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S|x_{N_A}, z_{N_A})\right]$

Dynamic Model

A2: Portfolio Choice to Cutoff Rule, Round 1 (Dynamic)

Portfolio Choice Problem:

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n;\overline{\lambda_2^*})\mathbb{1}(d_{1,n}=T), \text{ such that: } \sum_{n=1}^{N}\mathbb{1}(d_{1,n}=T)\leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n) = \int_{\widetilde{z}_n} \max\{P(y_n=S|x_n,\widetilde{z}_n),\overline{\lambda_2^*}\}dF_{z|x}(\widetilde{z}_n|x_n)$$

$$\mathcal{L}(d_{1,1},\ldots,d_{1,N};\lambda_1) = \sum_{n=1}^{N}\left[EV_{2,n}(x_n)\mathbb{1}(d_{1,n}=T)+\lambda_1\mathbb{1}(d_{1,n}=R)\right]+\lambda_1(N_T-N)$$

Cutoff Rule Solution:

$$\forall n\in\{1,\ldots,N\},\ d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n)>\lambda_1^* \\ \{T,R\} & \text{if } EV_{2,n}(x_n)=\lambda_1^* \\ R & \text{if } EV_{2,n}(x_n)<\lambda_1^* \end{cases}$$

Upshot: λ_1^* is cutoff level. Denote it hereafter as $L(y_1^*=S)\in\left[EV_{2,N_T+1},EV_{2,N_T}\right]$

Dynamic Model

A2: Dynamic Parameterization 1

Unrestricted CCPs:

$$P_{d_1} = P\left(d_1 = T \middle| x, t; \theta_T\right) = \frac{\exp(f_T(x,t)\theta_T)}{1 + \exp(f_T(x,t)\theta_T)}$$

$$P_{d_2} = P\left(d_2 = A \middle| x, z, t; \theta_A\right) = \frac{\exp(f_A(x,z,t)\theta_A)}{1 + \exp(f_A(x,z,t)\theta_A)}$$

Success Probability:

$$P_y = P\left(y = S \middle| x, z, t, k; \theta_S\right) = \frac{\exp(f_S(x,z,t,k)\theta_S)}{1 + \exp(f_S(x,z,t,k)\theta_S)}$$

Cutoff Level/Probability:

$$L_{y_1^*} = L\left(y_1^* = S \middle| t; \theta_{S_1^*}\right) = \exp(f_{S_1^*}(t)\theta_{S_1^*})$$

$$P_{y_2^*} = P\left(y_2^* = S \middle| t; \theta_{S_2^*}\right) = \frac{\exp(f_{S_2^*}(t)\theta_{S_2^*})}{1 + \exp(f_{S_2^*}(t)\theta_{S_2^*})}$$

Dynamic Model

A2: Dynamic Parameterization 2

$$\mathbb{E}[v_2(x,z,t,\overline{k},\epsilon_2)|x,t,\overline{k}] = \int_{\widetilde{z}} \sigma \log\left(\sum_{\widetilde{d}_2 \in \{A,R\}} e^{u_2(x,\widetilde{z},t,\overline{k},\widetilde{d}_2)/\sigma} \right) dF(\widetilde{z}|x,t)$$

$$= \int_{\widetilde{z}_1} \cdots \int_{\widetilde{z}_J} \left[\,\cdot\,\right] dF_1(\widetilde{z}_1|\widetilde{z}_2,\ldots,\widetilde{z}_J,x,t) \cdots dF_J(\widetilde{z}_J|x,t)$$

- **Continuous:** $f_{z|x,t}(z_j|x,t;\beta_{F_{z|x,t,j}},\sigma^2_{F_{z|x,t,j}}) = \frac{1}{\sqrt{2\pi\sigma^2_{F_{z|x,t,j}}}} \exp\left[\frac{-1}{2\sigma^2_{F_{z|x,t,j}}}(z_j - g(x,t)\beta_{F_{z|x,t,j}})^2\right]$
- **Binary:** $P(z_j = 1|x,t;\theta_{F_{z|x,t,j}}) = \frac{\exp\left(g(x,t)\theta_{F_{z|x,t,j}}\right)}{1+\exp\left(g(x,t)\theta_{F_{z|x,t,j}}\right)}$
- **Multinomial:** $P(z_j = \ell|x,t;\theta_{F_{z|x,t,j}}) = \frac{\exp\left(g(x,t)\theta^{\ell}_{F_{z|x,t,j}}\right)}{1+\sum_{i=1}^{L_j-1}\exp\left(g(x,t)\theta^{i}_{F_{z|x,t,j}}\right)}, \ \ell = 1,\ldots,L_j-1$

Dynamic Model

A2: Portfolio Choice to Cutoff Rule, Round 1 (Static)

Portfolio Choice Problem:

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} P(y_n = S|x_n)\mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = T) \leq N_T$$

$$\mathcal{L}(d_{1,1},\ldots,d_{1,N};\lambda_1) = \sum_{n=1}^{N} \left[P(y_n = S|x_n)\mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)\right] + \lambda_1(N_T - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1,\ldots,N\},\ d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } P(y_n = S|x_n) > \lambda_1^* \\ \{T,R\} & \text{if } P(y_n = S|x_n) = \lambda_1^* \\ R & \text{if } P(y_n = S|x_n) < \lambda_1^* \end{cases}$$

Upshot: λ_1^* is cutoff probability. Denote it hereafter as $P(y_1^* = S)$

- $P(y_1^* = S) \in \left[P(y_{N_T+1} = S|x_{N_T+1}), P(y_{N_T} = S|x_{N_T})\right]$

Static Model

A2: Dynamic Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{f\left(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_F\right)}_{\text{Density of } z^c|z^{\neg c}, x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}}$$

$$\prod_{n=N_T+1}^{N} \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^{N} \underbrace{f\left(z_n^{\neg c} | x_n, t_n; \theta_F\right)}_{\text{Density of } z^{\neg c}|x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- Success often unobservable for applicants from fewer than six years ago

▸▸ Maximum Likelihood Estimation

A2: Heckman (1979)-Corrected Dynamic Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{\Phi(x_n, t_n; \theta_H)}_{\text{Selection}} \underbrace{f\left(z_n | x_n, t_n, \frac{\phi(x_n, t_n; \theta_H)}{\Phi(x_n, t_n; \theta_H)}; \theta_F\right)}_{\text{Density of } z|x,t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)}}_{\text{Conditional choice probabilities}}$$

$$\underbrace{(1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probability}} \prod_{n=N_T+1}^{N} \underbrace{(1 - \Phi(x_n, t_n; \theta_H))}_{\text{Selection}} \underbrace{f\left(z_n^{\neg c} | x_n, t_n, \frac{-\phi(x_n, t_n; \theta_H)}{1 - \Phi(x_n, t_n; \theta_H)}; \theta_F\right)}_{\text{Density of } z^{\neg c}|x,t}$$

$$\underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^{N} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- *Committee Score* unobservable for applicants who didn't survive Round 1

Maximum Likelihood Estimation

A2: Empirical Considerations

- **Selection Effects:** Track success for all applicants, including non-matriculants
  - In selected samples, success parameters are biased down (e.g. quantitative GRE)
- **Potential Outcomes:** Track which graduate program all applicants attended, and add *Program Rank* as a control variable in success blocks of likelihood
  - In decision blocks containing success, fix *Program Rank* at this university's rank
  - If a variable only predicts success by helping applicant get into elite graduate program, then committee should not use that variable to rank applicants
- **Unobservables:** I may not observe all information that committee observes
  - **Example:** Recommendation letter has nuances that algorithms can't detect
  - Include subjective committee score in z as a proxy for this information

Maximum Likelihood Estimation

A2: Cutoff Rule with Matriculation

Portfolio Choice Problem:

$$\max_{(d_1,\dots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} P(y_n = S|w_n)P(m_n = M|w_n)\mathbb{1}(d_n = A),$$

$$\text{such that: } \sum_{n=1}^{N} P(m_n = M|w_n)\mathbb{1}(d_n = A) \leq N_M$$

$$\mathcal{L}(d_1,\dots,d_N;\lambda) = \sum_{n=1}^{N} \left[P(y_n = S|w_n)\mathbb{1}(d_n = A) + \lambda\mathbb{1}(d_n = R)\right] P(m_n = M|w_n) + \lambda(N_M - \overline{N_M})$$

Cutoff Rule Solution:

$$\forall n \in \{1,\dots,N\},\ d_n(w_n|\lambda^*) = \begin{cases} A & \text{if } P(y_n = S|w_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S|w_n) = \lambda^* \\ R & \text{if } P(y_n = S|w_n) < \lambda^* \end{cases}$$

Portfolio Choice with Matriculation

A2: Cutoff Rule with Matriculation (Large N)

Portfolio Choice Problem:

$$\max_{\delta\in\{\delta|\delta:\mathbb{R}^K\to[0,1]\}} \int_{w\in\mathbb{R}^K} P(y=S|w)P(m=M|w)\delta(w)f(w)dw,$$

$$\text{such that: } \int_{w\in\mathbb{R}^K} P(m=M|w)\delta(w)f(w)dw \leq N_M$$

$$\mathcal{L}(\delta;\lambda) = \int_{w\in\mathbb{R}^K} \left[P(y=S|w)\delta(w)+\lambda(1-\delta(w))\right]P(m=M|w)f(w)dw + \lambda(N_M-\overline{N_M})$$

Cutoff Rule Solution:

$$\forall w\in\mathbb{R}^K,\ \delta(w|\lambda^*) = \begin{cases} 1 & \text{if } P(y=S|w) > \lambda^* \\ [0,1] & \text{if } P(y=S|w) = \lambda^* \\ 0 & \text{if } P(y=S|w) < \lambda^* \end{cases}$$

Upshot: Point-identified cutoff probability λ^* is with respect to success probabilities

Portfolio Choice with Matriculation

A2: Cutoff Rule with Matriculation, Round 2

Portfolio Choice Problem:

$$\max_{(d_{2,1},\dots,d_{2,N_T})\in\{A,R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S|w_n)P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A),$$

$$\text{such that: } \sum_{n=1}^{N_T} P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A) \leq N_M$$

$$\mathcal{L}(d_{2,1},\dots,d_{2,N_T};\lambda_2)=\sum_{n=1}^{N_T}\left[P(y_n=S|w_n)\mathbb{1}(d_{2,n}=A)+\lambda_2\mathbb{1}(d_{2,n}=R)\right]P(m_n=M|w_n)+\lambda_2(N_M-\overline{N_M})$$

Cutoff Rule Solution:

$$\forall n \in \{1,\dots,N_T\},\ d_{2,n}(w_n|\lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S|w_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S|w_n) = \lambda_2^* \\ R & \text{if } P(y_n = S|w_n) < \lambda_2^* \end{cases}$$

Matriculation-Dynamic Model

A2: Cutoff Rule with Matriculation, Round 1

Portfolio Choice Problem:

$$\max_{(d_{1,1},\dots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n;\overline{\lambda_2^*})\mathbb{1}(d_{1,n}=T), \text{ such that: } \sum_{n=1}^{N}\mathbb{1}(d_{1,n}=T)\leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n) = \int_{\widetilde{z}_n} \max\{P(y_n=S|x_n,\widetilde{z}_n)-\overline{\lambda_2^*},0\}P(m_n=M|x_n,\widetilde{z}_n)dF_{z|x}(\widetilde{z}_n|x_n)$$

$$\mathcal{L}(d_{1,1},\dots,d_{1,N};\lambda_1) = \sum_{n=1}^{N}\left[EV_{2,n}(x_n)\mathbb{1}(d_{1,n}=T)+\lambda_1\mathbb{1}(d_{1,n}=R)\right]+\lambda_1(N_T-N)$$

Cutoff Rule Solution:

$$\forall n\in\{1,\dots,N\},\ d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n)>\lambda_1^* \\ \{T,R\} & \text{if } EV_{2,n}(x_n)=\lambda_1^* \\ R & \text{if } EV_{2,n}(x_n)<\lambda_1^* \end{cases}$$

Matriculation-Dynamic Model

A2: Matriculation-Dynamic Likelihood

Matriculation Probability:

$$P_m = P\left(m = M \middle| x, z, t; \theta_M\right) = \frac{\exp(f_M(x,z,t)\theta_M)}{1 + \exp(f_M(x,z,t)\theta_M)}$$

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_A} \underbrace{P_{m_n}(\theta_M)^{\mathbb{1}(m_n=M)}(1 - P_{m_n}(\theta_M))^{\mathbb{1}(m_n=D)}}_{\text{Matriculation probabilities}}$$

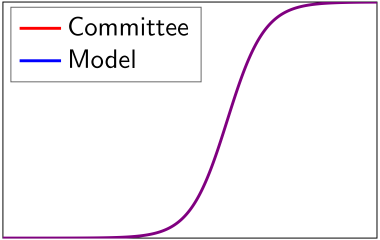
$$\prod_{n=1}^{N_T} \underbrace{f\left(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_F\right)}_{\text{Density of } z^c|z^{\neg c}, x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)}(1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}}$$

$$\prod_{n=N_T+1}^{N} \underbrace{\left(1 - P_{d_{1,n}}(\theta_T)\right)}_{\text{Conditional choice probability}} \prod_{n=1}^{N} \underbrace{f\left(z_n^{\neg c} | x_n, t_n; \theta_F\right)}_{\text{Density of } z^{\neg c}|x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

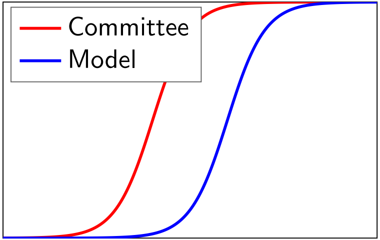
Maximum Likelihood Estimation

A2: How Deviation Gains Test Works

1. Write Bellman equations for committee's optimization problem
2. Estimate parameters. Rationality assumption distorts estimates
3. Re-estimate parameters without rationality, and use to solve model
4. Compare selections from model's decision rule to actual decision rule
5. To prevent overfitting, use perturbations of estimates for comparison



Optimal Committee



Suboptimal Committee

Deviation Gains Test

A2: Deviation Gains Algorithm

Dynamic/Static/Myopic:

1. Rank each t's applicants by $\mathbb{E}[v_2(x,z,t,\overline{k},\epsilon_2)|x,t,\overline{k}]$ with $\sigma=0$ for dynamic and myopic, or $P(y=S|x,t,\overline{k})$ for static. Transition top N_T, and reject the rest
2. Rank surviving applicants by $P(y=S|x,z,t,\overline{k})$, and accept top N_A survivors. Compute $\overline{P(y=S|x,z,t,\overline{k})}$ of acceptance set, and compare to committee's set

Matriculation-Dynamic/Static:

1. Rank each t's applicants by $\mathbb{E}[v_2(x,z,t,\overline{k},\epsilon_2)|x,t,\overline{k}]$ with $\sigma=0$ for dynamic, or $P(y=S|x,t,\overline{k})$ for static (best analogue). Transition top N_T, and reject the rest
2. Rank surviving applicants by $P(y=S|x,z,t,\overline{k})$ and accept them in descending order, each matriculating with probability $P(m=M|x,z,t)$, until N_M matriculate. Compute $\overline{P(y=S|x,z,t,\overline{k})}$ of matriculation set, and compare to committee's set

Deviation Gains Test

A2: Randomization Test

Theorem 5 (Proof)

Let d^R be the model's decision rule, and d^U the actual decision rule. By optimality:

$$\frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \overline{k}_n; \hat{\theta}_S^U\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d_{2,n}^R\left[P_{y_n}\left(\hat{\theta}_S^U\right)\right] = A\right\}$$
$$\geq \frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \overline{k}_n; \hat{\theta}_S^U\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d_{2,n}^U\left[P_{y_n}\left(\hat{\theta}_A^U\right)\right] = A\right\}$$

- But for perturbation $\tilde{\theta}_{S,b}^U$ of $\hat{\theta}_S^U$'s asymptotic distribution, can have:

$$\frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \overline{k}_n; \tilde{\theta}_{S,b}^U\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}_U\right)\right] = T, d_{2,n}^R\left[P_{y_n}\left(\hat{\theta}_S^U\right)\right] = A\right\}$$
$$< \frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \overline{k}_n; \tilde{\theta}_{S,b}^U\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}_U\right)\right] = T, d_{2,n}^U\left[P_{y_n}\left(\hat{\theta}_A^U\right)\right] = A\right\}$$

- For matriculation models, compare matriculants, not acceptances

Deviation Gains Test

A2: Summary of Simulated Results

- **Simulated Results:** Estimation results are well behaved
  - **Dynamic/Static:** Deviation gains significantly positive for simulated dataset where committee misweights applicant characteristics, but not for optimal simulated dataset
  - **Matriculation-Dynamic/Static:** Same as above
- Example of cutoff probability reversal if committee transitions few files in Round 1:
  - Three applicants: A, B, and C. Committee transitions two, accepts one
  - A is best and C is worst overall, but A is worst in objective variables
  - Committee rejects A in Round 1 and accepts B, who is worse than A
- By assuming h hours to read application file and w hourly wage, and calculating model's $\overline{P(\text{Success})}$ induced by all possible N_T, can put dollar value on success
  - Alternatively, what if committee uses AI to read every applicant's file?

Conclusion

A2: Extensions

- **Extension 1:** Uncover additional variables that predict matriculation but not failure
  - **Example:** Applicants with work experience more likely to succeed and matriculate
  - Can help committee find "people that we can afford" (*Moneyball*)
- **Extension 2:** Relax assumption that committee maximizes $P(\text{Success})$
  - Instead maximize combination of $P(\text{Success})$ and other objectives
  - **Example:** Committee may want diverse cohort (e.g. across fields): $P(\text{Success})^{\alpha} \times \text{Diversity}^{1-\alpha}$, where $\alpha \in [0, 1]$ can be estimated
- **Extension 3:** Waitlist and multiyear models (see online appendix)
  - Three-round models, and two-round models that repeat over infinite horizon
  - States contain info about past rounds/years, and committee values future rounds/years (memory includes matriculation probability and diversity)

Conclusion

Table 2: Economics Department Summary Statistics

(a) Objective Variables

	Full Sample		Transition = 1		Accept = 1		Matriculate = 1	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Transition	0.4204	0.4937	1.0000	0.0000	1.0000	0.0000	1.0000	0.0000
Waitlist	0.3324	0.4712	0.7686	0.4220	0.0743	0.2628	0.0896	0.2877
Accept	0.1103	0.3134	0.2625	0.4402	1.0000	0.0000	1.0000	0.0000
Matriculate	0.0339	0.1811	0.0807	0.2725	0.3074	0.4623	1.0000	0.0000
Open House	0.0353	0.1847	0.0817	0.2741	0.3267	0.4702	0.5522	0.5010
Attend Program	0.6024	0.4895	0.6752	0.4685	0.7704	0.4214	1.0000	0.0000
Complete Program	0.5578	0.4968	0.6210	0.4854	0.6654	0.4728	0.6709	0.4729
Top 100 Placement	0.1971	0.3979	0.2472	0.4316	0.3074	0.4623	0.3418	0.4773
Program Rank	88.9133	57.4823	78.5844	57.1361	63.3828	54.1845	37.0000	0.0000
Missing Program Rank	0.4199	0.4937	0.3422	0.4747	0.2490	0.4333	0.0000	0.0000
Placement Rank	127.7505	34.9306	123.0222	37.2570	117.2696	40.5746	117.4430	37.5728
Missing Placement Rank	0.4444	0.4970	0.3830	0.4864	0.3346	0.4728	0.3291	0.4729
2016	0.2052	0.4040	0.1696	0.3754	0.1790	0.3841	0.2025	0.4045
2017	0.2014	0.4011	0.1788	0.3833	0.1829	0.3873	0.1772	0.3843
2018	0.1928	0.3946	0.2686	0.4435	0.2062	0.4054	0.2532	0.4376
2019	0.2027	0.4021	0.2084	0.4064	0.2179	0.4136	0.2152	0.4136
Winsorized Age	25.9910	3.1479	26.2635	3.1823	26.0817	3.0652	26.3291	3.2806
Female	0.3912	0.4881	0.4147	0.4929	0.4397	0.4973	0.4177	0.4963
U.S.	0.2357	0.4245	0.2380	0.4261	0.2996	0.4590	0.3038	0.4628
International Asian	0.4809	0.4997	0.5230	0.4997	0.3424	0.4754	0.4304	0.4983
GRE Verbal Percentile	68.0129	22.5869	76.8421	18.3703	79.5703	18.3047	76.9744	19.4041
GRE Quantitative Percentile	88.4550	11.6330	93.1362	5.4135	93.5391	4.4614	92.3077	5.7554
GRE Analytic Percentile	50.0613	27.5744	56.3653	26.5466	60.3398	26.0880	57.4615	26.6481
Undergraduate GPA	3.6307	0.3026	3.6645	0.3003	3.7227	0.2610	3.7068	0.2880
TOEFL/IELTS	7.4376	0.5587	7.5641	0.5834	7.7011	0.6667	7.4464	0.6137
Elite U.S. University	0.1267	0.3327	0.1736	0.3790	0.2257	0.4188	0.1772	0.3843
Elite U.S. Liberal Arts College	0.0296	0.1696	0.0347	0.1832	0.0428	0.2028	0.0633	0.2450
Elite Foreign University	0.0386	0.1928	0.0490	0.2160	0.0350	0.1842	0.0253	0.1581
Other Foreign University	0.4049	0.4910	0.4188	0.4936	0.3619	0.4815	0.4430	0.4999
Missing Undergraduate Institution	0.2203	0.4145	0.1910	0.3933	0.1984	0.3996	0.1266	0.3346
Graduate Degree	0.5067	0.5001	0.5730	0.4949	0.5136	0.5008	0.5190	0.5028
Work Experience	0.5247	0.4995	0.5894	0.4922	0.6770	0.4685	0.7468	0.4376
#Professor Recommenders	2.6230	0.7371	2.6302	0.7165	2.5992	0.7118	2.5696	0.7104
#Prolific Recommenders	1.2602	1.0686	1.3912	1.0840	1.4825	1.0424	1.3038	1.0664
Missing GRE	0.0335	0.1800	0.0102	0.1006	0.0039	0.0624	0.0127	0.1125
Missing Undergraduate GPA	0.7853	0.4107	0.7487	0.4340	0.6770	0.4685	0.6835	0.4681
Missing TOEFL/IELTS	0.6076	0.4884	0.5935	0.4914	0.6615	0.4741	0.6456	0.4814
Missing Winsorized Age	0.0009	0.0293	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Observations	2329		979		257		79	

(b) Subjective Variables

	Full Sample		Transition = 1		Accept = 1		Matriculate = 1	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Committee Score	6.4070	1.8701	6.4070	1.8701	8.5003	0.8151	8.4854	0.7157
#Math Courses	5.7637	1.8154	5.8173	1.7904	5.6992	1.8640	5.6585	1.6219
Advanced Math	0.4860	0.5004	0.4909	0.5007	0.5680	0.4973	0.4651	0.5047
Letters Length/100	8.3474	3.5749	8.5333	3.8263	10.0514	4.2864	9.7088	3.4876
Letters Positive Terms (%)	11.9770	2.2993	11.9976	2.2077	12.2167	1.9255	12.5110	2.0904
Letters Negative Terms (%)	5.7286	1.3427	5.7216	1.3318	5.7446	1.2313	5.8938	1.2871
Letters Standout Terms (%)	0.6241	0.4020	0.6241	0.4029	0.7151	0.4509	0.7077	0.5109
Letters Ability Terms (%)	1.3326	0.6239	1.3457	0.5964	1.3796	0.5620	1.4334	0.6037
Letters Research Terms (%)	3.4837	1.3319	3.5208	1.3423	3.8548	1.1984	3.8395	1.2121
Letters Grindstone Terms (%)	1.8189	0.7734	1.8674	0.7808	1.9975	0.7588	2.1600	0.8093
Letters Teaching Terms (%)	1.7379	0.7710	1.7489	0.7759	1.7275	0.7536	1.8976	0.7577
Letters Communal Terms (%)	0.5946	0.3890	0.6019	0.3954	0.5494	0.3234	0.5630	0.3875
Letters Agentic Terms (%)	0.4768	0.3150	0.4842	0.3229	0.5039	0.3222	0.5597	0.3314
Letters Terms (%) PC1	-0.0579	1.6183	0.0000	1.5768	0.2176	1.3817	0.5697	1.5386
Letters Terms (%) PC2	-0.0271	1.1177	0.0000	1.1446	0.1002	1.1774	0.1699	1.2823
CV Length/100	4.3532	2.4237	4.3982	2.4758	4.9641	2.7175	4.8261	2.3473
CV Coding	2.8321	2.1135	2.7931	2.1000	2.8618	2.1129	2.5122	1.6450
CV Publication/Working Paper	0.2518	0.4346	0.2571	0.4377	0.3659	0.4836	0.3902	0.4939
Essay Length/100	4.5099	1.5741	4.5173	1.5805	4.7488	1.6373	4.9816	1.7932
Essay Specific Professor	0.4822	0.5003	0.5219	0.5003	0.5691	0.4972	0.6512	0.4822
Essay Economics Terms (%)	9.7284	2.6219	9.7216	2.6635	9.5303	2.7324	9.8941	3.2608
Essay Research Terms (%)	5.7975	2.0501	5.7402	2.0168	5.7313	1.8797	5.5231	1.8432
Essay Positive Terms (%)	11.8293	2.6518	11.7324	2.6650	11.4734	2.4592	11.3169	2.4455
Essay Negative Terms (%)	6.2844	1.9056	6.3233	1.9712	6.2889	2.0860	6.2651	1.9436
Missing Committee Score	0.5844	0.4929	0.0112	0.1055	0.0195	0.1384	0.0633	0.2450
Missing Application Pdf	0.8154	0.3881	0.6650	0.4722	0.5136	0.5008	0.4557	0.5012
Missing #Math Courses	0.8201	0.3842	0.6701	0.4704	0.5214	0.5005	0.4810	0.5028
Missing Letters	0.8175	0.3863	0.6691	0.4708	0.5136	0.5008	0.4557	0.5012
Missing CV	0.8210	0.3835	0.6742	0.4689	0.5214	0.5005	0.4810	0.5028
Missing Essay	0.8192	0.3849	0.6731	0.4693	0.5214	0.5005	0.4557	0.5012
Observations	2329		979		257		79	

Regression Results

A3: Economics Department Correlation Matrix

Table 3: Economics Department Correlation Matrix

	Trans	Wait	Acc	Mat	OH	Att	Com	T100	ProR	MProR	PlaR	MPlaR	2016	2017	2018	2019	Age
Transition	1																
Waitlist	0.8083	1															
Accept	0.4136	-0.1908	1														
Matriculate	0.2200	-0.0994	0.5320	1													
Open House	0.2192	-0.1104	0.5496	0.5400	1												
Attend Program	0.1266	0.0767	0.1209	0.1522	0.0833	1											
Complete Program	0.1085	0.0889	0.0763	0.0427	0.0154	0.9124	1										
Top 100 Placement	0.1073	0.0778	0.0977	0.0682	0.0498	0.4025	0.4412	1									
Program Rank	-0.1531	-0.0909	-0.1565	-0.1693	-0.0999	-0.8635	-0.7942	-0.4106	1								
Missing Program Rank	-0.1341	-0.0860	-0.1219	-0.1594	-0.0856	-0.9548	-0.8749	-0.4018	0.9044	1							
Placement Rank	-0.1153	-0.0802	-0.1057	-0.0553	-0.0666	-0.5176	-0.5673	-0.9004	0.4999	0.5032	1						
Missing Placement Rank	-0.1051	-0.0835	-0.0778	-0.0435	-0.0164	-0.9084	-0.9957	-0.4431	0.7924	0.8708	0.5698	1					
2016	-0.0752	-0.0883	-0.0229	-0.0013	-0.0126	-0.0194	0.0073	-0.0005	0.0077	0.0178	-0.0069	-0.0073	1				
2017	-0.0480	-0.0732	-0.0162	-0.0113	0.0162	0.0076	0.0138	0.0096	-0.0025	-0.0064	-0.0040	-0.0138	-0.2552	1			
2018	0.1637	0.1695	0.0120	0.0287	0.0009	0.0101	0.0078	0.0096	-0.0118	-0.0233	-0.0076	-0.0034	-0.2483	-0.2454	1		
2019	0.0121	-0.0050	0.0133	0.0058	-0.0045	0.0145	-0.0565	-0.0081	0.0123	0.0017	0.0166	0.0543	-0.2562	-0.2532	-0.2464	1	
Winsorized Age	0.0738	0.0157	0.0102	0.0201	0.0122	-0.0260	-0.0179	0.0398	0.0474	0.0304	-0.0148	0.0229	0.0102	-0.0091	-0.0128	0.0160	1
Female	0.0411	0.0255	0.0350	0.0102	0.0328	-0.0068	-0.0179	-0.0189	0.0074	0.0168	0.0007	0.0198	0.0066	-0.0163	0.0142	0.0358	-0.1468
U.S.	0.0046	-0.0393	0.0530	0.0301	0.0318	-0.1069	-0.1165	-0.0310	0.0900	0.1075	0.0538	0.1141	0.0083	0.0263	-0.0073	-0.0409	-0.1537
International Asian	0.0717	0.1739	-0.0976	-0.0189	-0.0210	0.1744	0.1736	0.0416	-0.1725	-0.1816	-0.0646	-0.1707	-0.0210	-0.0162	-0.0107	0.0407	-0.0407
GRE Verbal Percentile	0.3349	0.2346	0.1825	0.0746	0.1228	0.0524	0.0343	0.0479	-0.0681	-0.0536	-0.0548	-0.0358	-0.0272	-0.0158	0.0260	0.0343	-0.1056
GRE Quantitative Percentile	0.3448	0.2791	0.1559	0.0623	0.0605	0.1957	0.1930	0.0980	-0.2102	-0.1965	-0.1287	-0.1899	0.0061	0.0095	0.0116	-0.0255	-0.0622
GRE Analytic Percentile	0.1960	0.1198	0.1330	0.0505	0.0710	-0.0468	-0.0700	-0.0083	0.0359	0.0504	0.0058	0.0695	-0.0542	0.0113	0.0537	0.0379	-0.1533
Undergraduate GPA	0.0517	-0.0095	0.0747	0.0322	0.0440	0.0329	0.0179	-0.0166	-0.0382	-0.0336	0.0056	-0.0179	-0.0272	-0.0058	0.0260	0.0109	-0.0853
TOEFL/IELTS	0.1251	0.1060	0.0898	0.0017	0.0413	-0.0200	-0.0127	0.0342	-0.0053	0.0226	-0.0431	0.0108	-0.0018	0.0047	0.0176	-0.0092	-0.1177
Elite U.S. University	0.1203	0.0775	0.1048	0.0285	0.0769	-0.0151	-0.0196	-0.0167	-0.0171	0.0108	0.0124	0.0179	-0.0305	-0.0271	0.0070	0.0392	-0.0769
Elite U.S. Liberal Arts College	0.0256	-0.0020	0.0274	0.0372	0.0205	-0.0081	-0.0280	-0.0038	-0.0103	0.0104	-0.0012	0.0272	0.0178	0.0133	-0.0405	-0.0125	-0.0639
Elite Foreign University	0.0459	0.0691	-0.0066	-0.0130	-0.0062	0.0218	0.0305	0.0071	-0.0445	-0.0261	0.0027	-0.0314	0.0084	0.0215	-0.0246	-0.0290	-0.0129
Other Foreign University	0.0241	0.0431	-0.0309	0.0146	-0.0327	0.0571	0.0564	0.0311	-0.0260	-0.0496	-0.0454	-0.0512	-0.0228	-0.0085	-0.0772	0.1586	0.1709
Missing Undergraduate Institution	-0.0601	-0.0583	-0.0185	-0.0424	-0.0215	0.0169	0.0435	0.0180	-0.0124	-0.0198	-0.0227	-0.0458	0.0300	-0.0008	0.1946	-0.2576	-0.0011
Graduate Degree	0.1131	0.1254	0.0049	0.0046	0.0248	0.0529	0.0706	0.0226	-0.0399	-0.0548	-0.0328	-0.0698	0.0443	0.1379	-0.1556	-0.0729	0.3704
Work Experience	0.1103	0.0043	0.1074	0.0834	0.0484	-0.0723	-0.0789	0.0371	0.0722	0.0851	-0.0148	0.0795	-0.0017	-0.0002	-0.0056	0.0093	0.4141
#Professor Recommenders	0.0083	0.0374	-0.0114	-0.0136	-0.0257	0.0606	0.0466	-0.0073	-0.0472	-0.0547	0.0069	-0.0514	-0.0791	0.0186	0.0329	0.0390	-0.2398
#Prolific Recommenders	0.1044	0.0794	0.0733	0.0076	0.0374	0.1042	0.1084	0.0632	-0.1156	-0.1054	-0.0567	-0.1070	-0.0262	0.0330	0.0430	0.0062	0.0200
Missing GRE	-0.1102	-0.1318	-0.0579	-0.0217	-0.0358	-0.0731	-0.0649	-0.0322	0.0715	0.0786	0.0303	0.0641	0.0118	-0.0042	-0.0123	0.0071	0.0718
Missing Undergraduate GPA	-0.0759	-0.0361	-0.0929	-0.0464	-0.0501	0.0261	0.0292	0.0119	-0.0105	-0.0213	-0.0191	-0.0269	-0.0372	-0.0582	-0.0334	0.2064	0.1637
Missing TOEFL/IELTS	-0.0246	-0.0503	0.0389	0.0146	0.0438	-0.1211	-0.1296	-0.0660	0.1027	0.1244	0.0753	0.1260	-0.0248	-0.0372	-0.0241	0.0552	-0.1342
Missing Winsorized Age	-0.0250	-0.0231	-0.0103	-0.0055	-0.0063	-0.0361	-0.0329	-0.0145	0.0312	0.0345	0.0187	0.0328	0.0214	0.0218	-0.0143	-0.0148	0.0000
Committee Score	0.0000	-0.3453	0.5999	0.3028	0.3395	0.0873	0.0554	0.0493	-0.1117	-0.0842	-0.0463	-0.0594	-0.0390	-0.0215	0.0691	0.0682	-0.0598
#Math Courses	0.0107	0.0311	-0.0215	-0.0177	-0.0467	0.0368	0.0367	0.0132	-0.0467	-0.0422	-0.0177	-0.0375	0.0291	0.0473	0.0642	-0.1633	-0.0764
Advanced Math	0.2226	0.1304	0.2298	0.1071	0.1558	0.0463	0.0406	0.0031	-0.0742	-0.0510	-0.0116	-0.0420	-0.0182	-0.0752	0.0408	0.0398	-0.0349
Letters Length/100	0.0189	-0.1069	0.1781	0.0827	0.0792	0.0261	0.0189	0.0146	-0.0368	-0.0238	-0.0114	-0.0197	0.0075	-0.0049	-0.0155	0.0170	0.0376
Letters Positive Terms (%)	0.0161	-0.0760	0.0674	0.0780	0.0058	-0.0500	-0.0268	0.0708	0.0375	0.0280	-0.0454	0.0186	0.0464	-0.0001	-0.0522	0.0536	0.0432
Letters Negative Terms (%)	-0.0093	-0.0545	0.0077	0.0413	-0.0479	-0.0158	-0.0610	-0.0737	0.0245	0.0241	0.0592	0.0574	0.0186	0.0081	0.0824	-0.0272	0.0807
Letters Standout Terms (%)	0.0003	-0.1493	0.1464	0.0698	-0.0346	0.0222	0.0131	0.0878	-0.0031	-0.0252	-0.0422	-0.0153	0.0733	-0.0587	-0.0863	0.0900	0.0507
Letters Ability Terms (%)	0.0375	-0.0301	0.0487	0.0543	0.0006	-0.1785	-0.1573	-0.0137	0.1340	0.1765	0.0371	0.1502	0.0512	0.0014	0.0153	-0.0218	0.0866
Letters Research Terms (%)	0.0499	-0.0879	0.1800	0.0897	0.0944	0.0439	0.0375	0.0492	-0.0771	-0.0588	-0.0934	-0.0376	-0.0062	-0.0162	0.1051	0.0225	0.0258
Letters Grindstone Terms (%)	0.1125	-0.0320	0.1493	0.1481	0.0630	-0.1067	-0.1156	-0.0128	0.0728	0.1050	0.0347	0.1104	-0.0133	-0.0563	-0.0136	0.0294	0.1012
Letters Teaching Terms (%)	0.0258	-0.0022	-0.0087	0.0696	0.0224	-0.1522	-0.1563	0.0241	0.1552	0.1493	0.0234	0.1579	0.0595	-0.0270	-0.0398	0.0459	0.1921
Letters Communal Terms (%)	0.0338	0.0517	-0.0749	-0.0272	-0.0216	-0.0020	0.0357	-0.0500	0.0132	0.0135	0.0186	-0.0394	0.0324	0.0711	-0.0603	-0.0627	0.0382
Letters Agentic Terms (%)	0.0421	0.0589	0.0557	0.0884	0.0913	-0.0509	-0.0716	0.0030	0.0147	0.0225	0.0016	0.0686	0.0927	-0.0091	0.0022	-0.0789	-0.0172
Letters Terms (%) PC1	0.0130	-0.0283	0.0593	0.0870	0.0251	-0.0585	-0.0577	0.0109	0.0481	0.0525	0.0028	0.0547	0.0310	-0.0050	-0.0129	0.0043	0.0514
Letters Terms (%) PC2	0.0088	-0.0033	0.0396	0.0383	0.0466	-0.0062	-0.0170	-0.0190	-0.0053	0.0063	0.0007	0.0174	-0.0204	-0.0007	0.0628	-0.0096	0.0192
CV Length/100	0.0067	-0.0429	0.0959	0.0421	0.0772	0.0199	0.0202	0.0299	-0.0124	-0.0185	-0.0335	-0.0222	0.0320	-0.0248	-0.0129	-0.0118	0.0394
CV Coding	-0.0067	-0.0225	0.0102	-0.0320	-0.0333	0.0280	0.0338	0.0297	-0.0250	-0.0246	-0.0377	-0.0365	-0.0225	-0.0306	0.0089	0.0368	-0.0633
CV Publication/Working Paper	0.1587	0.0464	0.2207	0.1422	0.1135	0.0666	0.0560	0.0432	-0.0707	-0.0633	-0.0408	-0.0569	-0.0284	-0.0575	0.0197	0.0603	0.0894
Essay Length/100	0.0017	-0.0422	0.0591	0.0712	0.0195	0.0473	0.0287	0.0026	-0.0530	-0.0452	-0.0053	-0.0274	0.0204	-0.0137	-0.0110	0.0253	0.0388
Essay Specific Professor	0.2518	0.1247	0.2312	0.1775	0.1550	0.0489	0.0361	0.0306	-0.0633	-0.0470	-0.0396	-0.0374	-0.0440	-0.0641	0.0419	0.0752	0.0149
Essay Economics Terms (%)	-0.0009	0.0317	-0.0293	0.0156	-0.0042	-0.0358	-0.0452	-0.0431	0.0338	0.0270	0.0456	0.0426	-0.0102	0.0036	0.0191	-0.0257	-0.0088
Essay Research Terms (%)	-0.0101	0.0042	-0.0059	-0.0277	-0.0234	0.0838	0.0766	0.0144	-0.0690	-0.0720	-0.0305	-0.0802	-0.0296	0.0235	-0.0081	0.0445	-0.0767
Essay Positive Terms (%)	-0.0132	0.0211	-0.0442	-0.0405	-0.0379	-0.0035	0.0102	0.0332	0.0043	0.0060	-0.0127	-0.0110	-0.0123	-0.0010	0.0151	0.0029	-0.0450
Essay Negative Terms (%)	0.0074	0.0139	-0.0041	-0.0057	-0.0264	0.0107	0.0005	-0.0066	-0.0026	-0.0143	0.0085	-0.0030	0.0015	0.0056	-0.0294	0.0257	0.0215
Missing Committee Score	-0.9903	-0.7997	-0.4037	-0.1981	-0.2043	-0.1226	-0.1054	-0.1012	0.1477	0.1297	0.1106	0.1020	0.0748	0.0433	-0.1554	-0.0105	-0.0656
Missing Application Pdf	-0.3301	-0.2162	-0.2739	-0.1737	-0.1677	-0.0565	-0.0516	-0.0230	0.0840	0.0596	0.0243	0.0514	0.0528	0.0955	-0.0508	-0.1152	-0.0056
Missing #Math Courses	-0.3326	-0.2159	-0.2739	-0.1654	-0.1709	-0.0585	-0.0547	-0.0293	0.0872	0.0610	0.0309	0.0544	0.0498	0.0902	-0.0516	-0.1115	-0.0077
Missing Letters	-0.3273	-0.2105	-0.2771	-0.1755	-0.1700	-0.0545	-0.0491	-0.0230	0.0815	0.0574	0.0229	0.0489	0.0502	0.0986	-0.0509	-0.1130	-0.0081
Missing CV	-0.3261	-0.2095	-0.2752	-0.1661	-0.1568	-0.0590	-0.0528	-0.0248	0.0842	0.0615	0.0230	0.0525	0.0571	0.0893	-0.0528	-0.1156	-0.0142
Missing Essay	-0.3233	-0.2066	-0.2726	-0.1770	-0.1718	-0.0533	-0.0476	-0.0253	0.0819	0.0560	0.0253	0.0474	0.0508	0.0912	-0.0448	-0.1101	-0.0035
Observations	2329																

	Fem	US	IntAsn	GREV	GREQ	GREA	GPA	T/I	EUni	ELA	EFor	OFor	MUg	Grad	Work	Prof	Pro
Female	1																
U.S.	-0.0886	1															
International Asian	0.1108	-0.5345	1														
GRE Verbal Percentile	-0.0155	0.2891	-0.1077	1													
GRE Quantitative Percentile	-0.0398	-0.1080	0.2269	0.2114	1												
GRE Analytic Percentile	0.0022	0.3950	-0.2554	0.5519	-0.0386	1											
Undergraduate GPA	0.0693	0.0418	-0.0425	0.0835	0.0793	0.0863	1										
TOEFL/IELTS	0.0969	0.0119	-0.0524	0.2772	0.0337	0.2465	0.0031	1									
Elite U.S. University	0.0095	0.2082	-0.0410	0.1155	0.0303	0.1320	-0.0917	0.0035	1								
Elite U.S. Liberal Arts College	0.0260	0.1177	-0.0567	0.0805	0.0260	0.1074	0.0216	0.0113	-0.0665	1							
Elite Foreign University	0.0447	-0.1008	0.1325	0.0335	0.0642	-0.0303	0.0047	0.0366	-0.0764	-0.0350	1						
Other Foreign University	-0.0105	-0.4231	0.2355	-0.1253	0.0709	-0.2141	-0.0002	-0.0057	-0.3141	-0.1441	-0.1654	1					
Missing Undergraduate Institution	0.0092	-0.0364	-0.0761	-0.0414	-0.0056	-0.0322	0.0301	-0.0164	-0.2024	-0.0929	-0.1066	-0.4384	1				
Graduate Degree	-0.0010	-0.2492	0.1402	-0.1043	0.0039	-0.1313	-0.0877	0.0107	-0.1458	-0.0808	0.0597	0.1649	0.0831	1			
Work Experience	-0.0229	0.0870	-0.2765	0.0369	-0.1144	0.1031	0.0121	0.0128	0.0083	0.0294	-0.0367	-0.0399	0.0163	0.1219	1		
#Professor Recommenders	-0.0114	-0.0330	0.1402	-0.0051	0.1028	-0.0399	0.0557	-0.0180	-0.0592	0.0069	0.0270	0.0136	-0.0023	-0.0631	-0.2477	1	
#Prolific Recommenders	-0.0017	-0.2072	0.1686	-0.0282	0.1814	-0.1224	-0.0547	0.0052	-0.0311	-0.0497	0.0492	0.1487	-0.0014	0.1173	-0.0724	0.2047	1
Missing GRE	0.0122	-0.0303	-0.0502	-0.0081	-0.0129	0.0026	-0.0117	-0.0034	0.0080	-0.0325	0.0122	-0.0028	-0.0183	-0.0073	0.0434	-0.0375	-0.0163
Missing Undergraduate GPA	-0.0095	-0.3206	0.1579	-0.2161	0.0599	-0.3214	0.0000	-0.0211	-0.2096	-0.1183	0.0940	0.3226	0.0180	0.1345	-0.0097	0.0036	0.1459
Missing TOEFL/IELTS	-0.0207	0.4256	-0.2472	0.0917	-0.1772	0.2285	0.0072	0.0000	0.2876	0.1352	-0.0396	-0.3778	-0.1160	-0.1265	0.0345	-0.0031	-0.2002
Missing Winsorized Age	-0.0235	-0.0163	-0.0282	-0.0442	-0.0172	0.0000	0.0000	0.0000	-0.0112	-0.0051	-0.0059	0.0057	0.0198	0.0289	0.0279	0.0150	-0.0071
Committee Score	-0.0014	-0.0064	-0.0812	0.0408	0.0378	0.0481	0.0800	0.0726	0.0496	-0.0133	-0.0126	-0.0104	0.0234	-0.0609	0.0246	0.0279	0.0849
#Math Courses	-0.0104	0.0306	0.0164	0.0307	-0.0093	0.0315	-0.0128	0.0409	0.0339	-0.0032	-0.0111	-0.0406	-0.0053	-0.0392	0.0041	-0.0020	-0.0002
Advanced Math	-0.0054	-0.0328	0.0827	0.0181	0.1157	-0.0207	0.0099	0.0072	0.0701	0.0337	0.0150	0.0042	-0.0763	-0.0478	-0.0561	0.0363	0.1345
Letters Length/100	0.0367	0.0330	-0.0790	0.0531	-0.0110	0.0579	-0.0050	0.0487	0.0012	0.0635	0.0250	-0.0216	-0.0030	-0.0020	0.1001	-0.0420	0.0333
Letters Positive Terms (%)	-0.0145	0.1376	-0.0883	0.0349	0.0533	0.1164	-0.0099	-0.0573	-0.0018	0.1339	-0.0416	-0.0931	-0.0256	-0.0638	0.0769	-0.1049	-0.1867
Letters Negative Terms (%)	-0.0036	0.1392	-0.0830	0.0764	-0.0665	0.0632	-0.0612	-0.1270	0.0379	0.0199	0.0261	-0.0506	-0.0449	0.0424	0.0943	-0.1391	0.0507
Letters Standout Terms (%)	-0.0141	0.0120	-0.0394	0.0100	0.0078	0.0742	0.0477	0.0066	-0.1116	0.0846	0.0073	0.0514	-0.0690	-0.0211	0.0198	0.0226	0.0132
Letters Ability Terms (%)	-0.0015	0.1424	-0.1930	0.0423	0.0487	0.1273	0.0808	-0.0476	-0.0047	0.0583	0.0035	-0.1331	0.0026	0.0081	0.1310	-0.1811	-0.2310
Letters Research Terms (%)	0.0245	0.1273	-0.0539	0.1004	-0.0220	0.1052	-0.0448	0.0235	0.0296	0.0482	-0.0391	-0.0767	0.0477	0.0037	0.1626	-0.2141	-0.0945
Letters Grindstone Terms (%)	0.0839	0.1233	-0.1591	0.0209	-0.0701	0.0946	0.0185	-0.0524	0.0140	0.1687	-0.0368	-0.1797	0.0350	0.0451	0.2148	-0.1778	-0.1497
Letters Teaching Terms (%)	0.0369	0.0349	-0.1064	-0.0334	-0.0692	0.0481	0.0218	-0.0055	-0.0565	0.0274	-0.0166	0.0065	-0.0288	0.0670	0.1777	-0.0771	-0.1383
Letters Communal Terms (%)	-0.0136	-0.1655	0.0499	-0.0011	0.0538	0.0001	0.0203	-0.1205	-0.0344	0.0260	0.0362	-0.0034	0.0305	0.0770	-0.0646	0.1095	-0.0050
Letters Agentic Terms (%)	0.1191	0.1148	-0.0943	0.0638	-0.0830	0.0517	-0.0681	0.0347	0.0740	0.1425	-0.0270	-0.1247	0.0009	-0.0336	0.0006	-0.1529	-0.0772
Letters Terms (%) PC1	0.0207	0.0601	-0.0803	0.0219	-0.0120	0.0601	0.0043	-0.0258	-0.0053	0.0708	-0.0127	-0.0645	-0.0019	0.0041	0.0764	-0.0732	-0.0963
Letters Terms (%) PC2	0.0150	0.0516	-0.0323	0.0285	-0.0245	0.0213	-0.0207	-0.0088	0.0366	-0.0004	-0.0110	-0.0449	0.0199	0.0192	0.0790	-0.0774	-0.0236
CV Length/100	0.0249	0.0341	-0.0336	0.0120	-0.0114	0.0448	0.0014	0.0399	-0.0027	0.0241	-0.0241	0.0074	-0.0123	0.0314	0.1054	-0.0738	0.0126
CV Coding	-0.0004	-0.0209	0.0151	-0.0125	0.0021	-0.0083	-0.0131	0.0023	0.0122	0.0125	-0.0072	0.0100	-0.0158	0.0065	-0.0100	-0.0106	0.0452
CV Publication/Working Paper	-0.0215	-0.0475	0.0269	-0.0141	0.0741	-0.0197	-0.0054	-0.0400	0.0292	0.0109	-0.0006	0.0358	-0.0256	0.0282	0.0825	-0.0292	0.0846
Essay Length/100	-0.0133	-0.0100	0.0078	-0.0314	-0.0020	-0.0472	-0.0304	-0.0084	-0.0051	-0.0053	0.0103	0.0145	-0.0109	0.0979	0.0273	0.0028	0.0398
Essay Specific Professor	-0.0044	-0.0712	0.0804	-0.0046	0.1188	-0.0416	0.0123	0.0222	0.0562	-0.0091	-0.0067	0.0521	-0.0577	0.0065	-0.0138	0.0279	0.1057
Essay Economics Terms (%)	-0.0026	-0.0373	0.0596	-0.0069	0.0022	-0.0280	0.0313	-0.0115	0.0065	-0.0007	0.0222	-0.0436	0.0058	0.0017	-0.0049	-0.0217	0.0205
Essay Research Terms (%)	0.0498	-0.0432	0.0665	-0.0242	0.0132	-0.0050	0.0206	-0.0017	0.0157	0.0134	0.0160	-0.0295	-0.0003	-0.0478	-0.0345	0.0073	0.0095
Essay Positive Terms (%)	0.0208	-0.0019	0.0002	-0.0149	0.0057	0.0096	0.0287	0.0103	0.0155	-0.0262	-0.0164	-0.0213	-0.0069	-0.0358	-0.0021	0.0141	-0.0244
Essay Negative Terms (%)	-0.0251	-0.0063	0.0323	-0.0046	0.0103	-0.0007	-0.0384	-0.0316	0.0534	0.0249	0.0123	-0.0148	-0.0414	-0.0002	0.0003	-0.0086	0.0500
Missing Committee Score	-0.0471	-0.0078	-0.0793	-0.3358	-0.3437	-0.1972	-0.0516	-0.1281	-0.1189	-0.0273	-0.0479	-0.0179	0.0551	-0.1108	-0.1049	-0.0094	-0.1045
Missing Application Pdf	-0.0041	0.1078	-0.1112	0.0270	-0.1797	0.0709	-0.0220	0.0179	-0.0650	-0.0082	-0.0079	-0.0674	0.0874	0.0263	0.0656	-0.0632	-0.1400
Missing #Math Courses	-0.0025	0.1021	-0.1108	0.0188	-0.1817	0.0657	-0.0200	0.0155	-0.0670	-0.0039	-0.0105	-0.0646	0.0844	0.0275	0.0646	-0.0636	-0.1423
Missing Letters	-0.0063	0.1052	-0.1060	0.0263	-0.1765	0.0686	-0.0209	0.0219	-0.0674	-0.0092	-0.0091	-0.0632	0.0848	0.0274	0.0645	-0.0622	-0.1461
Missing CV	-0.0020	0.1142	-0.1198	0.0255	-0.1806	0.0760	-0.0193	0.0171	-0.0579	-0.0109	-0.0110	-0.0757	0.0915	0.0297	0.0556	-0.0596	-0.1431
Missing Essay	-0.0099	0.1058	-0.1106	0.0263	-0.1746	0.0683	-0.0222	0.0160	-0.0660	-0.0101	-0.0042	-0.0649	0.0881	0.0230	0.0668	-0.0601	-0.1415

	MGRE	MGPA	MT/I	MAge	Score	NMath	AMath	LetN	LPos	LNeg	LSta	LAbi	LRes	LGri	LTea	LCom	LAge
Missing GRE	1																
Missing Undergraduate GPA	0.0625	1															
Missing TOEFL/IELTS	0.0714	-0.2810	1														
Missing Winsorized Age	0.1575	0.0153	0.0236	1													
Committee Score	-0.0451	-0.0466	0.0188	0.0000	1												
#Math Courses	0.0170	-0.0632	0.0423	0.0004	-0.0232	1											
Advanced Math	-0.0417	-0.0261	-0.0030	-0.0092	0.1388	0.2255	1										
Letters Length/100	-0.0082	-0.0084	0.0034	0.0007	0.1631	0.0027	0.0502	1									
Letters Positive Terms (%)	0.1234	-0.0626	-0.0402		0.0587	-0.0070	-0.0916	0.0015	1								
Letters Negative Terms (%)	0.0151	-0.0109	0.1318		0.0330	0.0100	-0.0104	0.0546	0.1546	1							
Letters Standout Terms (%)	0.1382	0.0456	-0.0033		0.1475	0.0194	-0.1303	0.0427	0.4324	-0.0327	1						
Letters Ability Terms (%)	-0.0190	-0.0979	0.0358		0.0943	-0.0408	-0.1419	0.0162	0.5004	0.1896	0.2105	1					
Letters Research Terms (%)	-0.0446	0.0039	0.0552		0.1776	0.0127	0.0874	0.2071	0.0820	0.1569	-0.0520	0.0863	1				
Letters Grindstone Terms (%)	0.0905	-0.0876	0.0740		0.1127	-0.0523	-0.0447	0.0523	0.3502	0.1286	0.0678	0.3655	0.2281	1			
Letters Teaching Terms (%)	-0.0047	-0.0060	-0.0189		-0.0163	-0.0878	-0.1488	0.1290	0.3650	-0.0016	0.1177	0.3161	0.0976	0.2383	1		
Letters Communal Terms (%)	0.1483	-0.0116	-0.1435		-0.1044	-0.0440	-0.0754	-0.1683	0.2329	-0.0224	0.0070	0.0719	-0.1124	0.1381	0.0216	1	
Letters Agentic Terms (%)	0.0548	-0.1457	0.1325		0.0353	0.1087	0.0595	0.0830	0.3284	0.0143	0.1124	0.3027	0.0256	0.2276	0.2610	0.0207	1
Letters Terms (%) PC1	0.0261	-0.0426	0.0213	0.0004	0.0492	-0.0199	-0.1116	0.0803	0.8114	0.2386	0.3964	0.7486	0.2082	0.6157	0.5816	0.2083	0.5580
Letters Terms (%) PC2	-0.0271	-0.0068	0.0497	0.0003	0.0365	-0.0202	0.0655	0.1600	-0.2292	0.4519	-0.5572	-0.0082	0.7168	0.3168	0.0333	-0.2524	-0.1306
CV Length/100	-0.0228	-0.0163	-0.0019	0.0002	0.0962	0.0312	0.0677	0.2051	-0.0026	-0.0135	0.0985	0.0481	0.1497	0.0613	0.0839	-0.1494	0.1032
CV Coding	-0.0094	0.0346	0.0194	-0.0002	0.0740	0.0970	0.0863	0.1100	-0.0178	0.0284	0.0407	-0.0115	0.2123	0.0344	-0.0499	-0.0274	0.0278
CV Publication/Working Paper	-0.0289	0.0128	-0.0457	-0.0064	0.1599	-0.0381	0.2648	0.1565	0.1046	0.0451	0.1148	0.0639	0.2120	0.1178	0.1343	-0.0182	0.0086
Essay Length/100	-0.0057	0.0249	0.0041	0.0001	0.0474	-0.0164	0.0317	0.1083	-0.0279	0.0331	-0.0042	-0.0206	0.0524	-0.0092	-0.0672	-0.0119	-0.0283
Essay Specific Professor	-0.0406	0.0133	-0.0322	-0.0091	0.1428	0.0022	0.4462	0.0738	-0.0047	0.0772	-0.0022	-0.0215	0.0483	0.0270	0.0159	-0.0317	0.0294
Essay Economics Terms (%)	0.0164	-0.0006	0.0160	0.0000	0.0035	0.1185	0.0515	-0.0282	-0.0572	-0.0354	-0.0553	0.0057	0.1233	0.0995	0.0289	0.0579	0.0002
Essay Research Terms (%)	-0.0019	-0.0008	0.0049	-0.0003	0.0376	0.0664	0.0430	0.0499	-0.0016	0.0032	0.0595	-0.0279	0.2357	-0.0866	-0.0744	-0.0378	-0.0300
Essay Positive Terms (%)	0.0013	-0.0200	0.0169	-0.0005	-0.0277	0.0540	-0.0383	-0.1339	0.0887	-0.0558	-0.0026	-0.0028	-0.0436	-0.0223	0.0400	-0.0516	-0.0216
Essay Negative Terms (%)	-0.0132	0.0141	0.0353	0.0003	0.0270	0.0719	0.0827	-0.0128	0.0174	0.1960	0.0757	-0.0206	-0.0427	-0.0338	-0.0425	-0.0131	0.0363
Missing Committee Score	0.1134	0.0768	0.0216	0.0247	0.0000	-0.0153	-0.2077	-0.0137	-0.0089	0.0003	0.0162	-0.0200	-0.0288	-0.0786	0.0211	-0.0473	-0.0157
Missing Application Pdf	0.0578	-0.0143	0.0482	0.0140	-0.1544	0.0263	-0.6598	0.0467									
Missing #Math Courses	0.0623	-0.0108	0.0517	0.0137	-0.1618	0.0268	-0.6547	0.0402	-0.0241	-0.0133	0.0534	-0.0279	-0.0512	0.0813	0.0401	-0.0056	0.0053
Missing Letters	0.0571	-0.0142	0.0437	0.0139	-0.1570	0.0336	-0.6607	0.0470									
Missing CV	0.0558	-0.0205	0.0536	0.0137	-0.1568	0.0332	-0.6449	0.0394	-0.0109	-0.0393	-0.0461	0.0037	-0.0206	0.0649	0.0421	0.0770	0.1614
Missing Essay	0.0564	-0.0119	0.0429	0.0138	-0.1489	0.0146	-0.6606	0.0474	-0.0201	-0.0545	-0.0171	0.0164	-0.0382	0.0205	-0.0490	-0.0453	-0.0824

	LPC1	LPC2	CVN	CVCod	CVPap	EssN	EssProf	EssEcon	EssRes	EssPos	EssNeg	MScore	MPDF	MNMath	ML	MCV	MEss
Letters Terms (%) PC1	1																
Letters Terms (%) PC2	-0.0204	1															
CV Length/100	0.0820	0.0721	1														
CV Coding	0.0147	0.1218	0.2274	1													
CV Publication/Working Paper	0.1158	0.0865	0.2482	0.1413	1												
Essay Length/100	-0.0332	0.0460	0.0706	0.1168	0.0772	1											
Essay Specific Professor	-0.0084	0.0360	0.0795	0.1038	0.2923	0.0701	1										
Essay Economics Terms (%)	0.0166	0.1110	0.0016	0.0578	0.0391	-0.0593	0.0049	1									
Essay Research Terms (%)	-0.0307	0.0886	0.0185	0.1414	0.0621	0.0651	-0.0097	0.1593	1								
Essay Positive Terms (%)	0.0132	-0.0494	-0.0413	-0.0248	-0.0291	-0.1956	-0.0535	-0.0689	-0.0250	1							
Essay Negative Terms (%)	0.0140	-0.0083	0.0279	0.0566	0.0102	0.0716	0.0666	-0.1217	-0.1206	-0.0714	1						
Missing Committee Score	-0.0015	-0.0054	-0.0070	0.0098	-0.1443	0.0100	-0.2336	-0.0045	0.0027	0.0134	-0.0106	1					
Missing Application Pdf	0.0322	0.0218	0.0165	-0.0164	-0.4566	0.0042	-0.6494	-0.0023	-0.0250	-0.0327	0.0183	0.3105	1				
Missing #Math Courses	0.0343	0.0164	0.0076	-0.0239	-0.4531	0.0007	-0.6399	0.0044	-0.0279	-0.0345	0.0283	0.3149	0.9843	1			
Missing Letters	0.0324	0.0219	0.0199	-0.0202	-0.4599	0.0066	-0.6462	0.0039	-0.0208	-0.0261	0.0113	0.3076	0.9929	0.9769	1		
Missing CV	0.0419	0.0187	0.0168	-0.0168	-0.4653	0.0026	-0.6299	-0.0129	-0.0385	-0.0266	0.0242	0.3061	0.9814	0.9650	0.9740	1	
Missing Essay	0.0253	0.0200	0.0134	-0.0246	-0.4464	0.0042	-0.6578	-0.0023	-0.0253	-0.0331	0.0185	0.3034	0.9871	0.9710	0.9798	0.9709	1

▸▸ Regression Results

A3: Economics Department Rec Letters Principal Components

Table 4: Economics Department Rec Letters Principal Components

	PC1	PC2
Positive	0.5044	-0.1769
Negative	0.1479	0.3759
Standout	0.2115	-0.5082
Ability	0.4667	-0.0177
Research	0.1214	0.6418
Grindstone	0.3879	0.3075
Teaching	0.3708	0.0743
Communal	0.1284	-0.1983
Agentic	0.3768	-0.1324
Proportion of Variance	0.2763	0.1456

Regression Results

A3: Economics Department Logistic Regression Results

Table 5a: Economics Department Logistic Regression Results

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.3826	-0.5049	-0.2577	0.0230	0.0301	0.0885	1.8038	1.7607	-0.6618
	(0.2365)	(0.4703)	(0.6853)	(0.1708)	(0.1711)	(0.1887)	(4.2583)	(4.2504)	(3.6266)
2017	-0.0954	-1.3247***	0.1357	0.0829	0.0711	0.1460	2.7746	3.0235	0.7473
	(0.2564)	(0.4881)	(0.6732)	(0.1707)	(0.1721)	(0.1934)	(4.2295)	(4.2188)	(3.7157)
2018	-0.2206	-3.7609***	0.9476	0.0624	-0.0392	0.0142	2.7265	5.2685	4.9770
	(0.2648)	(0.6518)	(0.6200)	(0.1761)	(0.1807)	(0.2011)	(4.4414)	(4.5011)	(3.8786)
2019	-0.1059	-3.1670***	0.5909	0.0594	0.0144	0.0366	5.7085	6.6750	5.2241
	(0.2653)	(0.7114)	(0.6855)	(0.1821)	(0.1830)	(0.2052)	(4.6400)	(4.6266)	(4.1424)
Winsorized Age	-0.0218	-0.0080	-0.1567**	0.0240	0.0242	0.0420*	0.3596	0.3632	-0.1277
	(0.0342)	(0.0592)	(0.0778)	(0.0207)	(0.0205)	(0.0236)	(0.5457)	(0.5410)	(0.4839)
Female	0.3759**	0.8784**	-0.3445	-0.0880	-0.1026	-0.0256	1.8085	2.0921	-1.6375
	(0.1657)	(0.3813)	(0.3794)	(0.1126)	(0.1130)	(0.1252)	(2.8776)	(2.8750)	(2.5117)
U.S.	-0.4982*	-0.0681	1.0348*	0.1167	0.1396	0.1520	-0.6372	-0.9759	-0.7059
	(0.2624)	(0.4538)	(0.6098)	(0.1903)	(0.1909)	(0.2236)	(4.7990)	(4.7913)	(4.3464)
International Asian	-1.1569***	-1.1507***	1.1559**	0.1695	0.1949	-0.1005	-7.8212**	-8.2008**	3.3007
	(0.2129)	(0.4146)	(0.4904)	(0.1379)	(0.1395)	(0.1546)	(3.5528)	(3.5708)	(3.1577)
GRE Verbal Percentile	0.0264***	0.0079	-0.0107	0.0049	0.0030	0.0007	-0.1958**	-0.1470*	-0.0330
	(0.0054)	(0.0130)	(0.0118)	(0.0031)	(0.0032)	(0.0036)	(0.0787)	(0.0806)	(0.0735)
GRE Quantitative Percentile	0.1329***	0.0589**	-0.1033**	0.0245***	0.0184**	0.0047	-0.8498***	-0.7196***	-0.2428*
	(0.0202)	(0.0275)	(0.0464)	(0.0077)	(0.0075)	(0.0084)	(0.1567)	(0.1578)	(0.1460)
GRE Analytic Percentile	0.0076**	0.0094	-0.0119	-0.0020	-0.0027	-0.0021	0.0656	0.0821	0.0160
	(0.0039)	(0.0085)	(0.0095)	(0.0025)	(0.0025)	(0.0029)	(0.0669)	(0.0671)	(0.0618)
Undergraduate GPA	1.2726*	1.0509	-0.7209	-0.3976	-0.4722	-0.8687*	4.8040	6.9311	17.2597**
	(0.6975)	(1.2453)	(1.2760)	(0.3912)	(0.3848)	(0.4442)	(10.1461)	(10.0615)	(7.7507)
TOEFL/IELTS	0.4236	-0.1165	-0.4434	0.1778	0.1437	0.3240*	-2.2504	-1.2960	-7.6837*
	(0.2924)	(0.5976)	(0.5525)	(0.1571)	(0.1559)	(0.1757)	(4.3852)	(4.3431)	(3.9285)
Elite U.S. University	0.7328***	-0.1822	-0.4980	-0.0159	-0.0853	-0.1523	-2.3790	-0.3552	-0.3450
	(0.2774)	(0.5918)	(0.5799)	(0.2133)	(0.2133)	(0.2344)	(5.2934)	(5.2869)	(4.4943)
Elite U.S. Liberal Arts College	0.0167	0.5446	0.9907	0.0731	0.0803	-0.1010	-3.9470	-3.8047	-3.4006
	(0.4568)	(0.6896)	(0.8447)	(0.3370)	(0.3361)	(0.3936)	(8.8571)	(8.7916)	(8.2332)
Elite Foreign University	0.1244	-0.5507	-0.0673	0.1210	0.0751	0.0969	-2.7705	-1.5414	-2.2142
	(0.5319)	(0.9725)	(1.1224)	(0.3239)	(0.3259)	(0.3526)	(7.4748)	(7.4679)	(6.1380)
Other Foreign University	0.5323*	0.6328	0.7373	0.1483	0.1091	0.1587	-6.7584	-5.7186	-9.1260**
	(0.3022)	(0.6505)	(0.6347)	(0.1964)	(0.1970)	(0.2216)	(4.9679)	(4.9620)	(4.3486)
Missing Undergraduate Institution	0.3260	0.2482	-0.6851	0.2292	0.2082	0.1770	-7.9862	-7.5382	-8.1758*
	(0.3111)	(0.6016)	(0.7260)	(0.2017)	(0.2022)	(0.2345)	(5.1133)	(5.1129)	(4.5085)
Graduate Degree	0.1726	-0.2058	-0.2954	-0.0406	-0.0921	-0.1835	-3.2803	-2.0242	2.6215
	(0.1788)	(0.3845)	(0.5017)	(0.1231)	(0.1273)	(0.1370)	(3.1662)	(3.2087)	(2.7452)
Work Experience	0.7785***	0.7519**	1.1183**	0.2298*	0.1772	0.2839**	-3.5942	-2.2559	-6.4380**
	(0.1915)	(0.3400)	(0.5291)	(0.1253)	(0.1270)	(0.1422)	(3.2005)	(3.2301)	(2.9093)
#Professor Recommenders	-0.0377	-0.1797	0.3587	-0.0472	-0.0513	-0.0547	0.8765	0.9972	0.7371
	(0.1079)	(0.2434)	(0.3115)	(0.0749)	(0.0754)	(0.0862)	(2.0492)	(2.0495)	(1.7160)
#Prolific Recommenders	0.2208***	0.2081	-0.2993	0.1011*	0.0899*	0.0600	-1.4623	-1.1967	0.1726
	(0.0811)	(0.1721)	(0.2060)	(0.0528)	(0.0533)	(0.0585)	(1.3680)	(1.3670)	(1.1704)
Missing GRE	-1.5399	1.0141		-0.4515	-0.3258	0.0174	11.5202	8.5434	-3.7106
	(0.9821)	(1.1356)		(0.3510)	(0.3544)	(0.3797)	(8.6385)	(8.6602)	(6.1802)
Missing Undergraduate GPA	-0.3432	-0.2798	0.0798	-0.1010	-0.0639	-0.0513	2.4296	1.5256	-0.8710
	(0.2210)	(0.4413)	(0.4943)	(0.1565)	(0.1585)	(0.1756)	(3.9808)	(3.9842)	(3.3572)
Missing TOEFL/IELTS	0.4719**	0.3743	-0.0734	-0.1261	-0.1447	-0.0065	4.0029	4.3290	-2.3792
	(0.2062)	(0.4223)	(0.5891)	(0.1267)	(0.1282)	(0.1443)	(3.3209)	(3.3244)	(2.9657)
Committee Score		4.2419***	-0.2608		0.0943**	0.0059		-2.5088**	0.6884
		(0.4534)	(0.2250)		(0.0456)	(0.0488)		(1.1860)	(1.0132)
#Math Courses	-0.0949	-0.2706	0.0755	0.0626	0.0662	0.0008	-2.3828	-2.4550	0.9116
	(0.0773)	(0.2829)	(0.1680)	(0.0759)	(0.0766)	(0.0885)	(1.8033)	(1.8149)	(1.5082)
Advanced Math	0.6333**	1.3229	-1.3706**	-0.1932	-0.2078	-0.2275	3.0400	3.4570	1.9957
	(0.3036)	(0.9203)	(0.5621)	(0.2605)	(0.2639)	(0.2949)	(6.4446)	(6.4377)	(5.5819)
Letters Length/100	0.1358***	0.1084	-0.0910	0.0067	-0.0018	-0.0150	-0.2593	-0.0078	0.7206
	(0.0518)	(0.1424)	(0.0712)	(0.0335)	(0.0339)	(0.0388)	(0.8254)	(0.8325)	(0.7084)
Letters Terms (%) PC1	0.1507	0.2939	0.4716**	0.0362	0.0251	0.1260	1.6082	1.8374	-0.5612
	(0.0923)	(0.2912)	(0.2051)	(0.0732)	(0.0748)	(0.0915)	(2.0561)	(2.0540)	(1.8186)
Letters Terms (%) PC2	-0.0913	-0.4980	0.2724	-0.1226	-0.1199	-0.1420	0.7118	0.5616	0.1155
	(0.1464)	(0.4143)	(0.2604)	(0.1162)	(0.1177)	(0.1272)	(2.7992)	(2.8081)	(2.3450)
CV Length/100	0.0361	0.0559	-0.1986	0.0301	0.0274	0.0352	-1.0456	-1.0120	-0.8208
	(0.0617)	(0.1186)	(0.1337)	(0.0562)	(0.0563)	(0.0593)	(1.3224)	(1.3130)	(1.0252)
CV Coding	-0.0439	-0.2369*	-0.2911**	0.0716	0.0710	0.0992	-1.9601	-1.8941	-2.0142
	(0.0765)	(0.1381)	(0.1390)	(0.0532)	(0.0532)	(0.0654)	(1.5459)	(1.5358)	(1.3384)
CV Publication/Working Paper	0.6162*	0.8309	-0.1668	0.2940	0.2834	0.1310	-11.5166	-10.8849	-4.6204
	(0.3421)	(0.9326)	(0.5756)	(0.2998)	(0.3036)	(0.3383)	(7.2873)	(7.3131)	(6.7001)
Essay Length/100	0.1747*	0.2225	0.1776	-0.0008	-0.0008	-0.0711	-0.4758	-0.5351	1.8425
	(0.0933)	(0.3920)	(0.1536)	(0.0751)	(0.0764)	(0.0821)	(1.8648)	(1.8721)	(1.6240)
Essay Specific Professor	0.3842	0.4170	0.6845	0.2144	0.1774	0.2872	-6.8688	-5.9104	-7.5240
	(0.2908)	(0.7014)	(0.5119)	(0.2407)	(0.2410)	(0.2789)	(6.0223)	(5.9982)	(5.2718)
Essay Economics Terms (%)	-0.0169	0.0070	0.2576**	-0.1126**	-0.1150**	-0.0822	3.6540***	3.7277***	2.1702*
	(0.0547)	(0.0866)	(0.1159)	(0.0533)	(0.0531)	(0.0618)	(1.2540)	(1.2496)	(1.1179)
Essay Research Terms (%)	-0.0018	-0.0459	-0.1868	0.0697	0.0642	0.0150	-3.7084**	-3.6795**	-1.0575
	(0.0736)	(0.1690)	(0.1451)	(0.0633)	(0.0639)	(0.0674)	(1.6220)	(1.6209)	(1.3903)
Essay Positive Terms (%)	-0.0202	-0.0700	-0.0504	0.0724	0.0772*	0.0977*	-0.9256	-0.9929	-0.7266
	(0.0512)	(0.1044)	(0.1331)	(0.0453)	(0.0458)	(0.0542)	(1.1634)	(1.1496)	(0.9752)
Essay Negative Terms (%)	-0.0151	-0.0886	0.0019	-0.0386	-0.0454	-0.0472	0.8164	0.9436	0.9507
	(0.0770)	(0.1557)	(0.1071)	(0.0663)	(0.0663)	(0.0748)	(1.6317)	(1.6259)	(1.3985)
Missing Committee Score		-0.5168			-0.3458**	-0.2199		9.2234***	2.8589
		(0.6383)			(0.1397)	(0.1519)		(3.4582)	(3.0258)
Missing Application Pdf	-1.0107***	0.1926	-0.7407	0.1461	0.2686	0.2416	-5.1611	-8.4497	-6.8690
	(0.3188)	(0.8088)	(0.6084)	(0.2442)	(0.2479)	(0.2763)	(6.0328)	(6.1404)	(5.4325)
Open House			1.8384***						
			(0.4115)						
Program Rank						-0.0087***			0.1813***
						(0.0021)			(0.0404)
Missing Program Rank						-3.1432***			89.4721***
						(0.4077)			(5.6947)
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	257	2329	2329	2329	2329	2329	2329
Pseudo-R^2	0.2957	0.8108	0.2877	0.0308	0.0353	0.2430	0.0089	0.0097	0.1065

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Accept ($p < .1$):** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Quantitative Percentile (+), Work Experience (+), Committee Score (+), CV Coding (-)
- **Success ($p < .1$):** Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Essay Positive Terms (+), Program Rank (-), Missing Program Rank (-)

Acceptance, Matriculation, and Success AMEs

A3: Economics Department Logistic Regression AMEs

Table 5b: Economics Department Logistic Regression AMEs

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.0274	-0.0094	-0.0367	0.0035	0.0046	0.0112	0.8694	0.8485	-0.3038
	(0.0171)	(0.0085)	(0.0976)	(0.0262)	(0.0261)	(0.0238)	(2.0523)	(2.0482)	(1.6650)
2017	-0.0068	-0.0246***	0.0193	0.0127	0.0108	0.0184	1.3374	1.4571	0.3431
	(0.0184)	(0.0090)	(0.0958)	(0.0262)	(0.0263)	(0.0244)	(2.0386)	(2.0331)	(1.7054)
2018	-0.0158	-0.0699***	0.1348	0.0096	-0.0060	0.0018	1.3142	2.5391	2.2847
	(0.0189)	(0.0086)	(0.0857)	(0.0270)	(0.0276)	(0.0254)	(2.1405)	(2.1686)	(1.7791)
2019	-0.0076	-0.0589***	0.0841	0.0091	0.0022	0.0046	2.7516	3.2169	2.3982
	(0.0190)	(0.0109)	(0.0970)	(0.0279)	(0.0279)	(0.0259)	(2.2355)	(2.2284)	(1.8977)
Winsorized Age	-0.0016	-0.0001	-0.0223**	0.0037	0.0037	0.0053*	0.1734	0.1751	-0.0586
	(0.0024)	(0.0011)	(0.0108)	(0.0032)	(0.0031)	(0.0030)	(0.2629)	(0.2606)	(0.2222)
Female	0.0269**	0.0163**	-0.0490	-0.0135	-0.0157	-0.0032	0.8717	1.0082	-0.7517
	(0.0118)	(0.0068)	(0.0534)	(0.0173)	(0.0172)	(0.0158)	(1.3867)	(1.3851)	(1.1534)
U.S.	-0.0356*	-0.0013	0.1472*	0.0179	0.0213	0.0192	-0.3071	-0.4703	-0.3241
	(0.0188)	(0.0084)	(0.0864)	(0.0292)	(0.0291)	(0.0282)	(2.3132)	(2.3090)	(1.9951)
International Asian	-0.0828***	-0.0214***	0.1644**	0.0260	0.0298	-0.0127	-3.7699**	-3.9523**	1.5152
	(0.0148)	(0.0073)	(0.0691)	(0.0212)	(0.0213)	(0.0195)	(1.7114)	(1.7195)	(1.4490)
GRE Verbal Percentile	0.0019***	0.0001	-0.0015	0.0008	0.0005	0.0001	-0.0944**	-0.0708*	-0.0151
	(0.0004)	(0.0002)	(0.0017)	(0.0005)	(0.0005)	(0.0005)	(0.0379)	(0.0388)	(0.0337)
GRE Quantitative Percentile	0.0095***	0.0011**	-0.0147**	0.0038***	0.0028**	0.0006	-0.4096***	-0.3468***	-0.1114*
	(0.0014)	(0.0005)	(0.0063)	(0.0012)	(0.0011)	(0.0011)	(0.0757)	(0.0762)	(0.0670)
GRE Analytic Percentile	0.0005**	0.0002	-0.0017	-0.0003	-0.0004	-0.0003	0.0316	0.0395	0.0073
	(0.0003)	(0.0002)	(0.0013)	(0.0004)	(0.0004)	(0.0004)	(0.0322)	(0.0323)	(0.0283)
Undergraduate GPA	0.0910*	0.0195	-0.1026	-0.0610	-0.0721	-0.1096**	2.3156	3.3404	7.9232**
	(0.0499)	(0.0235)	(0.1820)	(0.0600)	(0.0587)	(0.0558)	(4.8911)	(4.8499)	(3.5564)
TOEFL/IELTS	0.0303	-0.0022	-0.0631	0.0273	0.0219	0.0409*	-1.0847	-0.6246	-3.5273*
	(0.0209)	(0.0111)	(0.0776)	(0.0241)	(0.0238)	(0.0221)	(2.1145)	(2.0935)	(1.8033)
Elite U.S. University	0.0524***	-0.0034	-0.0708	-0.0024	-0.0130	-0.0192	-1.1467	-0.1712	-0.1584
	(0.0198)	(0.0111)	(0.0816)	(0.0327)	(0.0326)	(0.0296)	(2.5514)	(2.5479)	(2.0632)
Elite U.S. Liberal Arts College	0.0012	0.0101	0.1409	0.0112	0.0123	-0.0127	-1.9025	-1.8336	-1.5611
	(0.0327)	(0.0126)	(0.1197)	(0.0517)	(0.0513)	(0.0496)	(4.2692)	(4.2369)	(3.7802)
Elite Foreign University	0.0089	-0.0102	-0.0096	0.0186	0.0115	0.0122	-1.3354	-0.7428	-1.0164
	(0.0381)	(0.0182)	(0.1597)	(0.0497)	(0.0497)	(0.0445)	(3.6022)	(3.5986)	(2.8173)
Other Foreign University	0.0381*	0.0118	0.1049	0.0228	0.0167	0.0200	-3.2576	-2.7560	-4.1894**
	(0.0216)	(0.0119)	(0.0890)	(0.0301)	(0.0301)	(0.0279)	(2.3945)	(2.3912)	(1.9955)
Missing Undergraduate Institution	0.0233	0.0046	-0.0975	0.0352	0.0318	0.0223	-3.8495	-3.6329	-3.7531*
	(0.0222)	(0.0111)	(0.1014)	(0.0309)	(0.0308)	(0.0296)	(2.4639)	(2.4635)	(2.0698)
Graduate Degree	0.0123	-0.0038	-0.0420	-0.0062	-0.0141	-0.0231	-1.5811	-0.9755	1.2034
	(0.0128)	(0.0071)	(0.0712)	(0.0189)	(0.0194)	(0.0172)	(1.5257)	(1.5461)	(1.2606)
Work Experience	0.0557***	0.0140**	0.1591**	0.0353*	0.0271	0.0358**	-1.7325	-1.0872	-2.9554**
	(0.0135)	(0.0064)	(0.0700)	(0.0192)	(0.0194)	(0.0178)	(1.5432)	(1.5570)	(1.3344)
#Professor Recommenders	-0.0027	-0.0033	0.0510	-0.0072	-0.0078	-0.0069	0.4225	0.4806	0.3384
	(0.0077)	(0.0045)	(0.0437)	(0.0115)	(0.0115)	(0.0109)	(0.9881)	(0.9881)	(0.7878)
#Prolific Recommenders	0.0158***	0.0039	-0.0426	0.0155*	0.0137*	0.0076	-0.7048	-0.5767	0.0792
	(0.0058)	(0.0031)	(0.0285)	(0.0081)	(0.0081)	(0.0074)	(0.6595)	(0.6589)	(0.5373)
Missing GRE	-0.1102	0.0188		-0.0693	-0.0497	0.0022	5.5529	4.1174	-1.7034
	(0.0706)	(0.0212)		(0.0538)	(0.0541)	(0.0479)	(4.1627)	(4.1727)	(2.8358)
Missing Undergraduate GPA	-0.0246	-0.0052	0.0114	-0.0155	-0.0098	-0.0065	1.1711	0.7352	-0.3999
	(0.0158)	(0.0082)	(0.0703)	(0.0240)	(0.0242)	(0.0221)	(1.9187)	(1.9201)	(1.5412)
Missing TOEFL/IELTS	0.0338**	0.0070	-0.0104	-0.0194	-0.0221	-0.0008	1.9295	2.0863	-1.0922
	(0.0147)	(0.0078)	(0.0839)	(0.0194)	(0.0195)	(0.0182)	(1.6007)	(1.6020)	(1.3613)
Committee Score		0.0788***	-0.0371		0.0144**	0.0007		-1.2091**	0.3160
		(0.0045)	(0.0322)		(0.0069)	(0.0062)		(0.5716)	(0.4649)
#Math Courses	-0.0068	-0.0050	0.0107	0.0096	0.0101	0.0001	-1.1485	-1.1832	0.4185
	(0.0055)	(0.0053)	(0.0239)	(0.0116)	(0.0117)	(0.0112)	(0.8692)	(0.8747)	(0.6922)
Advanced Math	0.0453**	0.0246	-0.1950**	-0.0296	-0.0317	-0.0287	1.4653	1.6660	0.9161
	(0.0216)	(0.0168)	(0.0793)	(0.0399)	(0.0402)	(0.0372)	(3.1061)	(3.1022)	(2.5621)
Letters Length/100	0.0097***	0.0020	-0.0129	0.0010	-0.0003	-0.0019	-0.1250	-0.0038	0.3308
	(0.0037)	(0.0026)	(0.0100)	(0.0051)	(0.0052)	(0.0049)	(0.3978)	(0.4012)	(0.3253)
Letters Terms (%) PC1	0.0108	0.0055	0.0671**	0.0056	0.0038	0.0159	0.7752	0.8855	-0.2576
	(0.0066)	(0.0054)	(0.0283)	(0.0112)	(0.0114)	(0.0115)	(0.9905)	(0.9892)	(0.8348)
Letters Terms (%) PC2	-0.0065	-0.0093	0.0387	-0.0188	-0.0183	-0.0179	0.3431	0.2706	0.0530
	(0.0105)	(0.0077)	(0.0371)	(0.0178)	(0.0179)	(0.0160)	(1.3495)	(1.3535)	(1.0765)
CV Length/100	0.0026	0.0010	-0.0282	0.0046	0.0042	0.0044	-0.5040	-0.4877	-0.3768
	(0.0044)	(0.0022)	(0.0190)	(0.0086)	(0.0086)	(0.0075)	(0.6374)	(0.6328)	(0.4706)
CV Coding	-0.0031	-0.0044*	-0.0414**	0.0110	0.0108	0.0125	-0.9448	-0.9129	-0.9246
	(0.0055)	(0.0026)	(0.0193)	(0.0082)	(0.0081)	(0.0082)	(0.7454)	(0.7403)	(0.6139)
CV Publication/Working Paper	0.0441*	0.0154	-0.0237	0.0451	0.0433	0.0165	-5.5511	-5.2459	-2.1210
	(0.0244)	(0.0173)	(0.0820)	(0.0460)	(0.0463)	(0.0427)	(3.5094)	(3.5215)	(3.0760)
Essay Length/100	0.0125*	0.0041	0.0253	-0.0001	-0.0001	-0.0090	-0.2293	-0.2579	0.8458
	(0.0067)	(0.0073)	(0.0219)	(0.0115)	(0.0117)	(0.0103)	(0.8988)	(0.9022)	(0.7451)
Essay Specific Professor	0.0275	0.0077	0.0974	0.0329	0.0271	0.0362	-3.3108	-2.8484	-3.4539
	(0.0208)	(0.0130)	(0.0724)	(0.0369)	(0.0368)	(0.0351)	(2.9044)	(2.8921)	(2.4207)
Essay Economics Terms (%)	-0.0012	0.0001	0.0366**	-0.0173**	-0.0175**	-0.0104	1.7613***	1.7965***	0.9962*
	(0.0039)	(0.0016)	(0.0162)	(0.0082)	(0.0081)	(0.0078)	(0.6048)	(0.6027)	(0.5134)
Essay Research Terms (%)	-0.0001	-0.0009	-0.0266	0.0107	0.0098	0.0019	-1.7875**	-1.7733**	-0.4855
	(0.0053)	(0.0031)	(0.0206)	(0.0097)	(0.0098)	(0.0085)	(0.7813)	(0.7808)	(0.6382)
Essay Positive Terms (%)	-0.0014	-0.0013	-0.0072	0.0111	0.0118*	0.0123*	-0.4461	-0.4785	-0.3335
	(0.0037)	(0.0019)	(0.0189)	(0.0069)	(0.0070)	(0.0068)	(0.5609)	(0.5541)	(0.4475)
Essay Negative Terms (%)	-0.0011	-0.0016	0.0003	-0.0059	-0.0069	-0.0060	0.3935	0.4548	0.4364
	(0.0055)	(0.0029)	(0.0152)	(0.0102)	(0.0101)	(0.0094)	(0.7865)	(0.7837)	(0.6420)
Missing Committee Score		-0.0096			-0.0528**	-0.0277		4.4451***	1.3124
		(0.0120)			(0.0212)	(0.0191)		(1.6677)	(1.3892)
Missing Application Pdf	-0.0723***	0.0036	-0.1054	0.0224	0.0410	0.0305	-2.4877	-4.0722	-3.1533
	(0.0227)	(0.0149)	(0.0858)	(0.0375)	(0.0379)	(0.0348)	(2.9090)	(2.9614)	(2.4962)
Open House			0.2615***						
			(0.0512)						
Program Rank						-0.0011***			0.0832***
						(0.0003)			(0.0185)
Missing Program Rank						-0.3965***			41.0729***
						(0.0512)			(2.6010)
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	257	2329	2329	2329	2329	2329	2329

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Accept ($p < .1$):** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Quantitative Percentile (+), Work Experience (+), Committee Score (+), CV Coding (-)
- **Success ($p < .1$):** Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Essay Positive Terms (+), Program Rank (-), Missing Program Rank (-)

▸▸ Acceptance, Matriculation, and Success AMEs

Table 6a: Economics Department Logistic Regression Results (Synthetic)

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.2091	-0.1311	0.1291	0.0997	0.0892	0.1020	-0.8042	-0.5831	-1.3927
	(0.2288)	(0.2577)	(0.6478)	(0.1812)	(0.1814)	(0.1970)	(4.5726)	(4.5702)	(4.1436)
2017	-0.3355	-0.5461**	0.6767	0.2827	0.2903	0.1565	-12.4495***	-12.2002***	-6.3639
	(0.2390)	(0.2627)	(0.6543)	(0.1761)	(0.1767)	(0.1925)	(4.4305)	(4.4347)	(3.9262)
2018	0.2889	-0.2993	0.2947	0.1627	0.2192	0.1272	-9.3709**	-9.2465**	-6.0098
	(0.2056)	(0.2212)	(0.5517)	(0.1806)	(0.1827)	(0.1972)	(4.4278)	(4.4749)	(3.9975)
2019	-0.1933	-0.3208	-0.0662	0.0116	0.0023	-0.1488	-5.3768	-4.7378	1.0400
	(0.2227)	(0.2490)	(0.6069)	(0.1927)	(0.1933)	(0.2061)	(4.4927)	(4.5026)	(3.9171)
Winsorized Age	-0.0508**	-0.0414*	0.0065	-0.0141	-0.0150	-0.0094	0.4466	0.4383	0.2496
	(0.0220)	(0.0243)	(0.0600)	(0.0170)	(0.0170)	(0.0185)	(0.4147)	(0.4141)	(0.3766)
Female	-0.1826	-0.1535	0.0691	0.1351	0.1379	0.2109*	-1.5097	-1.6305	-4.7144*
	(0.1446)	(0.1590)	(0.4006)	(0.1167)	(0.1168)	(0.1259)	(2.9357)	(2.9318)	(2.6789)
U.S.	0.0002	-0.0549	0.0998	0.0198	0.0241	-0.0123	2.8625	2.5697	4.2450
	(0.2240)	(0.2468)	(0.6663)	(0.1794)	(0.1801)	(0.1997)	(4.4691)	(4.4669)	(4.1868)
International Asian	-0.1230	-0.0646	-0.1205	0.2014	0.1972	0.1700	-0.7475	-0.9433	1.1783
	(0.1688)	(0.1838)	(0.5028)	(0.1394)	(0.1392)	(0.1541)	(3.3754)	(3.3732)	(3.1231)
GRE Verbal Percentile	-0.0025	-0.0001	0.0008	0.0004	0.0001	-0.0021	-0.0668	-0.0623	0.0181
	(0.0034)	(0.0036)	(0.0089)	(0.0026)	(0.0026)	(0.0029)	(0.0661)	(0.0663)	(0.0595)
GRE Quantitative Percentile	-0.0028	-0.0005	-0.0125	0.0145**	0.0143**	-0.0003	-0.5982***	-0.5916***	-0.0184
	(0.0075)	(0.0080)	(0.0195)	(0.0065)	(0.0065)	(0.0069)	(0.1570)	(0.1569)	(0.1376)
GRE Analytic Percentile	0.0041	0.0019	-0.0010	0.0009	0.0011	0.0019	0.0181	0.0156	-0.0008
	(0.0031)	(0.0033)	(0.0074)	(0.0024)	(0.0024)	(0.0026)	(0.0587)	(0.0587)	(0.0529)
Undergraduate GPA	-0.4230	-0.3157	0.9077	0.4075	0.3986	0.4627	-2.7870	-2.7400	-5.8113
	(0.3862)	(0.4536)	(1.8572)	(0.3984)	(0.3990)	(0.4326)	(9.4800)	(9.4624)	(8.8952)
TOEFL/IELTS	0.2329	0.2776	-0.2073	-0.0682	-0.0625	-0.0142	2.4237	2.2404	-0.4376
	(0.2154)	(0.2202)	(0.4100)	(0.1509)	(0.1507)	(0.1670)	(3.7070)	(3.7021)	(3.1595)
Elite U.S. University	-0.0473	-0.1157	-0.2546	0.3145	0.3119	0.3760*	-6.7562	-6.6352	-7.1987
	(0.2618)	(0.2943)	(0.7160)	(0.2054)	(0.2062)	(0.2230)	(5.3790)	(5.3875)	(4.8870)
Elite U.S. Liberal Arts College	-0.6642	-0.6053		0.0069	-0.0044	-0.0958	-2.2643	-2.2283	3.3486
	(0.5448)	(0.5744)		(0.3656)	(0.3659)	(0.3964)	(8.9098)	(8.8713)	(7.5578)
Elite Foreign University	-0.1797	-0.1798	0.5156	-0.0538	-0.0692	-0.0298	-1.0302	-0.9315	-0.2765
	(0.3900)	(0.4341)	(1.2147)	(0.3273)	(0.3294)	(0.3495)	(8.2453)	(8.2793)	(7.1254)
Other Foreign University	-0.0406	-0.1089	0.6613	0.1055	0.1125	0.0046	-6.0523	-6.2650	-0.6409
	(0.2078)	(0.2262)	(0.5235)	(0.1688)	(0.1689)	(0.1840)	(4.2123)	(4.2076)	(3.8417)
Missing Undergraduate Institution	-0.0012	0.0377	-0.6367	0.1037	0.0966	0.0297	-5.9887	-5.9936	-2.3404
	(0.2217)	(0.2413)	(0.5829)	(0.1844)	(0.1842)	(0.1977)	(4.5947)	(4.5865)	(4.1622)
Graduate Degree	-0.0846	-0.1354	-0.0100	0.1515	0.1611	0.1906	-1.4756	-1.6341	-2.0725
	(0.1458)	(0.1626)	(0.3834)	(0.1166)	(0.1165)	(0.1252)	(2.8643)	(2.8619)	(2.5933)
Work Experience	0.2547*	0.2149	-0.2181	0.0750	0.0805	0.1443	0.4084	0.3542	-2.7442
	(0.1463)	(0.1588)	(0.3573)	(0.1161)	(0.1160)	(0.1250)	(2.8682)	(2.8651)	(2.6268)
#Professor Recommenders	0.0194	-0.0565	0.1806	0.0038	0.0074	-0.0302	-0.2230	-0.1761	1.2584
	(0.0950)	(0.1021)	(0.2725)	(0.0759)	(0.0761)	(0.0794)	(1.9087)	(1.9104)	(1.7006)
#Prolific Recommenders	-0.0388	-0.0071	0.0168	0.0027	-0.0000	0.0325	0.3327	0.3170	-1.1079
	(0.0671)	(0.0713)	(0.1685)	(0.0521)	(0.0522)	(0.0575)	(1.3245)	(1.3242)	(1.2261)
Missing GRE	0.3002	0.0736		0.3136	0.3314	0.2674	-12.0035	-11.9570	-11.5860
	(0.3757)	(0.3872)		(0.2858)	(0.2857)	(0.3297)	(8.1036)	(8.0998)	(8.0641)
Missing Undergraduate GPA	0.1997	0.1720	0.2910	-0.1321	-0.1290	-0.0816	5.2305	5.1674	4.3351
	(0.1863)	(0.2016)	(0.5443)	(0.1391)	(0.1389)	(0.1534)	(3.6167)	(3.6124)	(3.3094)
Missing TOEFL/IELTS	-0.1740	-0.0792	-0.3009	0.1948	0.1815	0.2286*	-1.5099	-1.4121	-3.1183
	(0.1609)	(0.1741)	(0.4256)	(0.1280)	(0.1281)	(0.1380)	(3.1162)	(3.1192)	(2.7616)
Committee Score		0.0372	-0.1004		0.0465	0.0330		-1.6627	-0.4906
		(0.0423)	(0.1060)		(0.0505)	(0.0558)		(1.1655)	(1.1645)
#Math Courses	0.0632	0.0756	-0.1683	-0.1269	-0.1305*	-0.1764**	1.1826	1.2073	3.0509**
	(0.0687)	(0.0733)	(0.1601)	(0.0786)	(0.0780)	(0.0863)	(1.7287)	(1.7209)	(1.5010)
Advanced Math	0.0222	-0.0336	0.9873	-0.1639	-0.1630	-0.2842	1.1644	1.2290	6.1671
	(0.2590)	(0.2747)	(0.7209)	(0.3004)	(0.3013)	(0.3256)	(6.4828)	(6.4875)	(5.6351)
Letters Length/100	-0.0254	-0.0128	-0.0575	-0.1049***	-0.1057***	-0.1218***	1.6052**	1.6308**	2.1010***
	(0.0333)	(0.0361)	(0.0885)	(0.0381)	(0.0376)	(0.0428)	(0.7873)	(0.7869)	(0.6934)
Letters Terms (%) PC1	0.0280	-0.0073	0.0381	0.0574	0.0643	0.0566	1.1364	1.1816	1.0356
	(0.0779)	(0.0821)	(0.2097)	(0.0828)	(0.0834)	(0.0930)	(1.8933)	(1.9000)	(1.6894)
Letters Terms (%) PC2	-0.2328*	-0.2369*	-0.4921	0.0802	0.0718	0.1318	-2.8589	-2.7586	-5.0669**
	(0.1226)	(0.1255)	(0.3028)	(0.1205)	(0.1211)	(0.1250)	(2.6915)	(2.7005)	(2.1872)
CV Length/100	0.0971**	0.0795	0.1022	-0.0358	-0.0336	-0.0095	2.4879**	2.5172**	0.5354
	(0.0492)	(0.0508)	(0.1340)	(0.0548)	(0.0551)	(0.0641)	(1.2406)	(1.2458)	(1.0685)
CV Coding	-0.0198	-0.0473	-0.0834	0.0688	0.0720*	0.0795*	-1.6529	-1.6778	-1.0250
	(0.0411)	(0.0448)	(0.1069)	(0.0434)	(0.0434)	(0.0474)	(1.0435)	(1.0451)	(0.8904)
CV Publication/Working Paper	0.3079	0.3005	1.3491*	0.1111	0.1183	0.1111	2.7524	2.4246	5.5955
	(0.2852)	(0.2962)	(0.7475)	(0.3264)	(0.3272)	(0.3482)	(7.3505)	(7.3749)	(6.2211)
Essay Length/100	0.0679	0.0790	-0.0558	-0.0667	-0.0636	-0.0694	0.5523	0.4398	0.3231
	(0.0770)	(0.0831)	(0.1489)	(0.0780)	(0.0794)	(0.0871)	(1.7254)	(1.7321)	(1.3973)
Essay Specific Professor	-0.1756	-0.2494	-0.1115	-0.0547	-0.0524	0.0009	1.2639	1.2219	-3.1507
	(0.2564)	(0.2714)	(0.6302)	(0.2774)	(0.2798)	(0.3022)	(6.3588)	(6.3580)	(5.5747)
Essay Economics Terms (%)	-0.0123	-0.0094	-0.0406	-0.0091	-0.0082	-0.0222	0.2276	0.2960	0.6769
	(0.0542)	(0.0572)	(0.1334)	(0.0539)	(0.0547)	(0.0624)	(1.3179)	(1.3189)	(1.1799)
Essay Research Terms (%)	0.0303	0.0066	-0.0464	0.1042	0.1084	0.1176	-1.8679	-1.9069	-1.8170
	(0.0656)	(0.0702)	(0.1578)	(0.0703)	(0.0705)	(0.0723)	(1.5218)	(1.5245)	(1.2596)
Essay Positive Terms (%)	0.0180	0.0074	0.3699***	0.0871*	0.0868*	0.0947	-1.5542	-1.4775	-1.1629
	(0.0474)	(0.0484)	(0.1213)	(0.0511)	(0.0517)	(0.0609)	(1.1518)	(1.1568)	(1.1006)
Essay Negative Terms (%)	0.0333	0.0312	-0.0731	-0.0304	-0.0280	-0.0168	0.5011	0.4716	0.2911
	(0.0314)	(0.0351)	(0.0769)	(0.0359)	(0.0359)	(0.0382)	(0.8608)	(0.8614)	(0.7100)
Missing Committee Score		-4.3141***			0.2635**	0.4129***		-0.9440	-6.1957**
		(0.4312)			(0.1267)	(0.1388)		(3.0258)	(2.8364)
Missing Application Pdf	-0.9321***	-0.0588	0.9473	0.0574	-0.0560	-0.1100	4.5136	4.8399	5.6614
	(0.2534)	(0.2753)	(0.6793)	(0.2628)	(0.2678)	(0.2943)	(5.9238)	(6.0503)	(5.4160)
Open House			1.0643						
			(0.8268)						
Program Rank						-0.0049**			0.1176***
						(0.0019)			(0.0386)
Missing Program Rank						-2.5925***			94.5255***
						(0.3273)			(5.6183)
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	237	2329	2329	2329	2329	2329	2329
Pseudo-R^2	0.0552	0.2804	0.1402	0.0221	0.0245	0.1935	0.0040	0.0042	0.0953

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Accept ($p < .1$):** 2017 (-), Winsorized Age (-), Letters Terms (%) PC2, Missing Committee Score (-)
- **Success ($p < .1$):** Female (+), Elite U.S. University (+), Missing TOEFL/IELTS (+), #Math Courses (-), Letters Length/100 (-), CV Coding (+), Missing Committee Score (+), Program Rank (-), Missing Program Rank (-)

Acceptance, Matriculation, and Success AMEs

A3: Economics Department Logistic Regression AMEs (Synthetic)

Table 6b: Economics Department Logistic Regression AMEs (Synthetic)

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.0185	-0.0100	0.0201	0.0136	0.0122	0.0120	-0.3764	-0.2729	-0.6112
	(0.0203)	(0.0197)	(0.1005)	(0.0248)	(0.0248)	(0.0233)	(2.1401)	(2.1390)	(1.8189)
2017	-0.0297	-0.0417**	0.1052	0.0387	0.0396	0.0185	-5.8265***	-5.7099***	-2.7929
	(0.0212)	(0.0199)	(0.1005)	(0.0241)	(0.0241)	(0.0227)	(2.0748)	(2.0770)	(1.7228)
2018	0.0256	-0.0229	0.0458	0.0223	0.0299	0.0150	-4.3857**	-4.3275**	-2.6375
	(0.0182)	(0.0168)	(0.0856)	(0.0247)	(0.0250)	(0.0233)	(2.0732)	(2.0958)	(1.7555)
2019	-0.0171	-0.0245	-0.0103	0.0016	0.0003	-0.0176	-2.5164	-2.2174	0.4564
	(0.0197)	(0.0189)	(0.0944)	(0.0264)	(0.0264)	(0.0243)	(2.1013)	(2.1061)	(1.7192)
Winsorized Age	-0.0045**	-0.0032*	0.0010	-0.0019	-0.0020	-0.0011	0.2090	0.2051	0.1095
	(0.0020)	(0.0019)	(0.0093)	(0.0023)	(0.0023)	(0.0022)	(0.1941)	(0.1938)	(0.1654)
Female	-0.0162	-0.0117	0.0107	0.0185	0.0188	0.0249*	-0.7066	-0.7631	-2.0690*
	(0.0128)	(0.0121)	(0.0621)	(0.0160)	(0.0159)	(0.0149)	(1.3745)	(1.3728)	(1.1767)
U.S.	0.0000	-0.0042	0.0155	0.0027	0.0033	-0.0015	1.3397	1.2026	1.8630
	(0.0198)	(0.0189)	(0.1035)	(0.0246)	(0.0246)	(0.0236)	(2.0914)	(2.0905)	(1.8379)
International Asian	-0.0109	-0.0049	-0.0187	0.0276	0.0269	0.0201	-0.3498	-0.4415	0.5171
	(0.0150)	(0.0140)	(0.0782)	(0.0191)	(0.0190)	(0.0182)	(1.5798)	(1.5789)	(1.3706)
GRE Verbal Percentile	-0.0002	-0.0000	0.0001	0.0001	0.0000	-0.0002	-0.0313	-0.0291	0.0080
	(0.0003)	(0.0003)	(0.0014)	(0.0004)	(0.0004)	(0.0003)	(0.0309)	(0.0310)	(0.0261)
GRE Quantitative Percentile	-0.0002	-0.0000	-0.0019	0.0020**	0.0020**	-0.0000	-0.2800***	-0.2769***	-0.0081
	(0.0007)	(0.0006)	(0.0030)	(0.0009)	(0.0009)	(0.0008)	(0.0734)	(0.0734)	(0.0604)
GRE Analytic Percentile	0.0004	0.0001	-0.0002	0.0001	0.0002	0.0002	0.0085	0.0073	-0.0004
	(0.0003)	(0.0003)	(0.0011)	(0.0003)	(0.0003)	(0.0003)	(0.0275)	(0.0275)	(0.0232)
Undergraduate GPA	-0.0375	-0.0241	0.1411	0.0558	0.0544	0.0546	-1.3043	-1.2824	-2.5504
	(0.0342)	(0.0346)	(0.2875)	(0.0545)	(0.0545)	(0.0510)	(4.4373)	(4.4292)	(3.9043)
TOEFL/IELTS	0.0206	0.0212	-0.0322	-0.0093	-0.0085	-0.0017	1.1343	1.0486	-0.1921
	(0.0191)	(0.0168)	(0.0637)	(0.0207)	(0.0206)	(0.0197)	(1.7354)	(1.7331)	(1.3866)
Elite U.S. University	-0.0042	-0.0088	-0.0396	0.0431	0.0426	0.0444*	-3.1620	-3.1054	-3.1593
	(0.0232)	(0.0225)	(0.1109)	(0.0281)	(0.0282)	(0.0263)	(2.5179)	(2.5220)	(2.1436)
Elite U.S. Liberal Arts College	-0.0588	-0.0462		0.0009	-0.0006	-0.0113	-1.0597	-1.0429	1.4696
	(0.0483)	(0.0439)		(0.0500)	(0.0500)	(0.0468)	(4.1696)	(4.1517)	(3.3177)
Elite Foreign University	-0.0159	-0.0137	0.0801	-0.0074	-0.0095	-0.0035	-0.4822	-0.4360	-0.1214
	(0.0346)	(0.0332)	(0.1891)	(0.0448)	(0.0450)	(0.0412)	(3.8590)	(3.8749)	(3.1271)
Other Foreign University	-0.0036	-0.0083	0.1028	0.0144	0.0154	0.0005	-2.8325	-2.9321	-0.2813
	(0.0184)	(0.0173)	(0.0813)	(0.0231)	(0.0231)	(0.0217)	(1.9702)	(1.9680)	(1.6858)
Missing Undergraduate Institution	-0.0001	0.0029	-0.0989	0.0142	0.0132	0.0035	-2.8028	-2.8051	-1.0271
	(0.0196)	(0.0184)	(0.0902)	(0.0252)	(0.0251)	(0.0233)	(2.1499)	(2.1460)	(1.8264)
Graduate Degree	-0.0075	-0.0103	-0.0015	0.0207	0.0220	0.0225	-0.6906	-0.7648	-0.9095
	(0.0129)	(0.0124)	(0.0596)	(0.0160)	(0.0159)	(0.0147)	(1.3406)	(1.3395)	(1.1381)
Work Experience	0.0226*	0.0164	-0.0339	0.0103	0.0110	0.0170	0.1911	0.1658	-1.2044
	(0.0129)	(0.0121)	(0.0551)	(0.0159)	(0.0158)	(0.0147)	(1.3422)	(1.3408)	(1.1533)
#Professor Recommenders	0.0017	-0.0043	0.0281	0.0005	0.0010	-0.0036	-0.1044	-0.0824	0.5523
	(0.0084)	(0.0078)	(0.0423)	(0.0104)	(0.0104)	(0.0094)	(0.8933)	(0.8941)	(0.7465)
#Prolific Recommenders	-0.0034	-0.0005	0.0026	0.0004	-0.0000	0.0038	0.1557	0.1484	-0.4862
	(0.0059)	(0.0054)	(0.0262)	(0.0071)	(0.0071)	(0.0068)	(0.6197)	(0.6196)	(0.5384)
Missing GRE	0.0266	0.0056		0.0429	0.0453	0.0316	-5.6177	-5.5961	-5.0848
	(0.0333)	(0.0296)		(0.0391)	(0.0390)	(0.0389)	(3.7950)	(3.7934)	(3.5412)
Missing Undergraduate GPA	0.0177	0.0131	0.0452	-0.0181	-0.0176	-0.0096	2.4479	2.4184	1.9026
	(0.0165)	(0.0154)	(0.0847)	(0.0190)	(0.0190)	(0.0181)	(1.6942)	(1.6921)	(1.4526)
Missing TOEFL/IELTS	-0.0154	-0.0060	-0.0468	0.0267	0.0248	0.0270*	-0.7067	-0.6609	-1.3685
	(0.0143)	(0.0133)	(0.0659)	(0.0175)	(0.0175)	(0.0162)	(1.4590)	(1.4604)	(1.2133)
Committee Score		0.0028	-0.0156		0.0064	0.0039		-0.7782	-0.2153
		(0.0032)	(0.0165)		(0.0069)	(0.0066)		(0.5455)	(0.5111)
#Math Courses	0.0056	0.0058	-0.0262	-0.0174	-0.0178*	-0.0208**	0.5535	0.5650	1.3390**
	(0.0061)	(0.0056)	(0.0248)	(0.0107)	(0.0106)	(0.0101)	(0.8095)	(0.8059)	(0.6591)
Advanced Math	0.0020	-0.0026	0.1534	-0.0224	-0.0223	-0.0335	0.5450	0.5752	2.7066
	(0.0229)	(0.0210)	(0.1122)	(0.0411)	(0.0411)	(0.0384)	(3.0339)	(3.0362)	(2.4720)
Letters Length/100	-0.0022	-0.0010	-0.0089	-0.0144***	-0.0144***	-0.0144***	0.7512**	0.7632**	0.9221***
	(0.0029)	(0.0028)	(0.0137)	(0.0052)	(0.0051)	(0.0050)	(0.3694)	(0.3693)	(0.3050)
Letters Terms (%) PC1	0.0025	-0.0006	0.0059	0.0079	0.0088	0.0067	0.5318	0.5530	0.4545
	(0.0069)	(0.0063)	(0.0325)	(0.0113)	(0.0114)	(0.0110)	(0.8858)	(0.8889)	(0.7413)
Letters Terms (%) PC2	-0.0206*	-0.0181*	-0.0765*	0.0110	0.0098	0.0156	-1.3380	-1.2911	-2.2237**
	(0.0108)	(0.0095)	(0.0462)	(0.0165)	(0.0165)	(0.0147)	(1.2609)	(1.2651)	(0.9607)
CV Length/100	0.0086**	0.0061	0.0159	-0.0049	-0.0046	-0.0011	1.1643**	1.1781**	0.2350
	(0.0043)	(0.0039)	(0.0210)	(0.0075)	(0.0075)	(0.0076)	(0.5802)	(0.5825)	(0.4688)
CV Coding	-0.0018	-0.0036	-0.0130	0.0094	0.0098*	0.0094*	-0.7736	-0.7852	-0.4499
	(0.0036)	(0.0034)	(0.0167)	(0.0059)	(0.0059)	(0.0056)	(0.4887)	(0.4894)	(0.3909)
CV Publication/Working Paper	0.0273	0.0230	0.2097*	0.0152	0.0162	0.0131	1.2881	1.1348	2.4557
	(0.0252)	(0.0226)	(0.1128)	(0.0447)	(0.0447)	(0.0411)	(3.4400)	(3.4515)	(2.7307)
Essay Length/100	0.0060	0.0060	-0.0087	-0.0091	-0.0087	-0.0082	0.2585	0.2058	0.1418
	(0.0068)	(0.0063)	(0.0231)	(0.0107)	(0.0108)	(0.0103)	(0.8075)	(0.8107)	(0.6133)
Essay Specific Professor	-0.0156	-0.0190	-0.0173	-0.0075	-0.0072	0.0001	0.5915	0.5719	-1.3828
	(0.0227)	(0.0207)	(0.0978)	(0.0380)	(0.0382)	(0.0357)	(2.9760)	(2.9757)	(2.4463)
Essay Economics Terms (%)	-0.0011	-0.0007	-0.0063	-0.0012	-0.0011	-0.0026	0.1065	0.1385	0.2971
	(0.0048)	(0.0044)	(0.0207)	(0.0074)	(0.0075)	(0.0074)	(0.6168)	(0.6173)	(0.5179)
Essay Research Terms (%)	0.0027	0.0005	-0.0072	0.0143	0.0148	0.0139	-0.8742	-0.8925	-0.7975
	(0.0058)	(0.0054)	(0.0245)	(0.0096)	(0.0096)	(0.0085)	(0.7127)	(0.7141)	(0.5530)
Essay Positive Terms (%)	0.0016	0.0006	0.0575***	0.0119*	0.0118*	0.0112	-0.7274	-0.6915	-0.5104
	(0.0042)	(0.0037)	(0.0182)	(0.0070)	(0.0071)	(0.0072)	(0.5394)	(0.5418)	(0.4832)
Essay Negative Terms (%)	0.0030	0.0024	-0.0114	-0.0042	-0.0038	-0.0020	0.2345	0.2207	0.1278
	(0.0028)	(0.0027)	(0.0120)	(0.0049)	(0.0049)	(0.0045)	(0.4029)	(0.4032)	(0.3117)
Missing Committee Score		-0.3296***			0.0360**	0.0487***		-0.4418	-2.7191**
		(0.0330)			(0.0173)	(0.0162)		(1.4166)	(1.2447)
Missing Application Pdf	-0.0826***	-0.0045	0.1472	0.0079	-0.0076	-0.0130	2.1124	2.2652	2.4846
	(0.0225)	(0.0210)	(0.1064)	(0.0360)	(0.0366)	(0.0347)	(2.7720)	(2.8317)	(2.3765)
Open House			0.1654						
			(0.1278)						
Program Rank						-0.0006**			0.0516***
						(0.0002)			(0.0169)
Missing Program Rank						-0.3060***			41.4849***
						(0.0381)			(2.4608)
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	237	2329	2329	2329	2329	2329	2329

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Accept ($p < .1$):** 2017 (-), Winsorized Age (-), Letters Terms (%) PC2, Missing Committee Score (-)
- **Success ($p < .1$):** Female (+), Elite U.S. University (+), Missing TOEFL/IELTS (+), #Math Courses (-), Letters Length/100 (-), CV Coding (+), Missing Committee Score (+), Program Rank (-), Missing Program Rank (-)

⏩ Acceptance, Matriculation, and Success AMEs

A3: Economics Department Lasso Logistic Regression Results

Table 7a: Economics Department Lasso Logistic Regression Results

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.1107								
2017		-0.4928							
2018		-2.4456	0.0186						
2019		-1.7752							
Winsorized Age				0.0084	0.0047	0.0184			
Female	0.2674	0.5589							
U.S.	-0.1959							0.0477	
International Asian	-0.8686	-0.8651	0.1271				-0.5295	-0.7927	
GRE Verbal Percentile	0.0227	0.0047	-0.0015	0.0015			-0.0085		
GRE Quantitative Percentile	0.1027	0.0174	-0.0273	0.0179	0.0125		-0.2706	-0.2243	-0.0070
GRE Analytic Percentile	0.0056	0.0035							
Undergraduate GPA	0.9812	0.7135				-0.1316			0.2561
TOEFL/IELTS	0.3055		-0.2943	0.0300		0.0934	-0.6903	-0.5850	-2.2692
Elite U.S. University	0.4024								
Elite U.S. Liberal Arts College		0.1112							
Elite Foreign University		-0.4025							
Other Foreign University	0.0790	0.0940	0.1998						-0.2265
Missing Undergraduate Institution			-0.0036						
Graduate Degree	0.1178								
Work Experience	0.6006	0.4068	0.0404	0.0961	0.0470	0.1976			-2.0809
#Professor Recommenders									0.0922
#Prolific Recommenders	0.1534	0.0301	-0.0870	0.0526	0.0363		-0.0789	-0.0358	
Missing GRE	-0.6534			-0.0697					
Missing Undergraduate GPA	-0.2348	-0.1148							
Missing TOEFL/IELTS	0.1447			-0.1243	-0.1295		1.8402	2.2548	
Committee Score		3.4243			0.0353			-0.4899	
#Math Courses	-0.0553	-0.0354							
Advanced Math	0.4302	0.7352							
Letters Length/100	0.1196	0.0587							
Letters Terms (%) PC1	0.0746	0.1485	0.1258			0.0102			
Letters Terms (%) PC2	-0.0162	-0.2901							
CV Length/100	0.0219	0.0298		0.0014			-0.0286	-0.1179	
CV Coding	-0.0038	-0.1754		0.0165	0.0076		-0.2540	-0.4154	-0.0998
CV Publication/Working Paper	0.5563	0.6436		0.0892			-0.0987		
Essay Length/100	0.1202	0.0295							
Essay Specific Professor	0.3403	0.3958	0.0674						
Essay Economics Terms (%)	-0.0016			-0.0405	-0.0325	-0.0104	0.4473	0.6570	0.2117
Essay Research Terms (%)		-0.0278						-0.1506	
Essay Positive Terms (%)	-0.0049	-0.0382		0.0192	0.0140	0.0267			
Essay Negative Terms (%)		-0.0417							
Missing Committee Score		-0.8416			-0.2615	-0.0714		4.0146	0.7971
Missing Application Pdf	-1.0365								
Open House			1.0982						
Program Rank						-0.0082			0.1385
Missing Program Rank						-2.2557			19.4273
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	257	2329	2329	2329	2329	2329	2329
Pseudo-R^2	0.2841	0.8007	0.1281	0.0196	0.0213	0.2265	0.0206	0.0278	0.2693
CV Pseudo-R^2	0.2441	0.7605	0.0596	0.0082	0.0129	0.2199	0.0121	0.0183	0.2638

- **Accept:** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Verbal Percentile (+), GRE Quantitative Percentile (+), GRE Analytic Percentile (+), Undergrduate GPA (+), Elite U.S. Liberal Arts College (+), Elite Foreign University (-), Other Foreign University (+), Work Experience (+), # Prolific Recommenders (+), Missing Undergraduate GPA (-), Committee Score (+), #Math Courses (-), Advanced Math (+), Letters Length/100 (+), Letters Terms (%) PC1 (+), Letters Terms (%) PC2 (-), CV Length/100 (+), CV Coding (-), CV Publication/Working Paper (+), Essay Length/100 (+), Essay Specific Professor (+), Economics Research Terms (%) (-), Essay Positive Terms (%) (-), Essay Negative Terms (%) (-), Missing Committee Score (-)
- **Matriculate:** 2018 (+), International Asian (+), GRE Verbal Percentile (-), GRE Quantitative Percentile (-), TOEFL/IELTS (-), Other Foreign University (+), Missing Undergraduate Institution (-), Work Experience (+), #Prolific Recommenders (-), Letters Terms (%) PC1 (+), Essay Specific Professor (+), Open House (+)
- **T100 Placement:** Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Letters Terms (%) PC1 (+), Essay Economics Terms (%) (-), Essay Positive Terms (%) (+), Missing Committee Score (-), Program Rank (-), Missing Program Rank (-)

Acceptance, Matriculation, and Success AMEs

A3: Economics Department Static Estimates

Table 8: Economics Department Static Estimates

(a) Unrestricted

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-23.0601***	-3.4656***	-28.5278***	-1.2765***	-4.6847***	-0.5998***	-4.4553**	-0.5635**
	(2.0491)	(0.2371)	(6.9081)	(0.2933)	(1.6180)	(0.2054)	(2.1831)	(0.2752)
2016	-0.2896	-0.0435	-0.5294	-0.0237	0.0683	0.0087	0.0762	0.0096
	(0.1794)	(0.0268)	(0.5264)	(0.0230)	(0.1852)	(0.0237)	(0.1883)	(0.0238)
2017	-0.1533	-0.0230	-1.3366**	-0.0598**	0.1166	0.0149	0.1266	0.0160
	(0.1726)	(0.0260)	(0.5312)	(0.0235)	(0.1877)	(0.0240)	(0.1929)	(0.0244)
2018	1.5429***	0.2319***	-3.6854***	-0.1649***	0.0112	0.0014	0.0095	0.0012
	(0.1858)	(0.0251)	(0.6281)	(0.0209)	(0.1920)	(0.0246)	(0.2011)	(0.0254)
2019	0.2905	0.0437	-3.0765***	-0.1377***	0.0177	0.0023	0.0114	0.0014
	(0.1833)	(0.0273)	(0.6899)	(0.0257)	(0.1913)	(0.0245)	(0.1974)	(0.0250)
Winsorized Age	0.0628***	0.0094***	-0.0133	-0.0006	0.0439*	0.0056*	0.0428*	0.0054*
	(0.0218)	(0.0032)	(0.0612)	(0.0027)	(0.0229)	(0.0029)	(0.0235)	(0.0030)
Female	0.3919***	0.0589***	0.8909**	0.0399***	-0.0342	-0.0044	-0.0422	-0.0053
	(0.1109)	(0.0164)	(0.3672)	(0.0154)	(0.1232)	(0.0158)	(0.1246)	(0.0158)
U.S.	-0.1124	-0.0169	0.0293	0.0013	0.1634	0.0209	0.1597	0.0202
	(0.1894)	(0.0285)	(0.4978)	(0.0223)	(0.2198)	(0.0281)	(0.2219)	(0.0280)
International Asian	0.0087	0.0013	-1.1484***	-0.0514***	-0.0894	-0.0114	-0.0921	-0.0116
	(0.1396)	(0.0210)	(0.4133)	(0.0172)	(0.1489)	(0.0191)	(0.1540)	(0.0195)
GRE Verbal Percentile	0.0326***	0.0049***	0.0039	0.0002	0.0015	0.0002	0.0006	0.0001
	(0.0034)	(0.0005)	(0.0131)	(0.0006)	(0.0034)	(0.0004)	(0.0036)	(0.0005)
GRE Quantitative Percentile	0.1829***	0.0275***	0.0402	0.0018	0.0065	0.0008	0.0028	0.0004
	(0.0163)	(0.0018)	(0.0326)	(0.0015)	(0.0076)	(0.0010)	(0.0080)	(0.0010)
GRE Analytic Percentile	0.0152***	0.0023***	0.0078	0.0004	-0.0011	-0.0001	-0.0022	-0.0003
	(0.0026)	(0.0004)	(0.0090)	(0.0004)	(0.0028)	(0.0004)	(0.0029)	(0.0004)
TOEFL/IELTS	-0.0097	-0.0015	-0.1286	-0.0058	0.3267*	0.0418*	0.3375*	0.0427*
	(0.1632)	(0.0245)	(0.5621)	(0.0250)	(0.1721)	(0.0220)	(0.1752)	(0.0221)
Elite U.S. University	0.9066***	0.1363***	-0.5317	-0.0238	-0.0506	-0.0065	-0.0941	-0.0119
	(0.2103)	(0.0312)	(0.6438)	(0.0293)	(0.2283)	(0.0292)	(0.2319)	(0.0293)
Elite U.S. Liberal Arts College	0.3268	0.0491	0.3253	0.0146	-0.0812	-0.0104	-0.0866	-0.0110
	(0.3402)	(0.0510)	(0.8010)	(0.0356)	(0.3863)	(0.0495)	(0.3922)	(0.0496)
Elite Foreign University	0.0518	0.0078	-0.9107	-0.0408	0.0910	0.0117	0.0958	0.0121
	(0.2959)	(0.0445)	(0.9288)	(0.0424)	(0.3413)	(0.0437)	(0.3439)	(0.0435)
Other Foreign University	0.1161	0.0174	0.2293	0.0103	0.2023	0.0259	0.1627	0.0206
	(0.1969)	(0.0296)	(0.6888)	(0.0306)	(0.2140)	(0.0274)	(0.2168)	(0.0274)
Missing Undergraduate Institution	-0.4337**	-0.0652**	0.0505	0.0023	0.1941	0.0249	0.1768	0.0224
	(0.1996)	(0.0300)	(0.6688)	(0.0299)	(0.2292)	(0.0293)	(0.2321)	(0.0294)
Graduate Degree	1.0229***	0.1537***	-0.3419	-0.0153	-0.1377	-0.0176	-0.1684	-0.0213
	(0.1262)	(0.0177)	(0.3883)	(0.0171)	(0.1312)	(0.0168)	(0.1367)	(0.0172)
Work Experience	0.7870***	0.1183***	0.5914*	0.0265*	0.3144**	0.0403**	0.2740*	0.0347*
	(0.1256)	(0.0179)	(0.3412)	(0.0154)	(0.1361)	(0.0173)	(0.1424)	(0.0179)
#Professor Recommenders	-0.0211	-0.0032	-0.1896	-0.0085	-0.0570	-0.0073	-0.0622	-0.0079
	(0.0839)	(0.0126)	(0.2574)	(0.0113)	(0.0846)	(0.0108)	(0.0859)	(0.0109)
#Prolific Recommenders	0.1325***	0.0199***	0.1903	0.0085	0.0559	0.0072	0.0667	0.0084
	(0.0513)	(0.0077)	(0.1757)	(0.0077)	(0.0574)	(0.0073)	(0.0582)	(0.0073)
Missing GRE	-1.8142***	-0.2727***	1.6533	0.0740	-0.0087	-0.0011	0.0247	0.0031
	(0.5268)	(0.0772)	(1.8092)	(0.0811)	(0.3716)	(0.0476)	(0.3772)	(0.0477)
Missing TOEFL/IELTS	0.0947	0.0142	0.4511	0.0202	-0.0080	-0.0010	-0.0036	-0.0005
	(0.1295)	(0.0194)	(0.4238)	(0.0189)	(0.1419)	(0.0182)	(0.1435)	(0.0181)
Committee Score			3.7089***	0.1660***			-0.0006	-0.0001
			(0.3984)	(0.0065)			(0.0486)	(0.0062)
#Math Courses			-0.2494	-0.0112			0.0007	0.0001
			(0.2386)	(0.0108)			(0.0879)	(0.0111)
Advanced Math			1.1003	0.0492			-0.2220	-0.0281
			(0.7916)	(0.0347)			(0.2926)	(0.0370)
Letters Length/100			0.0770	0.0034			-0.0135	-0.0017
			(0.1080)	(0.0048)			(0.0385)	(0.0049)
Letters Terms (%) PC1			0.2451	0.0110			0.1278	0.0162
			(0.2489)	(0.0112)			(0.0911)	(0.0115)
Letters Terms (%) PC2			-0.3767	-0.0169			-0.1364	-0.0172
			(0.3450)	(0.0153)			(0.1257)	(0.0159)
CV Length/100			0.0391	0.0017			0.0347	0.0044
			(0.1060)	(0.0047)			(0.0592)	(0.0075)
CV Coding			-0.2228*	-0.0100*			0.1024	0.0130
			(0.1301)	(0.0060)			(0.0651)	(0.0082)
CV Publication/Working Paper			0.7611	0.0341			0.1440	0.0182
			(0.8128)	(0.0364)			(0.3359)	(0.0425)
Essay Length/100			0.2047	0.0092			-0.0718	-0.0091
			(0.3483)	(0.0157)			(0.0818)	(0.0103)
Essay Specific Professor			0.4125	0.0185			0.2894	0.0366
			(0.6138)	(0.0275)			(0.2770)	(0.0350)
Essay Economics Terms (%)			0.0154	0.0007			-0.0854	-0.0108
			(0.0814)	(0.0036)			(0.0614)	(0.0077)
Essay Research Terms (%)			-0.0751	-0.0034			0.0146	0.0018
			(0.1492)	(0.0067)			(0.0672)	(0.0085)
Essay Positive Terms (%)			-0.0689	-0.0031			0.0977*	0.0124*
			(0.0968)	(0.0043)			(0.0539)	(0.0068)
Essay Negative Terms (%)			-0.0863	-0.0039			-0.0441	-0.0056
			(0.1368)	(0.0061)			(0.0745)	(0.0094)
Missing Committee Score							-0.2179	-0.0276
							(0.1515)	(0.0191)
Missing Application Pdf			0.2614	0.0117			0.2498	0.0316
			(0.7005)	(0.0311)			(0.2756)	(0.0348)
Program Rank					-0.0084***	-0.0011***	-0.0086***	-0.0011***
					(0.0020)	(0.0003)	(0.0021)	(0.0003)
Missing Program Rank					-3.1385***	-0.4018***	-3.1438***	-0.3976***
					(0.4051)	(0.0517)	(0.4076)	(0.0514)
Obs | Param | LL | AIC	2329	133	-2964.5	6195.0				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Restricted

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	-0.8781***	29.36%	-0.4389***	39.20%	-1.2521***	-0.9262**	
	(0.1628)	(23.20%, 36.38%)	(0.1432)	(32.75%, 46.05%)	(0.4466)	(0.4540)	
2016	-0.0091	29.17%	0.0554	40.53%	-0.0178	0.0421	
	(0.2466)	(22.21%, 37.27%)	(0.1767)	(33.55%, 47.92%)	(0.2477)	(0.1771)	
2017	0.3482*	37.06%	0.1035	41.69%	0.3447*	0.0698	
	(0.1963)	(32.10%, 42.29%)	(0.1727)	(34.83%, 48.90%)	(0.1964)	(0.1725)	
2018	0.3936**	38.12%	0.0040	39.30%	0.4390**	-0.0912	
	(0.1856)	(33.98%, 42.44%)	(0.1908)	(31.80%, 47.33%)	(0.1859)	(0.1872)	
2019	0.2163	34.04%	0.0175	39.62%	0.2258	-0.0617	
	(0.2017)	(28.67%, 39.85%)	(0.1791)	(32.54%, 47.16%)	(0.2018)	(0.1778)	
Winsorized Age					0.0018	0.0000	
					(0.0012)	(0.0016)	
Female					0.0118	0.0233	
					(0.0073)	(0.0146)	
U.S.					-0.0034	0.0022	
					(0.0058)	(0.0131)	
International Asian					-0.0006	-0.0302	
					(0.0041)	(0.0187)	
GRE Verbal Percentile					0.0010*	0.0001	
					(0.0005)	(0.0003)	
GRE Quantitative Percentile					0.0055*	0.0011	
					(0.0032)	(0.0011)	
GRE Analytic Percentile					0.0005*	0.0002	
					(0.0003)	(0.0002)	
TOEFL/IELTS					-0.0009	-0.0015	
					(0.0047)	(0.0144)	
Elite U.S. University					0.0252	-0.0131	
					(0.0155)	(0.0178)	
Elite U.S. Liberal Arts College					0.0087	0.0095	
					(0.0113)	(0.0216)	
Elite Foreign University					0.0013	-0.0229	
					(0.0086)	(0.0272)	
Other Foreign University					0.0038	0.0075	
					(0.0062)	(0.0180)	
Missing Undergraduate Institution					-0.0119	0.0030	
					(0.0084)	(0.0173)	
Graduate Degree					0.0306*	-0.0103	
					(0.0175)	(0.0117)	
Work Experience					0.0235*	0.0168	
					(0.0141)	(0.0130)	
#Professor Recommenders					-0.0004	-0.0057	
					(0.0025)	(0.0074)	
#Prolific Recommenders					0.0039	0.0052	
					(0.0026)	(0.0053)	
Missing GRE					-0.0566	0.0458	
					(0.0361)	(0.0532)	
Missing TOEFL/IELTS					0.0029	0.0123	
					(0.0041)	(0.0127)	
Committee Score						0.0961*	
						(0.0491)	
#Math Courses						-0.0064	
						(0.0069)	
Advanced Math						0.0287	
						(0.0244)	
Letters Length/100						0.0019	
						(0.0029)	
Letters Terms (%) PC1						0.0063	
						(0.0072)	
Letters Terms (%) PC2						-0.0103	
						(0.0104)	
CV Length/100						0.0010	
						(0.0027)	
CV Coding						-0.0054	
						(0.0043)	
CV Publication/Working Paper						0.0207	
						(0.0245)	
Essay Length/100						0.0050	
						(0.0089)	
Essay Specific Professor						0.0106	
						(0.0168)	
Essay Economics Terms (%)						0.0002	
						(0.0021)	
Essay Research Terms (%)						-0.0019	
						(0.0039)	
Essay Positive Terms (%)						-0.0014	
						(0.0025)	
Essay Negative Terms (%)						-0.0023	
						(0.0038)	
Missing Committee Score						-0.2328**	
						(0.1162)	
Missing Application Pdf						0.0079	
						(0.0185)	
Program Rank					-0.0085***	-0.0078***	
					(0.0021)	(0.0020)	
Missing Program Rank					-3.0030***	-3.1171***	
					(0.4080)	(0.4041)	
Sigma							0.0064*
							(0.0035)
Obs | Param | LL | AIC	2329	80	-2992.0	6144.1			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

▸▸ Restricted Static Estimates

A3: Economics Department Dynamic Estimates

Table 9: Economics Department Dynamic Estimates

(a) Unrestricted

	Committee Score	Transition		Accept		Success	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-2.7966	-23.0601***	-3.4656***	-28.5278***	-1.2765***	-4.4553**	-0.5635**
	(2.1751)	(2.0491)	(0.2371)	(6.9081)	(0.2933)	(2.1831)	(0.2752)
2016	0.0915	-0.2896	-0.0435	-0.5294	-0.0237	0.0762	0.0096
	(0.1856)	(0.1794)	(0.0268)	(0.5264)	(0.0230)	(0.1883)	(0.0238)
2017	0.2354	-0.1533	-0.0230	-1.3366**	-0.0598**	0.1266	0.0160
	(0.2130)	(0.1726)	(0.0260)	(0.5312)	(0.0235)	(0.1929)	(0.0244)
2018	0.5999***	1.5429***	0.2319***	-3.6854***	-0.1649***	0.0095	0.0012
	(0.1777)	(0.1858)	(0.0251)	(0.6281)	(0.0209)	(0.2011)	(0.0254)
2019	0.6118***	0.2905	0.0437	-3.0765***	-0.1377***	0.0114	0.0014
	(0.1907)	(0.1833)	(0.0273)	(0.6899)	(0.0257)	(0.1974)	(0.0250)
Winsorized Age	-0.0621***	0.0628***	0.0094***	-0.0133	-0.0006	0.0428*	0.0054*
	(0.0235)	(0.0218)	(0.0032)	(0.0612)	(0.0027)	(0.0235)	(0.0030)
Female	-0.0543	0.3919***	0.0589***	0.8909**	0.0399***	-0.0422	-0.0053
	(0.1125)	(0.1109)	(0.0164)	(0.3672)	(0.0154)	(0.1246)	(0.0158)
U.S.	-0.5071**	-0.1124	-0.0169	0.0293	0.0013	0.1597	0.0202
	(0.2043)	(0.1894)	(0.0285)	(0.4978)	(0.0223)	(0.2219)	(0.0280)
International Asian	-0.7530***	0.0087	0.0013	-1.1484***	-0.0514***	-0.0921	-0.0116
	(0.1452)	(0.1396)	(0.0210)	(0.4133)	(0.0172)	(0.1540)	(0.0195)
GRE Verbal Percentile	0.0066*	0.0326***	0.0049***	0.0039	0.0002	0.0006	0.0001
	(0.0036)	(0.0034)	(0.0005)	(0.0131)	(0.0006)	(0.0036)	(0.0005)
GRE Quantitative Percentile	0.0447***	0.1829***	0.0275***	0.0402	0.0018	0.0028	0.0004
	(0.0111)	(0.0163)	(0.0018)	(0.0326)	(0.0015)	(0.0080)	(0.0010)
GRE Analytic Percentile	0.0029	0.0152***	0.0023***	0.0078	0.0004	-0.0022	-0.0003
	(0.0027)	(0.0026)	(0.0004)	(0.0090)	(0.0004)	(0.0029)	(0.0004)
TOEFL/IELTS	0.4407***	-0.0097	-0.0015	-0.1286	-0.0058	0.3375*	0.0427*
	(0.1696)	(0.1632)	(0.0245)	(0.5621)	(0.0250)	(0.1752)	(0.0221)
Elite U.S. University	0.5507***	0.9066***	0.1363***	-0.5317	-0.0238	-0.0941	-0.0119
	(0.1994)	(0.2103)	(0.0312)	(0.6438)	(0.0293)	(0.2319)	(0.0293)
Elite U.S. Liberal Arts College	-0.0711	0.3268	0.0491	0.3253	0.0146	-0.0866	-0.0110
	(0.3683)	(0.3402)	(0.0510)	(0.8010)	(0.0356)	(0.3922)	(0.0496)
Elite Foreign University	0.2902	0.0518	0.0078	-0.9107	-0.0408	0.0958	0.0121
	(0.2991)	(0.2959)	(0.0445)	(0.9288)	(0.0424)	(0.3439)	(0.0435)
Other Foreign University	0.4797**	0.1161	0.0174	0.2293	0.0103	0.1627	0.0206
	(0.2022)	(0.1969)	(0.0296)	(0.6888)	(0.0306)	(0.2168)	(0.0274)
Missing Undergraduate Institution	0.5097**	-0.4337**	-0.0652**	0.0505	0.0023	0.1768	0.0224
	(0.2157)	(0.1996)	(0.0300)	(0.6688)	(0.0299)	(0.2321)	(0.0294)
Graduate Degree	-0.0757	1.0229***	0.1537***	-0.3419	-0.0153	-0.1684	-0.0213
	(0.1279)	(0.1262)	(0.0177)	(0.3883)	(0.0171)	(0.1367)	(0.0172)
Work Experience	0.3458**	0.7870***	0.1183***	0.5914*	0.0265*	0.2740*	0.0347*
	(0.1345)	(0.1256)	(0.0179)	(0.3412)	(0.0154)	(0.1424)	(0.0179)
#Professor Recommenders	0.0618	-0.0211	-0.0032	-0.1896	-0.0085	-0.0622	-0.0079
	(0.0769)	(0.0839)	(0.0126)	(0.2574)	(0.0113)	(0.0859)	(0.0109)
#Prolific Recommenders	0.1667***	0.1325***	0.0199***	0.1903	0.0085	0.0667	0.0084
	(0.0548)	(0.0513)	(0.0077)	(0.1757)	(0.0077)	(0.0582)	(0.0073)
Missing GRE	-1.6087*	-1.8142***	-0.2727***	1.6533	0.0740	0.0247	0.0031
	(0.9422)	(0.5268)	(0.0772)	(1.8092)	(0.0811)	(0.3772)	(0.0477)
Missing TOEFL/IELTS	0.3199**	0.0947	0.0142	0.4511	0.0202	-0.0036	-0.0005
	(0.1319)	(0.1295)	(0.0194)	(0.4238)	(0.0189)	(0.1435)	(0.0181)
Committee Score				3.7089***	0.1660***	-0.0006	-0.0001
				(0.3984)	(0.0065)	(0.0486)	(0.0062)
#Math Courses	-0.0790			-0.2494	-0.0112	0.0007	0.0001
	(0.0519)			(0.2386)	(0.0108)	(0.0879)	(0.0111)
Advanced Math	0.2979			1.1003	0.0492	-0.2220	-0.0281
	(0.1889)			(0.7916)	(0.0347)	(0.2926)	(0.0370)
Letters Length/100	0.0858***			0.0770	0.0034	-0.0135	-0.0017
	(0.0245)			(0.1080)	(0.0048)	(0.0385)	(0.0049)
Letters Terms (%) PC1	0.1027*			0.2451	0.0110	0.1278	0.0162
	(0.0551)			(0.2489)	(0.0112)	(0.0911)	(0.0115)
Letters Terms (%) PC2	-0.0208			-0.3767	-0.0169	-0.1364	-0.0172
	(0.0823)			(0.3450)	(0.0153)	(0.1257)	(0.0159)
CV Length/100	0.0513			0.0391	0.0017	0.0347	0.0044
	(0.0329)			(0.1060)	(0.0047)	(0.0592)	(0.0075)
CV Coding	0.0381			-0.2228*	-0.0100*	0.1024	0.0130
	(0.0424)			(0.1301)	(0.0060)	(0.0651)	(0.0082)
CV Publication/Working Paper	0.5482***			0.7611	0.0341	0.1440	0.0182
	(0.1912)			(0.8128)	(0.0364)	(0.3359)	(0.0425)
Essay Length/100	0.0927			0.2047	0.0092	-0.0718	-0.0091
	(0.0597)			(0.3483)	(0.0157)	(0.0818)	(0.0103)
Essay Specific Professor	0.1411			0.4125	0.0185	0.2894	0.0366
	(0.1744)			(0.6138)	(0.0275)	(0.2770)	(0.0350)
Essay Economics Terms (%)	0.0287			0.0154	0.0007	-0.0854	-0.0108
	(0.0312)			(0.0814)	(0.0036)	(0.0614)	(0.0077)
Essay Research Terms (%)	0.0396			-0.0751	-0.0034	0.0146	0.0018
	(0.0426)			(0.1492)	(0.0067)	(0.0672)	(0.0085)
Essay Positive Terms (%)	0.0171			-0.0689	-0.0031	0.0977*	0.0124*
	(0.0345)			(0.0968)	(0.0043)	(0.0539)	(0.0068)
Essay Negative Terms (%)	0.0441			-0.0863	-0.0039	-0.0441	-0.0056
	(0.0443)			(0.1368)	(0.0061)	(0.0745)	(0.0094)
Missing Committee Score						-0.2179	-0.0276
						(0.1515)	(0.0191)
Missing Application Pdf	-0.5376***			0.2614	0.0117	0.2498	0.0316
	(0.1841)			(0.7005)	(0.0311)	(0.2756)	(0.0348)
Program Rank						-0.0086***	-0.0011***
						(0.0021)	(0.0003)
Missing Program Rank						-3.1438***	-0.3976***
						(0.4076)	(0.0514)
$\sqrt{MSE}$	1.6467***						
	(0.0370)						
Obs | Param | LL | AIC	2329	582	-14135.6	29435.1			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Restricted

	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-5.6768**	-0.9860***	0.3731	-0.5503***	36.58%	-0.7153*	
	(2.3401)	(0.1000)	(0.3067, 0.4539)	(0.1561)	(29.81%, 43.92%)	(0.3751)	
2016	0.0001	0.0171	0.3795	0.0269	37.21%	0.0155	
	(0.1827)	(0.1265)	(0.3025, 0.4761)	(0.2038)	(29.20%, 45.99%)	(0.2039)	
2017	0.4867**	0.1006	0.4126	0.1561	40.27%	0.1404	
	(0.2144)	(0.1085)	(0.3563, 0.4778)	(0.1786)	(34.52%, 46.30%)	(0.1795)	
2018	0.9329***	0.0277	0.3836	0.0473	37.68%	-0.0004	
	(0.1967)	(0.1208)	(0.3271, 0.4497)	(0.1943)	(31.79%, 43.97%)	(0.2015)	
2019	0.6228***	0.0411	0.3887	0.0667	38.14%	0.0232	
	(0.1963)	(0.1142)	(0.3268, 0.4624)	(0.1852)	(31.89%, 44.81%)	(0.1863)	
Winsorized Age	-0.0454*					-0.0004	
	(0.0246)					(0.0009)	
Female	-0.1026					0.0114	
	(0.1055)					(0.0091)	
U.S.	-0.3132					0.0035	
	(0.2098)					(0.0081)	
International Asian	-0.3624**					-0.0051	
	(0.1577)					(0.0070)	
GRE Verbal Percentile	0.0110***					0.0002	
	(0.0036)					(0.0003)	
GRE Quantitative Percentile	0.0580***					0.0007	
	(0.0107)					(0.0008)	
GRE Analytic Percentile	0.0026					0.0001	
	(0.0028)					(0.0001)	
TOEFL/IELTS	0.3161*					0.0007	
	(0.1703)					(0.0073)	
Elite U.S. University	0.5975***					-0.0108	
	(0.1775)					(0.0108)	
Elite U.S. Liberal Arts College	-0.4500					0.0047	
	(0.3224)					(0.0151)	
Elite Foreign University	0.2329					-0.0086	
	(0.2727)					(0.0136)	
Other Foreign University	0.4356**					0.0048	
	(0.1932)					(0.0112)	
Missing Undergraduate Institution	0.3922*					-0.0001	
	(0.2038)					(0.0098)	
Graduate Degree	-0.0531					-0.0022	
	(0.1498)					(0.0063)	
Work Experience	0.0819					0.0088	
	(0.1415)					(0.0079)	
#Professor Recommenders	0.1025					-0.0005	
	(0.0784)					(0.0037)	
#Prolific Recommenders	0.0536					-0.0002	
	(0.0598)					(0.0026)	
Missing GRE	-1.2274					0.0216	
	(0.9140)					(0.0228)	
Missing TOEFL/IELTS	0.2149*					0.0036	
	(0.1252)					(0.0061)	
Committee Score						0.0514	
						(0.0355)	
#Math Courses	-0.1443					-0.0029	
	(0.0920)					(0.0047)	
Advanced Math	0.4000					-0.0022	
	(0.2955)					(0.0108)	
Letters Length/100	0.0906**					0.0005	
	(0.0439)					(0.0020)	
Letters Terms (%) PC1	0.2131**					0.0053	
	(0.0942)					(0.0056)	
Letters Terms (%) PC2	-0.1480					-0.0045	
	(0.1963)					(0.0067)	
CV Length/100	0.0283					0.0013	
	(0.0724)					(0.0021)	
CV Coding	0.0306					-0.0044	
	(0.0981)					(0.0040)	
CV Publication/Working Paper	1.1465***					0.0006	
	(0.4362)					(0.0134)	
Essay Length/100	0.2719***					0.0080	
	(0.1044)					(0.0090)	
Essay Specific Professor	0.1374					-0.0060	
	(0.2212)					(0.0136)	
Essay Economics Terms (%)	0.0664					0.0004	
	(0.0550)					(0.0015)	
Essay Research Terms (%)	0.1364*					-0.0002	
	(0.0821)					(0.0030)	
Essay Positive Terms (%)	0.0422					-0.0004	
	(0.0718)					(0.0029)	
Essay Negative Terms (%)	0.1003					0.0008	
	(0.0766)					(0.0033)	
Missing Committee Score						-0.2001	
						(0.1292)	
Missing Application Pdf	-0.7069**					-0.0267	
	(0.2763)					(0.0180)	
Program Rank						-0.0080***	
						(0.0020)	
Missing Program Rank						-3.0773***	
						(0.4051)	
$\sqrt{MSE}$	1.7300***						
	(0.0475)						
Sigma							0.0020
							(0.0013)
Obs | Param | LL | AIC	2329	529	-14131.0	29319.9			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Dynamic Estimates

A3: Economics Department Myopic Estimates

Table 10: Economics Department Myopic Estimates

(a) Unrestricted

	Committee Score	Transition		Accept		Success	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	4.0405***	-23.0601***	-3.4656***	-28.5278***	-1.2765***	-4.4553**	-0.5635**
	(0.9990)	(2.0491)	(0.2371)	(6.9081)	(0.2933)	(2.1831)	(0.2752)
2016	0.1701	-0.2896	-0.0435	-0.5294	-0.0237	0.0762	0.0096
	(0.1966)	(0.1794)	(0.0268)	(0.5264)	(0.0230)	(0.1883)	(0.0238)
2017	0.4355**	-0.1533	-0.0230	-1.3366**	-0.0598**	0.1266	0.0160
	(0.2214)	(0.1726)	(0.0260)	(0.5312)	(0.0235)	(0.1929)	(0.0244)
2018	0.8410***	1.5429***	0.2319***	-3.6854***	-0.1649***	0.0095	0.0012
	(0.1873)	(0.1858)	(0.0251)	(0.6281)	(0.0209)	(0.2011)	(0.0254)
2019	0.7560***	0.2905	0.0437	-3.0765***	-0.1377***	0.0114	0.0014
	(0.1987)	(0.1833)	(0.0273)	(0.6899)	(0.0257)	(0.1974)	(0.0250)
Winsorized Age		0.0628***	0.0094***	-0.0133	-0.0006	0.0428*	0.0054*
		(0.0218)	(0.0032)	(0.0612)	(0.0027)	(0.0235)	(0.0030)
Female		0.3919***	0.0589***	0.8909**	0.0399***	-0.0422	-0.0053
		(0.1109)	(0.0164)	(0.3672)	(0.0154)	(0.1246)	(0.0158)
U.S.		-0.1124	-0.0169	0.0293	0.0013	0.1597	0.0202
		(0.1894)	(0.0285)	(0.4978)	(0.0223)	(0.2219)	(0.0280)
International Asian		0.0087	0.0013	-1.1484***	-0.0514***	-0.0921	-0.0116
		(0.1396)	(0.0210)	(0.4133)	(0.0172)	(0.1540)	(0.0195)
GRE Verbal Percentile		0.0326***	0.0049***	0.0039	0.0002	0.0006	0.0001
		(0.0034)	(0.0005)	(0.0131)	(0.0006)	(0.0036)	(0.0005)
GRE Quantitative Percentile		0.1829***	0.0275***	0.0402	0.0018	0.0028	0.0004
		(0.0163)	(0.0018)	(0.0326)	(0.0015)	(0.0080)	(0.0010)
GRE Analytic Percentile		0.0152***	0.0023***	0.0078	0.0004	-0.0022	-0.0003
		(0.0026)	(0.0004)	(0.0090)	(0.0004)	(0.0029)	(0.0004)
TOEFL/IELTS		-0.0097	-0.0015	-0.1286	-0.0058	0.3375*	0.0427*
		(0.1632)	(0.0245)	(0.5621)	(0.0250)	(0.1752)	(0.0221)
Elite U.S. University		0.9066***	0.1363***	-0.5317	-0.0238	-0.0941	-0.0119
		(0.2103)	(0.0312)	(0.6438)	(0.0293)	(0.2319)	(0.0293)
Elite U.S. Liberal Arts College		0.3268	0.0491	0.3253	0.0146	-0.0866	-0.0110
		(0.3402)	(0.0510)	(0.8010)	(0.0356)	(0.3922)	(0.0496)
Elite Foreign University		0.0518	0.0078	-0.9107	-0.0408	0.0958	0.0121
		(0.2959)	(0.0445)	(0.9288)	(0.0424)	(0.3439)	(0.0435)
Other Foreign University		0.1161	0.0174	0.2293	0.0103	0.1627	0.0206
		(0.1969)	(0.0296)	(0.6888)	(0.0306)	(0.2168)	(0.0274)
Missing Undergraduate Institution		-0.4337**	-0.0652**	0.0505	0.0023	0.1768	0.0224
		(0.1996)	(0.0300)	(0.6688)	(0.0299)	(0.2321)	(0.0294)
Graduate Degree		1.0229***	0.1537***	-0.3419	-0.0153	-0.1684	-0.0213
		(0.1262)	(0.0177)	(0.3883)	(0.0171)	(0.1367)	(0.0172)
Work Experience		0.7870***	0.1183***	0.5914*	0.0265*	0.2740*	0.0347*
		(0.1256)	(0.0179)	(0.3412)	(0.0154)	(0.1424)	(0.0179)
#Professor Recommenders		-0.0211	-0.0032	-0.1896	-0.0085	-0.0622	-0.0079
		(0.0839)	(0.0126)	(0.2574)	(0.0113)	(0.0859)	(0.0109)
#Prolific Recommenders		0.1325***	0.0199***	0.1903	0.0085	0.0667	0.0084
		(0.0513)	(0.0077)	(0.1757)	(0.0077)	(0.0582)	(0.0073)
Missing GRE		-1.8142***	-0.2727***	1.6533	0.0740	0.0247	0.0031
		(0.5268)	(0.0772)	(1.8092)	(0.0811)	(0.3772)	(0.0477)
Missing TOEFL/IELTS		0.0947	0.0142	0.4511	0.0202	-0.0036	-0.0005
		(0.1295)	(0.0194)	(0.4238)	(0.0189)	(0.1435)	(0.0181)
Committee Score				3.7089***	0.1660***	-0.0006	-0.0001
				(0.3984)	(0.0065)	(0.0486)	(0.0062)
#Math Courses	-0.0804			-0.2494	-0.0112	0.0007	0.0001
	(0.0520)			(0.2386)	(0.0108)	(0.0879)	(0.0111)
Advanced Math	0.3560*			1.1003	0.0492	-0.2220	-0.0281
	(0.1909)			(0.7916)	(0.0347)	(0.2926)	(0.0370)
Letters Length/100	0.1225***			0.0770	0.0034	-0.0135	-0.0017
	(0.0242)			(0.1080)	(0.0048)	(0.0385)	(0.0049)
Letters Terms (%) PC1	0.0974*			0.2451	0.0110	0.1278	0.0162
	(0.0536)			(0.2489)	(0.0112)	(0.0911)	(0.0115)
Letters Terms (%) PC2	-0.0925			-0.3767	-0.0169	-0.1364	-0.0172
	(0.0856)			(0.3450)	(0.0153)	(0.1257)	(0.0159)
CV Length/100	0.0588*			0.0391	0.0017	0.0347	0.0044
	(0.0323)			(0.1060)	(0.0047)	(0.0592)	(0.0075)
CV Coding	0.0551			-0.2228*	-0.0100*	0.1024	0.0130
	(0.0417)			(0.1301)	(0.0060)	(0.0651)	(0.0082)
CV Publication/Working Paper	0.5324***			0.7611	0.0341	0.1440	0.0182
	(0.1975)			(0.8128)	(0.0364)	(0.3359)	(0.0425)
Essay Length/100	0.0763			0.2047	0.0092	-0.0718	-0.0091
	(0.0567)			(0.3483)	(0.0157)	(0.0818)	(0.0103)
Essay Specific Professor	0.1442			0.4125	0.0185	0.2894	0.0366
	(0.1767)			(0.6138)	(0.0275)	(0.2770)	(0.0350)
Essay Economics Terms (%)	0.0042			0.0154	0.0007	-0.0854	-0.0108
	(0.0314)			(0.0814)	(0.0036)	(0.0614)	(0.0077)
Essay Research Terms (%)	0.0304			-0.0751	-0.0034	0.0146	0.0018
	(0.0446)			(0.1492)	(0.0067)	(0.0672)	(0.0085)
Essay Positive Terms (%)	0.0175			-0.0689	-0.0031	0.0977*	0.0124*
	(0.0373)			(0.0968)	(0.0043)	(0.0539)	(0.0068)
Essay Negative Terms (%)	0.0429			-0.0863	-0.0039	-0.0441	-0.0056
	(0.0460)			(0.1368)	(0.0061)	(0.0745)	(0.0094)
Missing Committee Score						-0.2179	-0.0276
						(0.1515)	(0.0191)
Missing Application Pdf	-0.4006**			0.2614	0.0117	0.2498	0.0316
	(0.1850)			(0.7005)	(0.0311)	(0.2756)	(0.0348)
Program Rank						-0.0086***	-0.0011***
						(0.0021)	(0.0003)
Missing Program Rank						-3.1438***	-0.3976***
						(0.4076)	(0.0514)
$\sqrt{MSE}$	1.7525***						
	(0.0403)						
Obs | Param | LL | AIC	2329	297	-14470.7	29535.4			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Restricted

	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	3.4796***	-0.9771***	0.3764	-0.5522***	36.54%	-1.2859**	
	(1.2648)	(0.0960)	(0.3118, 0.4544)	(0.1553)	(29.80%, 43.84%)	(0.5090)	
2016	0.0978	-0.0005	0.3762	-0.0048	36.42%	-0.0164	
	(0.1990)	(0.1320)	(0.2992, 0.4731)	(0.2135)	(28.21%, 45.51%)	(0.2130)	
2017	0.7100***	0.1077	0.4192	0.1614	40.35%	0.1384	
	(0.2332)	(0.1059)	(0.3701, 0.4749)	(0.1760)	(35.00%, 45.95%)	(0.1772)	
2018	1.0737***	0.0146	0.3820	0.0286	37.20%	-0.0305	
	(0.1730)	(0.1219)	(0.3237, 0.4506)	(0.1968)	(31.07%, 43.76%)	(0.1993)	
2019	0.8095***	0.0005	0.3766	0.0004	36.54%	-0.0654	
	(0.1948)	(0.1254)	(0.3086, 0.4596)	(0.2030)	(29.43%, 44.30%)	(0.2045)	
Winsorized Age						0.0010	
						(0.0007)	
Female						0.0132**	
						(0.0063)	
U.S.						0.0010	
						(0.0048)	
International Asian						-0.0075	
						(0.0051)	
GRE Verbal Percentile						0.0008**	
						(0.0003)	
GRE Quantitative Percentile						0.0039**	
						(0.0017)	
GRE Analytic Percentile						0.0004**	
						(0.0002)	
TOEFL/IELTS						0.0009	
						(0.0042)	
Elite U.S. University						0.0105	
						(0.0071)	
Elite U.S. Liberal Arts College						0.0064	
						(0.0086)	
Elite Foreign University						-0.0012	
						(0.0077)	
Other Foreign University						0.0066	
						(0.0059)	
Missing Undergraduate Institution						-0.0034	
						(0.0053)	
Graduate Degree						0.0194**	
						(0.0090)	
Work Experience						0.0189**	
						(0.0089)	
#Professor Recommenders						-0.0014	
						(0.0022)	
#Prolific Recommenders						0.0016	
						(0.0016)	
Missing GRE						-0.0449	
						(0.0282)	
Missing TOEFL/IELTS						0.0012	
						(0.0035)	
Committee Score						0.0686**	
						(0.0302)	
#Math Courses	-0.0627					-0.0011	
	(0.0689)					(0.0071)	
Advanced Math	0.3298					0.0023	
	(0.2533)					(0.0161)	
Letters Length/100	0.1467***					0.0023	
	(0.0317)					(0.0029)	
Letters Terms (%) PC1	0.1155					0.0016	
	(0.0772)					(0.0087)	
Letters Terms (%) PC2	-0.1020					-0.0069	
	(0.1152)					(0.0089)	
CV Length/100	0.0628					0.0022	
	(0.0407)					(0.0026)	
CV Coding	0.0466					-0.0051	
	(0.0580)					(0.0036)	
CV Publication/Working Paper	0.7501***					-0.0009	
	(0.2653)					(0.0155)	
Essay Length/100	0.1355					0.0083	
	(0.0858)					(0.0131)	
Essay Specific Professor	0.1387					0.0023	
	(0.2579)					(0.0205)	
Essay Economics Terms (%)	-0.0024					-0.0018	
	(0.0396)					(0.0021)	
Essay Research Terms (%)	0.0663					0.0023	
	(0.0596)					(0.0044)	
Essay Positive Terms (%)	0.0403					0.0011	
	(0.0476)					(0.0033)	
Essay Negative Terms (%)	0.0459					-0.0009	
	(0.0575)					(0.0048)	
Missing Committee Score						-0.1177	
						(0.1482)	
Missing Application Pdf	-0.9529***					-0.0359**	
	(0.2274)					(0.0157)	
Program Rank						-0.0078***	
						(0.0020)	
Missing Program Rank						-3.0747***	
						(0.4058)	
$\sqrt{MSE}$	1.8070***						
	(0.0436)						
Sigma							0.0027**
							(0.0012)
Obs | Param | LL | AIC	2329	244	-14536.1	29560.2			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Dynamic Estimates

Table 11: Economics Department Matriculation-Dynamic Estimates

(a) Unrestricted

	Committee Score	Transition		Accept		Success		Matriculate	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-2.7966	-23.0601***	-3.4656***	-28.5278***	-1.2765***	-4.4553**	-0.5635**	21.0916***	3.3374***
	(2.1751)	(2.0491)	(0.2371)	(6.9081)	(0.2933)	(2.1831)	(0.2752)	(6.9868)	(1.0163)
2016	0.0915	-0.2896	-0.0435	-0.5294	-0.0237	0.0762	0.0096	0.3206	0.0507
	(0.1856)	(0.1794)	(0.0268)	(0.5264)	(0.0230)	(0.1883)	(0.0238)	(0.6530)	(0.1030)
2017	0.2354	-0.1533	-0.0230	-1.3366**	-0.0598**	0.1266	0.0160	0.9971	0.1578
	(0.2130)	(0.1726)	(0.0260)	(0.5312)	(0.0235)	(0.1929)	(0.0244)	(0.6410)	(0.1008)
2018	0.5999***	1.5429***	0.2319***	-3.6854***	-0.1649***	0.0095	0.0012	1.6496***	0.2610***
	(0.1777)	(0.1858)	(0.0251)	(0.6281)	(0.0209)	(0.2011)	(0.0254)	(0.5920)	(0.0886)
2019	0.6118***	0.2905	0.0437	-3.0765***	-0.1377***	0.0114	0.0014	1.0916*	0.1727*
	(0.1907)	(0.1833)	(0.0273)	(0.6899)	(0.0257)	(0.1974)	(0.0250)	(0.6190)	(0.0971)
Winsorized Age	-0.0621***	0.0628***	0.0094***	-0.0133	-0.0006	0.0428*	0.0054*	-0.1396*	-0.0221*
	(0.0235)	(0.0218)	(0.0032)	(0.0612)	(0.0027)	(0.0235)	(0.0030)	(0.0738)	(0.0113)
Female	-0.0543	0.3919***	0.0589***	0.8909**	0.0399***	-0.0422	-0.0053	-0.4562	-0.0722
	(0.1125)	(0.1109)	(0.0164)	(0.3672)	(0.0154)	(0.1246)	(0.0158)	(0.3550)	(0.0556)
U.S.	-0.5071**	-0.1124	-0.0169	0.0293	0.0013	0.1597	0.0202	0.9160	0.1449
	(0.2043)	(0.1894)	(0.0285)	(0.4978)	(0.0223)	(0.2219)	(0.0280)	(0.5700)	(0.0896)
International Asian	-0.7530***	0.0087	0.0013	-1.1484***	-0.0514***	-0.0921	-0.0116	1.1667**	0.1846***
	(0.1452)	(0.1396)	(0.0210)	(0.4133)	(0.0172)	(0.1540)	(0.0195)	(0.4531)	(0.0689)
GRE Verbal Percentile	0.0066*	0.0326***	0.0049***	0.0039	0.0002	0.0006	0.0001	-0.0043	-0.0007
	(0.0036)	(0.0034)	(0.0005)	(0.0131)	(0.0006)	(0.0036)	(0.0005)	(0.0115)	(0.0018)
GRE Quantitative Percentile	0.0447***	0.1829***	0.0275***	0.0402	0.0018	0.0028	0.0004	-0.1358***	-0.0215***
	(0.0111)	(0.0163)	(0.0018)	(0.0326)	(0.0015)	(0.0080)	(0.0010)	(0.0444)	(0.0064)
GRE Analytic Percentile	0.0029	0.0152***	0.0023***	0.0078	0.0004	-0.0022	-0.0003	-0.0185**	-0.0029**
	(0.0027)	(0.0026)	(0.0004)	(0.0090)	(0.0004)	(0.0029)	(0.0004)	(0.0087)	(0.0013)
TOEFL/IELTS	0.4407***	-0.0097	-0.0015	-0.1286	-0.0058	0.3375*	0.0427*	-0.5140	-0.0813
	(0.1696)	(0.1632)	(0.0245)	(0.5621)	(0.0250)	(0.1752)	(0.0221)	(0.5175)	(0.0807)
Elite U.S. University	0.5507***	0.9066***	0.1363***	-0.5317	-0.0238	-0.0941	-0.0119	-0.4342	-0.0687
	(0.1994)	(0.2103)	(0.0312)	(0.6438)	(0.0293)	(0.2319)	(0.0293)	(0.5475)	(0.0858)
Elite U.S. Liberal Arts College	-0.0711	0.3268	0.0491	0.3253	0.0146	-0.0866	-0.0110	1.0020	0.1586
	(0.3683)	(0.3402)	(0.0510)	(0.8010)	(0.0356)	(0.3922)	(0.0496)	(0.7518)	(0.1185)
Elite Foreign University	0.2902	0.0518	0.0078	-0.9107	-0.0408	0.0958	0.0121	-0.2759	-0.0437
	(0.2991)	(0.2959)	(0.0445)	(0.9288)	(0.0424)	(0.3439)	(0.0435)	(0.9777)	(0.1548)
Other Foreign University	0.4797**	0.1161	0.0174	0.2293	0.0103	0.1627	0.0206	0.4536	0.0718
	(0.2022)	(0.1969)	(0.0296)	(0.6888)	(0.0306)	(0.2168)	(0.0274)	(0.6285)	(0.0992)
Missing Undergraduate Institution	0.5097**	-0.4337**	-0.0652**	0.0505	0.0023	0.1768	0.0224	-0.8208	-0.1299
	(0.2157)	(0.1996)	(0.0300)	(0.6688)	(0.0299)	(0.2321)	(0.0294)	(0.6805)	(0.1058)
Graduate Degree	-0.0757	1.0229***	0.1537***	-0.3419	-0.0153	-0.1684	-0.0213	-0.0653	-0.0103
	(0.1279)	(0.1262)	(0.0177)	(0.3883)	(0.0171)	(0.1367)	(0.0172)	(0.4719)	(0.0746)
Work Experience	0.3458**	0.7870***	0.1183***	0.5914*	0.0265*	0.2740*	0.0347*	0.9166**	0.1450**
	(0.1345)	(0.1256)	(0.0179)	(0.3412)	(0.0154)	(0.1424)	(0.0179)	(0.4411)	(0.0667)
#Professor Recommenders	0.0618	-0.0211	-0.0032	-0.1896	-0.0085	-0.0622	-0.0079	0.3588	0.0568
	(0.0769)	(0.0839)	(0.0126)	(0.2574)	(0.0113)	(0.0859)	(0.0109)	(0.2789)	(0.0433)
#Prolific Recommenders	0.1667***	0.1325***	0.0199***	0.1903	0.0085	0.0667	0.0084	-0.2845	-0.0450
	(0.0548)	(0.0513)	(0.0077)	(0.1757)	(0.0077)	(0.0582)	(0.0073)	(0.1875)	(0.0288)
Missing GRE	-1.6087*	-1.8142***	-0.2727***	1.6533	0.0740	0.0247	0.0031		
	(0.9422)	(0.5268)	(0.0772)	(1.8092)	(0.0811)	(0.3772)	(0.0477)		
Missing TOEFL/IELTS	0.3199**	0.0947	0.0142	0.4511	0.0202	-0.0036	-0.0005	-0.0091	-0.0014
	(0.1319)	(0.1295)	(0.0194)	(0.4238)	(0.0189)	(0.1435)	(0.0181)	(0.5591)	(0.0885)
Committee Score				3.7089***	0.1660***	-0.0006	-0.0001	-0.1637	-0.0259
				(0.3984)	(0.0065)	(0.0486)	(0.0062)	(0.2099)	(0.0336)
#Math Courses	-0.0790			-0.2494	-0.0112	0.0007	0.0001	-0.0182	-0.0029
	(0.0519)			(0.2386)	(0.0108)	(0.0879)	(0.0111)	(0.1671)	(0.0265)
Advanced Math	0.2979			1.1003	0.0492	-0.2220	-0.0281	-1.1143**	-0.1763**
	(0.1889)			(0.7916)	(0.0347)	(0.2926)	(0.0370)	(0.5578)	(0.0877)
Letters Length/100	0.0858***			0.0770	0.0034	-0.0135	-0.0017	-0.0744	-0.0118
	(0.0245)			(0.1080)	(0.0048)	(0.0385)	(0.0049)	(0.0711)	(0.0113)
Letters Terms (%) PC1	0.1027*			0.2451	0.0110	0.1278	0.0162	0.3997**	0.0632**
	(0.0551)			(0.2489)	(0.0112)	(0.0911)	(0.0115)	(0.1903)	(0.0287)
Letters Terms (%) PC2	-0.0208			-0.3767	-0.0169	-0.1364	-0.0172	0.2897	0.0458
	(0.0823)			(0.3450)	(0.0153)	(0.1257)	(0.0159)	(0.2460)	(0.0388)
CV Length/100	0.0513			0.0391	0.0017	0.0347	0.0044	-0.1109	-0.0175
	(0.0329)			(0.1060)	(0.0047)	(0.0592)	(0.0075)	(0.1105)	(0.0175)
CV Coding	0.0381			-0.2228*	-0.0100*	0.1024	0.0130	-0.2901**	-0.0459**
	(0.0424)			(0.1301)	(0.0060)	(0.0651)	(0.0082)	(0.1176)	(0.0181)
CV Publication/Working Paper	0.5482***			0.7611	0.0341	0.1440	0.0182	-0.3451	-0.0546
	(0.1912)			(0.8128)	(0.0364)	(0.3359)	(0.0425)	(0.5102)	(0.0812)
Essay Length/100	0.0927			0.2047	0.0092	-0.0718	-0.0091	0.0882	0.0140
	(0.0597)			(0.3483)	(0.0157)	(0.0818)	(0.0103)	(0.1386)	(0.0220)
Essay Specific Professor	0.1411			0.4125	0.0185	0.2894	0.0366	0.6812	0.1078
	(0.1744)			(0.6138)	(0.0275)	(0.2770)	(0.0350)	(0.4976)	(0.0773)
Essay Economics Terms (%)	0.0287			0.0154	0.0007	-0.0854	-0.0108	0.2244**	0.0355**
	(0.0312)			(0.0814)	(0.0036)	(0.0614)	(0.0077)	(0.1075)	(0.0169)
Essay Research Terms (%)	0.0396			-0.0751	-0.0034	0.0146	0.0018	-0.1904	-0.0301
	(0.0426)			(0.1492)	(0.0067)	(0.0672)	(0.0085)	(0.1440)	(0.0226)
Essay Positive Terms (%)	0.0171			-0.0689	-0.0031	0.0977*	0.0124*	-0.0170	-0.0027
	(0.0345)			(0.0968)	(0.0043)	(0.0539)	(0.0068)	(0.1151)	(0.0182)
Essay Negative Terms (%)	0.0441			-0.0863	-0.0039	-0.0441	-0.0056	-0.0176	-0.0028
	(0.0443)			(0.1368)	(0.0061)	(0.0745)	(0.0094)	(0.1043)	(0.0165)
Missing Committee Score						-0.2179	-0.0276		
						(0.1515)	(0.0191)		
Missing Application Pdf	-0.5376***			0.2614	0.0117	0.2498	0.0316	-0.7440	-0.1177
	(0.1841)			(0.7005)	(0.0311)	(0.2756)	(0.0348)	(0.5803)	(0.0919)
Program Rank						-0.0086***	-0.0011***		
						(0.0021)	(0.0003)		
Missing Program Rank						-3.1438***	-0.3976***		
						(0.4076)	(0.0514)		
$\sqrt{MSE}$	1.6467***								
	(0.0370)								
Obs | Param | LL | AIC	2329	621	-14259.1	29760.1					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

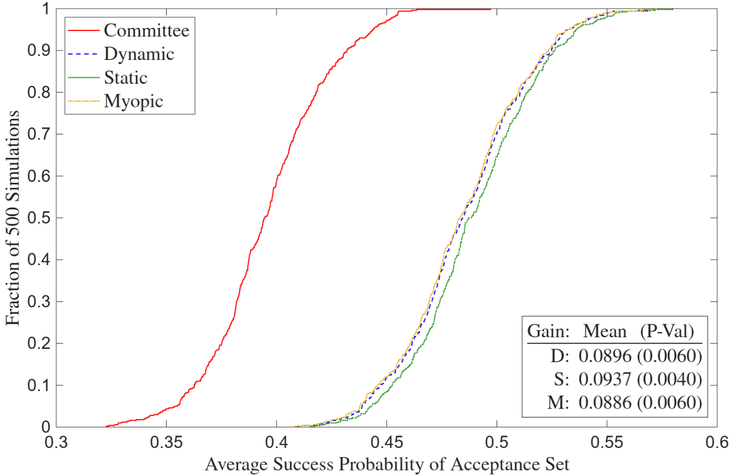
(b) Restricted

	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Matriculate	Sigma
Intercept	-6.5631***	-4.9411***	0.0071	-0.5602***	36.35%	-0.8176**	1.5446	
	(2.5333)	(0.5082)	(0.0026, 0.0194)	(0.1594)	(29.47%, 43.84%)	(0.3643)	(5.3407)	
2016	0.2912	-0.0124	0.0071	0.0811	38.25%	0.0776	-1.2678*	
	(0.2079)	(0.1800)	(0.0028, 0.0180)	(0.1961)	(31.05%, 46.00%)	(0.1966)	(0.7295)	
2017	0.6424***	0.3850**	0.0105	0.1816	40.65%	0.1644	-0.3048	
	(0.2488)	(0.1788)	(0.0041, 0.0267)	(0.1820)	(35.15%, 46.39%)	(0.1821)	(0.8727)	
2018	0.8514***	0.2894	0.0095	0.0102	36.59%	-0.0518	1.3060*	
	(0.1924)	(0.1890)	(0.0038, 0.0240)	(0.2053)	(30.01%, 43.71%)	(0.2096)	(0.7318)	
2019	0.5883***	0.2802	0.0095	0.0456	37.41%	-0.0066	0.7687	
	(0.2154)	(0.2153)	(0.0035, 0.0255)	(0.1960)	(31.04%, 44.26%)	(0.1975)	(0.8962)	
Winsorized Age	-0.0569**					-0.0001	-0.0312	
	(0.0252)					(0.0011)	(0.0547)	
Female	-0.0776					0.0116	0.0927	
	(0.1152)					(0.0083)	(0.2112)	
U.S.	-0.3707*					-0.0033	0.4696	
	(0.2114)					(0.0084)	(0.4504)	
International Asian	-0.5051***					-0.0103	0.5968*	
	(0.1631)					(0.0083)	(0.3430)	
GRE Verbal Percentile	0.0082**					0.0000	0.0098	
	(0.0037)					(0.0002)	(0.0075)	
GRE Quantitative Percentile	0.0704***					0.0010	-0.0610*	
	(0.0105)					(0.0009)	(0.0338)	
GRE Analytic Percentile	0.0016					0.0001	0.0021	
	(0.0029)					(0.0002)	(0.0066)	
TOEFL/IELTS	0.2848					-0.0023	-0.0629	
	(0.1746)					(0.0101)	(0.2753)	
Elite U.S. University	0.6391***					-0.0098	-0.0637	
	(0.1892)					(0.0142)	(0.3925)	
Elite U.S. Liberal Arts College	-0.3528					0.0127	-0.2757	
	(0.3489)					(0.0174)	(0.6783)	
Elite Foreign University	0.1906					-0.0187	0.4536	
	(0.2875)					(0.0195)	(0.5403)	
Other Foreign University	0.5511***					0.0038	0.2515	
	(0.2049)					(0.0148)	(0.3764)	
Missing Undergraduate Institution	0.5458***					0.0033	-0.3893	
	(0.2109)					(0.0140)	(0.4067)	
Graduate Degree	-0.1826					-0.0069	0.4784	
	(0.1589)					(0.0082)	(0.3581)	
Work Experience	0.0765					0.0080	0.5662	
	(0.1486)					(0.0075)	(0.3731)	
#Professor Recommenders	0.0824					-0.0021	0.2152	
	(0.0822)					(0.0054)	(0.1592)	
#Prolific Recommenders	0.0919					-0.0002	-0.1769*	
	(0.0590)					(0.0032)	(0.1041)	
Missing GRE	-1.3931					0.0140		
	(0.8753)					(0.0192)		
Missing TOEFL/IELTS	0.2632**					0.0023	0.0851	
	(0.1311)					(0.0066)	(0.2494)	
Committee Score						0.0649**	0.5117***	
						(0.0299)	(0.1497)	
#Math Courses	-0.1660					-0.0048	-0.1664	
	(0.1024)					(0.0057)	(0.1619)	
Advanced Math	0.7224*					0.0257	0.0920	
	(0.4161)					(0.0208)	(0.5297)	
Letters Length/100	0.1095**					0.0013	-0.0122	
	(0.0471)					(0.0022)	(0.0646)	
Letters Terms (%) PC1	0.2635***					0.0089	0.4342***	
	(0.0988)					(0.0071)	(0.1642)	
Letters Terms (%) PC2	-0.3026*					-0.0108	-0.2680	
	(0.1764)					(0.0091)	(0.2167)	
CV Length/100	0.0263					0.0004	-0.0601	
	(0.0752)					(0.0023)	(0.0819)	
CV Coding	0.0417					-0.0049	-0.0663	
	(0.1086)					(0.0044)	(0.1017)	
CV Publication/Working Paper	1.3411***					0.0190	-0.2251	
	(0.3435)					(0.0192)	(0.5320)	
Essay Length/100	0.2771*					0.0065	0.1436	
	(0.1475)					(0.0103)	(0.1533)	
Essay Specific Professor	0.4040					0.0133	0.6561	
	(0.3655)					(0.0168)	(0.4106)	
Essay Economics Terms (%)	0.0881					0.0010	0.0869	
	(0.0633)					(0.0022)	(0.0809)	
Essay Research Terms (%)	0.1320					0.0005	-0.1728	
	(0.0843)					(0.0030)	(0.1284)	
Essay Positive Terms (%)	0.0538					-0.0006	0.0066	
	(0.0761)					(0.0025)	(0.0966)	
Essay Negative Terms (%)	0.0781					-0.0021	-0.0494	
	(0.0830)					(0.0035)	(0.1032)	
Missing Committee Score						-0.1920		
						(0.1290)		
Missing Application Pdf	-0.4098					0.0124	-1.5977***	
	(0.2846)					(0.0125)	(0.2218)	
Program Rank						-0.0079***		
						(0.0020)		
Missing Program Rank						-3.0808***		
						(0.4053)		
$\sqrt{MSE}$	1.7632***							
	(0.0469)							
Sigma								0.0020**
								(0.0010)
Obs | Param | LL | AIC	2329	568	-14260.9	29657.8				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

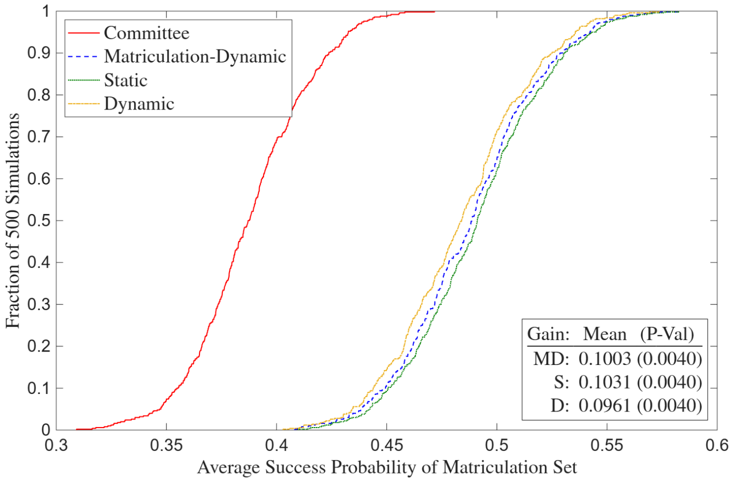
Restricted Matriculation-Dynamic Estimates

A3: Economics Department Deviation Gains



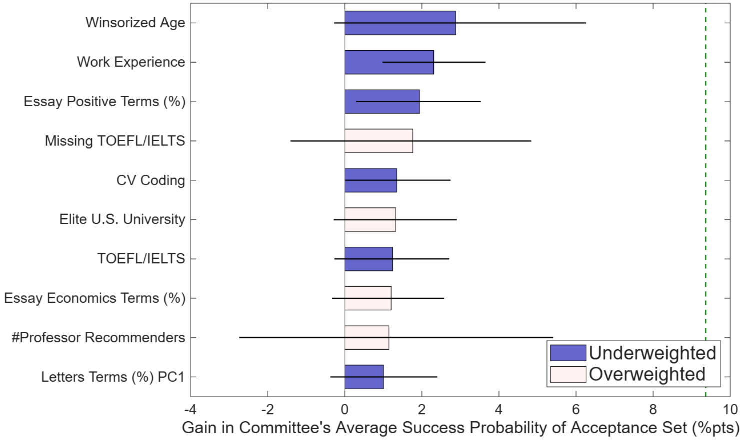
Deviation Gains

A3: Economics Department Deviation Gains (Matriculation)



Deviation Gains (Matriculation)

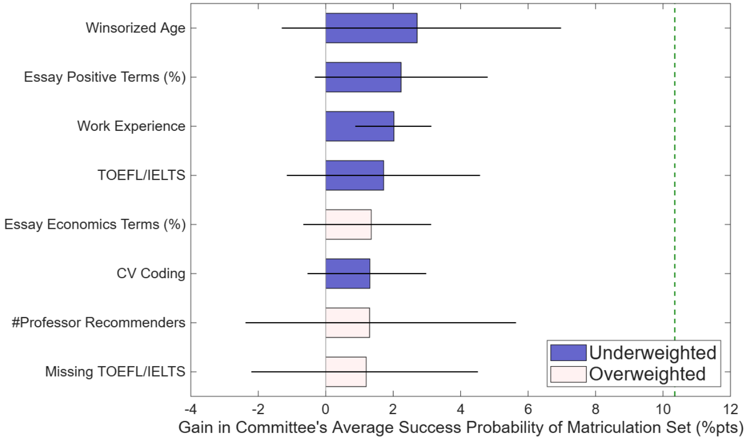
A3: Economics Department Gains Attribution



1. Experience
2. English Proficiency
3. Essay/Letters Subtext
4. Industry Background
5. Undergraduate

Deviation Gains Attribution

A3: Economics Department Gains Attribution (Matriculation)



1. Experience
2. English Proficiency
3. Essay Subtext
4. Industry Background

Deviation Gains Attribution (Matriculation)

A3: Economics Department Static Attribution

Table 12: Economics Department Static Attribution

	Gain (%pt)	Sample Mean	Committee Mean	1-Variable Optimal Mean	Joint Optimal Mean	1-Variable Difference (SD)	Joint Difference (SD)
Winsorized Age	2.8788	25.9910	26.2840	29.9572	29.5331	1.1674	1.0326
Work Experience	2.3065	0.5247	0.7276	0.9844	0.9650	0.5141	0.4752
Essay Positive Terms (%)	1.9410	11.7499	11.6521	12.5376	12.1183	0.7858	0.4137
Missing TOEFL/IELTS	1.7635	0.6076	0.6654	0.0739	0.4086	-1.2110	-0.5258
CV Coding	1.3478	2.8001	2.7820	3.3399	2.9415	0.6244	0.1785
Elite U.S. University	1.3179	0.1267	0.2335	0.0078	0.0389	-0.6784	-0.5848
TOEFL/IELTS	1.2400	7.4376	7.5675	7.7759	7.6827	0.5957	0.3294
Essay Economics Terms (%)	1.2043	9.7228	9.7423	9.0284	9.5388	-0.6411	-0.1828
#Professor Recommenders	1.1455	2.6230	2.6148	1.3852	2.1362	-1.6680	-0.6493
Letters Terms (%) PC1	1.0074	-0.0106	0.1374	0.4969	0.1387	0.5202	0.0018

Deviation Gains Attribution

A3: Economics Department Matriculation-Static Attribution

Table 13: Economics Department Matriculation-Static Attribution

	Gain (%pt)	Sample Mean	Committee Mean	1-Variable Optimal Mean	Joint Optimal Mean	1-Variable Difference (SD)	Joint Difference (SD)
Winsorized Age	2.7099	25.9910	26.3284	30.9159	29.7737	1.4580	1.0950
Essay Positive Terms (%)	2.2339	11.7499	11.6107	12.9136	12.2763	1.1534	0.5905
Work Experience	2.0220	0.5247	0.7938	0.9949	0.9799	0.4077	0.3725
TOEFL/IELTS	1.7179	7.4376	7.4939	7.8501	7.6128	1.0238	0.3397
Essay Economics Terms (%)	1.3503	9.7228	9.9994	9.0856	9.5738	-0.8271	-0.3821
CV Coding	1.3100	2.8001	2.6661	3.3395	2.8782	0.7618	0.2374
#Professor Recommenders	1.3005	2.6230	2.6171	1.4173	2.2255	-1.6321	-0.5312
Missing TOEFL/IELTS	1.1999	0.6076	0.6858	0.0442	0.4368	-1.3075	-0.5099

Deviation Gains Attribution (Matriculation)

A3: Graduate School Means by Program

Table 14a: Graduate School Means by Program

	Economics	Government	History	Linguistics	Psychology
Accept	0.1121	0.0430	0.0986	0.0678	0.0535
Covid Year	0.2478	0.2792	0.3029	0.3460	0.3004
Winsorized Age	26.2782	27.9633	26.8478	28.0692	25.8230
Female	0.4201	0.4182	0.4232	0.6133	0.7490
U.S. African/Hispanic/Native	0.0289	0.0766	0.1159	0.0665	0.1728
U.S. Asian	0.0313	0.0421	0.0304	0.0353	0.0576
International Asian	0.5322	0.2510	0.1058	0.2714	0.1523
International Other	0.2517	0.2967	0.2203	0.3256	0.1584
GRE Verbal Percentile	71.8677	81.3486	82.6054	73.5451	75.6192
GRE Quantitative Percentile	86.2163	63.3428	49.3295	57.8455	58.0000
GRE Analytic Percentile	52.6351	72.1905	75.0460	63.6824	69.2077
Undergraduate GPA	3.6534	3.5675	3.6531	3.6567	3.5695
TOEFL/IELTS	7.5232	7.5422	7.4252	7.5444	7.3158
Elite U.S. University	0.1516	0.1847	0.2565	0.1954	0.2840
Elite U.S. Liberal Arts College	0.0423	0.0672	0.0696	0.0231	0.0535
Elite Foreign University	0.0717	0.0578	0.0478	0.0488	0.0206
Other Foreign University	0.5736	0.3953	0.2333	0.5102	0.1852
Graduate Degree	0.7478	0.7553	0.6652	0.7544	0.4259
Work Experience	0.5938	0.7463	0.6638	0.7544	0.7593
#Professor Recommenders	2.5173	2.3469	2.6261	2.4355	2.1831
#Prolific Recommenders	1.0760	0.6912	0.5043	0.7069	0.3128
#Math Courses	6.6396				
Advanced Math	0.4827				
Missing GRE	0.0255	0.4612	0.6217	0.6839	0.4650
Missing Undergraduate GPA	0.6607	0.4827	0.3072	0.5726	0.2490
Missing TOEFL/IELTS	0.5958	0.7714	0.8449	0.7707	0.8827
Observations	2078	2231	690	737	486

- **Economics Highest:** Accept, International Asian, GRE Quantitative Percentile, Elite Foreign University, Other Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, U.S. African/Hispanic/Native, GRE Verbal Percentile, GRE Analytic Percentile, Elite U.S. University, Work Experience, Missing GRE, Missing TOEFL/IELTS

Insufficiency of Comparing AMEs

A3: Graduate School Means by Program, Accept $= 1$

Table 14b: Graduate School Means by Program, Accept $= 1$

	Economics	Government	History	Linguistics	Psychology
Covid Year	0.1888	0.2188	0.2059	0.2200	0.2308
Winsorized Age	25.0815	27.7292	27.2794	27.1000	25.7692
Female	0.4850	0.5000	0.5588	0.5800	0.6154
U.S. African/Hispanic/Native	0.0386	0.1458	0.1765	0.0400	0.0769
U.S. Asian	0.0429	0.0312	0.0147	0.0600	0.1154
International Asian	0.4163	0.1667	0.0882	0.3600	0.0769
International Other	0.2575	0.2292	0.3235	0.2000	0.1538
GRE Verbal Percentile	84.3980	88.8725	86.6328	77.9161	85.7729
GRE Quantitative Percentile	91.9289	68.3708	54.8113	64.5504	62.3462
GRE Analytic Percentile	69.0355	82.7222	78.0879	68.8958	74.6169
Undergraduate GPA	3.7336	3.6404	3.7204	3.7004	3.6341
TOEFL/IELTS	7.6419	7.6050	7.4730	7.6237	7.3178
Elite U.S. University	0.2833	0.3125	0.3382	0.2200	0.5385
Elite U.S. Liberal Arts College	0.1030	0.1458	0.1176	0.0200	0.1154
Elite Foreign University	0.1030	0.0521	0.0588	0.1000	0.0385
Other Foreign University	0.3863	0.2396	0.2647	0.4000	0.1154
Graduate Degree	0.5837	0.6771	0.7206	0.7400	0.4615
Work Experience	0.6309	0.8125	0.7647	0.7000	0.9231
#Professor Recommenders	2.5494	2.3750	2.8235	2.7200	2.2308
#Prolific Recommenders	1.2833	0.9167	0.6765	1.0800	0.4615
#Math Courses	7.2575				
Advanced Math	0.6781				
Missing GRE	0.0086	0.2917	0.6324	0.5800	0.1923
Missing Undergraduate GPA	0.4893	0.3125	0.3088	0.4800	0.1923
Missing TOEFL/IELTS	0.6567	0.8854	0.8529	0.7600	0.8846
Observations	233	96	68	50	26

- **Economics Highest:** International Asian, GRE Quantitative Percentile, Undergraduate GPA, TOEFL/IELTS, Elite Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, Winsorized Age, Female, U.S. African/Hispanic/Native, Work Experience, Missing GRE, Missing TOEFL/IELTS

Insufficiency of Comparing AMEs

A3: Graduate School Correlation Matrix

Table 15: Graduate School Correlation Matrix

	Covid	Age	Fem	USAHN	USAsn	IntAsn	IntOth	GREV	GREQ	GREA	GPA	T/I	EUSUni	EUSLA	EFor	OthFor	Grad	Work	Prof	Pro	MGRE	MGPA	MT/I
Covid Year	1.00																						
Winsorized Age	0.01	1.00																					
Female	0.02	-0.13	1.00																				
U.S. African/Hispanic/Native	0.00	-0.04	0.04	1.00																			
U.S. Asian	-0.00	-0.06	0.05	-0.05	1.00																		
International Asian	-0.05	-0.07	0.02	-0.19	-0.14	1.00																	
International Other	0.01	0.18	-0.06	-0.17	-0.12	-0.42	1.00																
GRE Verbal Percentile	0.03	-0.06	-0.04	0.00	0.04	-0.12	-0.15	1.00															
GRE Quantitative Percentile	-0.02	-0.16	-0.10	-0.18	0.00	0.44	-0.10	0.16	1.00														
GRE Analytic Percentile	0.05	-0.06	0.03	0.08	0.08	-0.37	-0.08	0.60	-0.20	1.00													
Undergraduate GPA	-0.02	-0.29	0.07	-0.13	-0.02	0.02	0.00	0.06	0.12	0.05	1.00												
TOEFL/IELTS	-0.01	-0.07	0.01	-0.02	0.01	0.04	0.00	0.24	0.12	0.19	0.00	1.00											
Elite U.S. University	0.01	-0.12	0.02	0.12	0.14	-0.12	-0.22	0.16	-0.00	0.19	-0.01	-0.02	1.00										
Elite U.S. Liberal Arts College	-0.02	-0.02	0.04	0.04	0.04	-0.08	-0.09	0.12	-0.01	0.13	-0.02	-0.00	-0.11	1.00									
Elite Foreign University	-0.00	-0.04	-0.00	-0.06	-0.04	0.09	0.09	0.02	0.08	-0.02	0.01	0.04	-0.12	-0.06	1.00								
Other Foreign University	-0.02	0.20	-0.04	-0.23	-0.14	0.34	0.39	-0.26	0.20	-0.39	0.02	0.02	-0.43	-0.21	-0.22	1.00							
Graduate Degree	0.00	0.41	-0.08	-0.13	-0.09	0.21	0.15	-0.14	0.05	-0.20	-0.22	0.03	-0.21	-0.10	0.05	0.37	1.00						
Work Experience	0.01	0.33	0.05	0.05	0.01	-0.18	0.11	0.01	-0.15	0.11	-0.08	-0.03	-0.02	0.03	-0.03	-0.01	0.10	1.00					
#Professor Recommenders	-0.02	-0.19	-0.04	-0.04	0.01	0.11	-0.09	0.01	0.09	-0.04	0.14	0.03	-0.02	0.02	-0.00	-0.02	-0.03	-0.18	1.00				
#Prolific Recommenders	0.01	-0.06	-0.02	-0.07	-0.01	0.24	-0.10	-0.01	0.25	-0.11	0.04	0.10	-0.01	-0.01	0.06	0.08	0.17	-0.09	0.19	1.00			
Missing GRE	0.09	0.15	0.05	0.11	-0.01	-0.22	0.17	0.09	-0.33	0.17	-0.09	-0.10	-0.03	-0.04	-0.03	-0.01	0.02	0.10	-0.08	-0.20	1.00		
Missing Undergraduate GPA	-0.01	0.18	-0.04	-0.24	-0.15	0.36	0.42	-0.24	0.21	-0.38	0.02	0.04	-0.46	-0.23	0.23	0.83	0.37	-0.02	-0.03	0.10	-0.01	1.00	
Missing TOEFL/IELTS	0.05	-0.07	0.03	0.16	0.11	-0.32	-0.22	0.16	-0.24	0.32	-0.02	-0.02	0.29	0.14	-0.04	-0.58	-0.20	0.05	-0.07	-0.05	0.12	-0.58	1.00
Observations	6222																						

Insufficiency of Comparing AMEs

A3: Graduate School Logistic Regression Results

Table 16a: Graduate School Logistic Regression Results

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3966**	-0.4332*	-0.5752*	-0.5501	-0.1766
	(0.2010)	(0.2612)	(0.3479)	(0.4129)	(0.5545)
Winsorized Age	-0.0995***	0.0454*	0.0380	0.0222	-0.0520
	(0.0358)	(0.0271)	(0.0327)	(0.0532)	(0.0899)
Female	0.4079**	0.6723***	0.6539**	0.1311	-0.5060
	(0.1681)	(0.2299)	(0.2971)	(0.3452)	(0.5606)
U.S. African/Hispanic/Native	0.4221	1.0081***	1.1601***	0.0120	-0.8263
	(0.5258)	(0.3545)	(0.4426)	(0.8411)	(1.1611)
U.S. Asian	-0.5394	-0.5669	-0.3260	0.8638	1.1711
	(0.4199)	(0.6504)	(1.2512)	(0.7201)	(0.7276)
International Asian	-0.8445***	-0.1853	0.9405	0.4890	-0.7827
	(0.2945)	(0.4146)	(0.7233)	(0.6729)	(0.9323)
International Other	0.5034	0.3781	1.3523***	-0.1061	0.6239
	(0.3172)	(0.4131)	(0.5086)	(0.7483)	(1.1070)
GRE Verbal Percentile	0.0155***	0.0281*	0.0753**	-0.0092	0.0607***
	(0.0060)	(0.0160)	(0.0344)	(0.0154)	(0.0218)
GRE Quantitative Percentile	0.1547***	0.0161*	0.0210*	0.0211	-0.0138
	(0.0186)	(0.0086)	(0.0116)	(0.0150)	(0.0159)
GRE Analytic Percentile	0.0179***	0.0199**	0.0030	0.0175	-0.0202
	(0.0042)	(0.0094)	(0.0134)	(0.0127)	(0.0218)
Undergraduate GPA	2.2212***	0.6968	1.3648**	1.2645	1.7772*
	(0.7219)	(0.5476)	(0.5513)	(0.9703)	(0.9487)
TOEFL/IELTS	0.3878	0.6456	1.0963**	0.9655*	-0.4656
	(0.2883)	(0.4708)	(0.5526)	(0.5456)	(0.6112)
Elite U.S. University	0.8850***	0.3943	0.6541*	-0.6057	1.4303**
	(0.2960)	(0.3135)	(0.3802)	(0.4985)	(0.6192)
Elite U.S. Liberal Arts	1.1173***	0.6276*	1.0698**	-0.6088	2.3151**
	(0.3813)	(0.3754)	(0.5399)	(1.4186)	(1.0642)
Elite Foreign University	1.0061*	0.4759	1.7715**	1.6195*	0.3889
	(0.6060)	(0.8224)	(0.7938)	(0.8587)	(1.9889)
Other Foreign University	0.5442	0.5777	1.3733**	0.4188	-1.0737
	(0.5765)	(0.7686)	(0.6992)	(0.7157)	(1.3016)
Graduate Degree	0.0467	0.0733	0.3128	0.4049	0.4892
	(0.2238)	(0.2583)	(0.3280)	(0.4896)	(0.6399)
Work Experience	0.3932**	0.4027	0.4972	0.1495	1.8201**
	(0.1889)	(0.2993)	(0.3295)	(0.3778)	(0.8564)
#Professor Recommenders	0.0340	-0.0179	0.5511**	0.4781*	-0.0981
	(0.1317)	(0.1339)	(0.2554)	(0.2662)	(0.2660)
#Prolific Recommenders	0.2635***	0.3025**	0.1880	0.3284**	0.5901*
	(0.0854)	(0.1262)	(0.1576)	(0.1544)	(0.3145)
#Math Courses	0.1689***				
	(0.0596)				
Advanced Math	0.2319				
	(0.1790)				
Missing GRE	0.1899	-0.3259	0.6282	-0.0831	-1.1514**
	(0.7646)	(0.3100)	(0.4275)	(0.3761)	(0.5702)
Missing Undergraduate GPA	-0.4745	-0.4917	-1.5492**	-1.1097*	0.4690
	(0.4891)	(0.5440)	(0.6893)	(0.6481)	(1.1831)
MissingTOEFL/IELTS	-0.3786	0.6188	0.6745	-0.0440	-1.7224*
	(0.2432)	(0.5657)	(0.5190)	(0.5099)	(0.9271)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R^2	0.2677	0.1525	0.1982	0.1871	0.2475

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, #Prolific Recommenders, #Math Courses
- **Economics Negative ($p < .1$):** Covid Year, Winsorized Age, International Asian

Insufficiency of Comparing AMEs

A3: Graduate School Logistic Regression AMEs

Table 16b: Graduate School Logistic Regression AMEs

Depdendent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0303**	-0.0164	-0.0431*	-0.0298	-0.0075	-0.0218***
	(0.0153)	(0.0100)	(0.0261)	(0.0224)	(0.0238)	(0.0084)
Winsorized Age	-0.0076***	0.0017*	0.0028	0.0012	-0.0022	0.0016*
	(0.0027)	(0.0010)	(0.0025)	(0.0029)	(0.0038)	(0.0009)
Female	0.0312**	0.0254***	0.0490**	0.0071	-0.0216	0.0207***
	(0.0129)	(0.0088)	(0.0220)	(0.0187)	(0.0239)	(0.0075)
U.S. African/Hispanic/Native	0.0323	0.0381***	0.0869***	0.0007	-0.0352	0.0384***
	(0.0402)	(0.0137)	(0.0332)	(0.0455)	(0.0487)	(0.0119)
U.S. Asian	-0.0413	-0.0215	-0.0244	0.0468	0.0499	0.0062
	(0.0322)	(0.0247)	(0.0936)	(0.0391)	(0.0313)	(0.0185)
International Asian	-0.0646***	-0.0070	0.0704	0.0265	-0.0334	0.0069
	(0.0224)	(0.0157)	(0.0543)	(0.0363)	(0.0393)	(0.0132)
International Other	0.0385	0.0143	0.1013***	-0.0057	0.0266	0.0238*
	(0.0243)	(0.0157)	(0.0385)	(0.0406)	(0.0471)	(0.0132)
GRE Verbal Percentile	0.0012***	0.0011*	0.0056**	-0.0005	0.0026***	0.0015***
	(0.0005)	(0.0006)	(0.0026)	(0.0008)	(0.0010)	(0.0005)
GRE Quantitative Percentile	0.0118***	0.0006*	0.0016*	0.0011	-0.0006	0.0006**
	(0.0014)	(0.0003)	(0.0009)	(0.0008)	(0.0007)	(0.0003)
GRE Analytic Percentile	0.0014***	0.0008**	0.0002	0.0009	-0.0009	0.0007**
	(0.0003)	(0.0004)	(0.0010)	(0.0007)	(0.0009)	(0.0003)
Undergraduate GPA	0.1699***	0.0264	0.1022**	0.0684	0.0757*	0.0488***
	(0.0551)	(0.0208)	(0.0423)	(0.0522)	(0.0418)	(0.0176)
TOEFL/IELTS	0.0297	0.0244	0.0821**	0.0523*	-0.0198	0.0326***
	(0.0219)	(0.0179)	(0.0417)	(0.0301)	(0.0259)	(0.0123)
Elite U.S. University	0.0677***	0.0149	0.0490*	-0.0328	0.0610**	0.0233**
	(0.0225)	(0.0119)	(0.0281)	(0.0264)	(0.0271)	(0.0098)
Elite U.S. Liberal Arts College	0.0855***	0.0238*	0.0801**	-0.0330	0.0987**	0.0420***
	(0.0288)	(0.0143)	(0.0402)	(0.0761)	(0.0480)	(0.0136)
Elite Foreign University	0.0770*	0.0180	0.1327**	0.0877*	0.0166	0.0479**
	(0.0463)	(0.0311)	(0.0591)	(0.0481)	(0.0850)	(0.0241)
Other Foreign University	0.0416	0.0219	0.1028*	0.0227	-0.0458	0.0273
	(0.0441)	(0.0291)	(0.0527)	(0.0391)	(0.0556)	(0.0221)
Graduate Degree	0.0036	0.0028	0.0234	0.0219	0.0208	0.0106
	(0.0171)	(0.0098)	(0.0247)	(0.0262)	(0.0273)	(0.0089)
Work Experience	0.0301**	0.0152	0.0372	0.0081	0.0776**	0.0221**
	(0.0144)	(0.0113)	(0.0248)	(0.0205)	(0.0368)	(0.0089)
#Professor Recommenders	0.0026	-0.0007	0.0413**	0.0259*	-0.0042	0.0063
	(0.0101)	(0.0051)	(0.0190)	(0.0142)	(0.0113)	(0.0047)
#Prolific Recommenders	0.0202***	0.0114**	0.0141	0.0178**	0.0251*	0.0133***
	(0.0065)	(0.0048)	(0.0118)	(0.0082)	(0.0130)	(0.0038)
#Math Courses	0.0129***					
	(0.0045)					
Advanced Math	0.0177					
	(0.0137)					
Missing GRE	0.0145	-0.0123	0.0470	-0.0045	-0.0491*	-0.0062
	(0.0585)	(0.0118)	(0.0319)	(0.0204)	(0.0252)	(0.0088)
Missing Undergraduate GPA	-0.0363	-0.0186	-0.1160**	-0.0601*	0.0200	-0.0291
	(0.0375)	(0.0206)	(0.0524)	(0.0355)	(0.0505)	(0.0179)
Missing TOEFL/IELTS	-0.0290	0.0234	0.0505	-0.0024	-0.0734*	0.0108
	(0.0188)	(0.0215)	(0.0388)	(0.0276)	(0.0417)	(0.0135)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic | DF | P-Value	115.8837	23	0.0000			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, #Prolific Recommenders, #Math Courses
- **Economics Negative ($p < .1$):** Covid Year, Winsorized Age, International Asian

Insufficiency of Comparing AMEs

A3: Graduate School Logistic Regression Results (Synthetic)

Table 17a: Graduate School Logistic Regression Results (Synthetic)

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.1997	-0.0323	0.2028	-0.4791	-0.5946
	(0.1633)	(0.2215)	(0.3078)	(0.3477)	(0.4520)
Winsorized Age	-0.0509**	-0.0198	0.0576*	-0.0164	0.1203***
	(0.0201)	(0.0174)	(0.0307)	(0.0357)	(0.0400)
Female	0.1941	0.0923	0.1389	0.2595	0.2119
	(0.1402)	(0.2004)	(0.2526)	(0.3255)	(0.4310)
U.S. African/Hispanic/Native	0.3053	-0.1457	-0.2379	-1.6401	0.5413
	(0.3724)	(0.4303)	(0.4670)	(1.0720)	(0.5306)
U.S. Asian	0.5663	0.6016	-0.7934	-0.4305	0.7122
	(0.3642)	(0.4452)	(1.1675)	(0.7936)	(0.8863)
International Asian	-0.1428	0.1630	0.5780	-0.4965	0.1868
	(0.2169)	(0.2803)	(0.3740)	(0.4275)	(0.6866)
International Other	-0.1231	0.1395	0.1097	-0.1602	-1.0807
	(0.2311)	(0.3046)	(0.3237)	(0.3832)	(0.7583)
GRE Verbal Percentile	0.0167***	0.0247**	0.0164	-0.0029	0.0132
	(0.0048)	(0.0117)	(0.0127)	(0.0100)	(0.0184)
GRE Quantitative Percentile	0.0233***	-0.0027	0.0026	0.0304**	0.0097
	(0.0090)	(0.0051)	(0.0088)	(0.0137)	(0.0099)
GRE Analytic Percentile	0.0042	0.0004	-0.0073	-0.0042	0.0069
	(0.0031)	(0.0059)	(0.0092)	(0.0101)	(0.0103)
Undergraduate GPA	0.2292	0.6327	1.3552**	-1.3171**	-0.1002
	(0.4745)	(0.4985)	(0.5796)	(0.5584)	(0.6152)
TOEFL/IELTS	-0.4673**	0.5141	-0.7648	0.2446	-0.7354
	(0.2157)	(0.4185)	(0.6814)	(0.7498)	(0.8025)
Elite U.S. University	-0.4850*	-0.2243	0.0572	0.4764	0.3941
	(0.2560)	(0.3258)	(0.3421)	(0.4788)	(0.4782)
Elite U.S. Liberal Arts College	0.2429	0.1759	0.2358	0.4860	0.8652
	(0.3049)	(0.4042)	(0.4964)	(0.9631)	(0.7261)
Elite Foreign University	-0.1622	-1.1929	0.7404	0.4801	3.0530***
	(0.3015)	(0.7484)	(0.5668)	(0.7841)	(0.8673)
Other Foreign University	-0.1087	0.2043	0.2721	0.1731	0.3023
	(0.1979)	(0.2732)	(0.3391)	(0.4292)	(0.5926)
Graduate Degree	-0.0198	0.1770	-0.1601	-0.8698***	-0.2769
	(0.1624)	(0.2439)	(0.2765)	(0.3320)	(0.3814)
Work Experience	0.2525*	-0.0391	-0.0736	-0.2650	-1.1936**
	(0.1441)	(0.2396)	(0.2629)	(0.3417)	(0.4805)
#Professor Recommenders	-0.1059	0.0911	-0.1828	0.2181	-0.4923**
	(0.0909)	(0.1138)	(0.1473)	(0.1817)	(0.1956)
#Prolific Recommenders	0.0039	0.2189**	-0.1743	0.0673	0.5134**
	(0.0656)	(0.1108)	(0.1723)	(0.1642)	(0.2213)
#Math Courses	0.0366				
	(0.0421)				
Advanced Math	0.1961				
	(0.1416)				
Missing GRE	-0.1993	-0.2029	-0.0011	-0.0433	-0.6040
	(0.4328)	(0.2195)	(0.2689)	(0.3503)	(0.4017)
Missing Undergraduate GPA	0.1531	0.0895	-0.2455	0.6298**	-0.3574
	(0.1548)	(0.2291)	(0.3015)	(0.3125)	(0.4342)
Missing TOEFL/IELTS	-0.2950*	0.4898	0.5752	0.1346	-2.0781***
	(0.1640)	(0.3480)	(0.5466)	(0.4345)	(0.7633)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R^2	0.0524	0.0366	0.0753	0.1046	0.2230

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** GRE Verbal Percentile, GRE Quantitative Percentile, Work Experience
- **Economics Negative ($p < .1$):** Winsorized Age, TOEFL/IELTS, Elite U.S. University, Missing TOEFL/IELTS

Insufficiency of Comparing AMEs

A3: Graduate School Logistic Regression AMEs (Synthetic)

Table 17b: Graduate School Logistic Regression AMEs (Synthetic)

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0204	-0.0014	0.0192	-0.0298	-0.0345	-0.0089
	(0.0167)	(0.0099)	(0.0291)	(0.0216)	(0.0262)	(0.0086)
Winsorized Age	-0.0052**	-0.0009	0.0054*	-0.0010	0.0070***	0.0004
	(0.0021)	(0.0008)	(0.0029)	(0.0022)	(0.0023)	(0.0008)
Female	0.0198	0.0041	0.0131	0.0162	0.0123	0.0099
	(0.0143)	(0.0089)	(0.0239)	(0.0204)	(0.0251)	(0.0078)
U.S. African/Hispanic/Native	0.0312	-0.0065	-0.0225	-0.1021	0.0314	-0.0137
	(0.0381)	(0.0192)	(0.0441)	(0.0673)	(0.0306)	(0.0147)
U.S. Asian	0.0579	0.0268	-0.0750	-0.0268	0.0413	0.0241
	(0.0372)	(0.0199)	(0.1101)	(0.0495)	(0.0516)	(0.0179)
International Asian	-0.0146	0.0073	0.0546	-0.0309	0.0108	0.0075
	(0.0222)	(0.0125)	(0.0353)	(0.0263)	(0.0399)	(0.0112)
International Other	-0.0126	0.0062	0.0104	-0.0100	-0.0627	-0.0014
	(0.0236)	(0.0136)	(0.0306)	(0.0237)	(0.0443)	(0.0106)
GRE Verbal Percentile	0.0017***	0.0011**	0.0015	-0.0002	0.0008	0.0008**
	(0.0005)	(0.0005)	(0.0012)	(0.0006)	(0.0010)	(0.0004)
GRE Quantitative Percentile	0.0024***	-0.0001	0.0003	0.0019**	0.0006	0.0003
	(0.0009)	(0.0002)	(0.0008)	(0.0009)	(0.0006)	(0.0002)
GRE Analytic Percentile	0.0004	0.0000	-0.0007	-0.0003	0.0004	0.0000
	(0.0003)	(0.0003)	(0.0009)	(0.0006)	(0.0006)	(0.0002)
Undergraduate GPA	0.0234	0.0282	0.1281**	-0.0820**	-0.0058	0.0130
	(0.0485)	(0.0223)	(0.0553)	(0.0358)	(0.0357)	(0.0158)
TOEFL/IELTS	-0.0478**	0.0229	-0.0723	0.0152	-0.0427	0.0050
	(0.0221)	(0.0187)	(0.0645)	(0.0466)	(0.0469)	(0.0187)
Elite U.S. University	-0.0496*	-0.0100	0.0054	0.0297	0.0229	0.0036
	(0.0261)	(0.0145)	(0.0324)	(0.0297)	(0.0274)	(0.0108)
Elite U.S. Liberal Arts College	0.0248	0.0078	0.0223	0.0303	0.0502	0.0217
	(0.0312)	(0.0180)	(0.0469)	(0.0602)	(0.0418)	(0.0160)
Elite Foreign University	-0.0166	-0.0532	0.0700	0.0299	0.1771***	0.0158
	(0.0308)	(0.0337)	(0.0537)	(0.0490)	(0.0484)	(0.0179)
Other Foreign University	-0.0111	0.0091	0.0257	0.0108	0.0175	0.0116
	(0.0202)	(0.0122)	(0.0321)	(0.0267)	(0.0344)	(0.0103)
Graduate Degree	-0.0020	0.0079	-0.0151	-0.0542***	-0.0161	-0.0093
	(0.0166)	(0.0109)	(0.0261)	(0.0206)	(0.0220)	(0.0085)
Work Experience	0.0258*	-0.0017	-0.0070	-0.0165	-0.0692**	-0.0093
	(0.0148)	(0.0107)	(0.0248)	(0.0213)	(0.0275)	(0.0086)
#Professor Recommenders	-0.0108	0.0041	-0.0173	0.0136	-0.0286***	-0.0027
	(0.0093)	(0.0051)	(0.0140)	(0.0114)	(0.0110)	(0.0042)
#Prolific Recommenders	0.0004	0.0098*	-0.0165	0.0042	0.0298**	0.0083*
	(0.0067)	(0.0050)	(0.0164)	(0.0102)	(0.0128)	(0.0045)
#Math Courses	0.0037					
	(0.0043)					
Advanced Math	0.0201					
	(0.0145)					
Missing GRE	-0.0204	-0.0090	-0.0001	-0.0027	-0.0350	-0.0098
	(0.0443)	(0.0098)	(0.0254)	(0.0218)	(0.0239)	(0.0079)
Missing Undergraduate GPA	0.0157	0.0040	-0.0232	0.0392*	-0.0207	0.0019
	(0.0158)	(0.0102)	(0.0285)	(0.0201)	(0.0254)	(0.0083)
Missing TOEFL/IELTS	-0.0302*	0.0218	0.0544	0.0084	-0.1205***	0.0063
	(0.0168)	(0.0156)	(0.0517)	(0.0272)	(0.0434)	(0.0130)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic | DF | P-Value	37.4658	23	0.0290			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** GRE Verbal Percentile, GRE Quantitative Percentile, Work Experience
- **Economics Negative ($p < .1$):** Winsorized Age, TOEFL/IELTS, Elite U.S. University, Missing TOEFL/IELTS

Insufficiency of Comparing AMEs

A3: Graduate School Lasso Logistic Regression Results

Table 18a: Graduate School Lasso Logistic Regression Results

Depdendent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3330	-0.3123	-0.3498	-0.2378	
Winsorized Age	-0.0788	0.0266	0.0200		
Female	0.3352	0.5096	0.5068		-0.0654
U.S. African/Hispanic/Native	0.2638	0.8041	0.7157		
U.S. Asian	-0.2802	-0.2918		0.0413	0.1155
International Asian	-0.6744	-0.0726		0.0465	
International Other	0.4729	0.1946	0.7240	-0.1087	
GRE Verbal Percentile	0.0139	0.0252	0.0272		0.0343
GRE Quantitative Percentile	0.1349	0.0119	0.0187	0.0154	
GRE Analytic Percentile	0.0166	0.0208		0.0080	
Undergraduate GPA	1.9977	0.4621	0.7414	0.1714	
TOEFL/IELTS	0.4009	0.3295	0.5199	0.5151	
Elite U.S. University	0.6886	0.2709	0.3267		0.4606
Elite U.S. Liberal Arts College	0.8956	0.5199	0.5985		0.4470
Elite Foreign University	0.3724		0.0414	0.4343	
Other Foreign University					
Graduate Degree			0.0978		
Work Experience	0.2835	0.3447	0.3273		0.5559
#Professor Recommenders			0.3739	0.2322	
#Prolific Recommenders	0.2243	0.2362	0.1651	0.2213	0.0632
#Math Courses	0.1493				
Advanced Math	0.2272				
Missing GRE		-0.2304		-0.0527	-0.5416
Missing Undergraduate GPA					
Missing TOEFL/IELTS	-0.2364	0.2852	0.0796		
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R^2	0.2642	0.1456	0.1589	0.1486	0.1362
CV Pseudo-R^2	0.2297	0.0912	0.0592	0.0850	0.0537

- **Economics Positive:** Female, U.S. African/Hispanic/Native, International Other, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, TOEFL/IELTS, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, # Prolific Recommenders, #Math Courses, Advanced Math
- **Economics Negative:** Covid Year, Winsorized Age, U.S. Asian, International Asian, Missing TOEFL/IELTS

Insufficiency of Comparing AMEs

A4: Simulated Wald Test Results (Same)

Table 19a: Simulated Wald Test Results (Same)

	Dependent Variable = Accept				Dependent Variable = Success			
	Economics		Non-Economics		Economics		Non-Economics	
	Coef	AME	Coef	AME	Coef	AME	Coef	AME
Intercept	-16.3848***	-1.5822***	-28.5524***	-1.3677***	-10.4586***	-1.9105***	-10.2610***	-1.8514***
	(1.1315)	(0.0956)	(1.4963)	(0.0500)	(0.7236)	(0.1070)	(0.4815)	(0.0628)
GRE Quantitative Percentile	0.1509***	0.0146***	0.3245***	0.0155***	0.1058***	0.0193***	0.1154***	0.0208***
	(0.0131)	(0.0012)	(0.0179)	(0.0007)	(0.0088)	(0.0014)	(0.0070)	(0.0011)
Gender	0.4714***	0.0455***	0.7880***	0.0377***	-0.1384	-0.0253	-0.0497	-0.0090
	(0.1475)	(0.0140)	(0.1696)	(0.0081)	(0.1095)	(0.0200)	(0.0895)	(0.0161)
Demographic 1	-0.6287***	-0.0607***	-1.8488***	-0.0886***	-0.3486**	-0.0637**	-0.3079***	-0.0556***
	(0.2088)	(0.0200)	(0.2445)	(0.0115)	(0.1467)	(0.0267)	(0.1123)	(0.0202)
Demographic 2	0.1531	0.0148	0.3703*	0.0177*	0.1756	0.0321	0.0282	0.0051
	(0.2072)	(0.0200)	(0.1929)	(0.0091)	(0.1558)	(0.0284)	(0.1069)	(0.0193)
Advanced Math	0.2889*	0.0279*			0.3267***	0.0597***		
	(0.1612)	(0.0156)			(0.1112)	(0.0202)		
Committee Score	0.2241***	0.0216***	0.3994***	0.0191***	0.1026***	0.0187***	0.0757***	0.0137***
	(0.0413)	(0.0039)	(0.0506)	(0.0023)	(0.0287)	(0.0052)	(0.0228)	(0.0041)
Non-Economics 1			0.8376***	0.0401***			1.9483***	0.3515***
			(0.2297)	(0.0107)			(0.1236)	(0.0205)
Observations | Pseudo-R^2	5000	0.3251			5000	0.1431		
Parameters | LL | AIC	14	-1130.4	2288.8		14	-2691.4	5410.8	
Wald Statistic | DF | P-Value	3.1018	5	0.6843		3.0042	5	0.6993	

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Acceptance, Matriculation, and Success AMEs

A4: Simulated Wald Test Results (Different)

Table 19b: Simulated Wald Test Results (Different)

	Dependent Variable = Accept				Dependent Variable = Success			
	Economics		Non-Economics		Economics		Non-Economics	
	Coef	AME	Coef	AME	Coef	AME	Coef	AME
Intercept	-16.3291***	-1.5784***	-8.4753***	-0.3642***	-10.4581***	-1.9104***	-3.2236***	-0.5997***
	(1.1274)	(0.0955)	(0.8673)	(0.0412)	(0.7237)	(0.1070)	(0.4018)	(0.0728)
GRE Quantitative Percentile	0.1502***	0.0145***	0.0402***	0.0017***	0.1058***	0.0193***	0.0034	0.0006
	(0.0131)	(0.0012)	(0.0128)	(0.0006)	(0.0088)	(0.0014)	(0.0062)	(0.0012)
Gender	0.4702***	0.0454***	0.5942***	0.0255***	-0.1372	-0.0251	0.0375	0.0070
	(0.1473)	(0.0140)	(0.1820)	(0.0079)	(0.1095)	(0.0200)	(0.0868)	(0.0161)
Demographic 1	-0.6246***	-0.0604***	-0.2541	-0.0109	-0.3544**	-0.0647**	-0.1152	-0.0214
	(0.2085)	(0.0200)	(0.2393)	(0.0103)	(0.1467)	(0.0267)	(0.1103)	(0.0205)
Demographic 2	0.1555	0.0150	0.0081	0.0003	0.1713	0.0313	-0.0562	-0.0105
	(0.2070)	(0.0200)	(0.2109)	(0.0091)	(0.1557)	(0.0284)	(0.1030)	(0.0192)
Advanced Math	0.2886*	0.0279*			0.3284***	0.0600***		
	(0.1611)	(0.0156)			(0.1112)	(0.0202)		
Committee Score	0.2241***	0.0217***	0.3817***	0.0164***	0.1026***	0.0187***	0.1808***	0.0336***
	(0.0413)	(0.0039)	(0.0528)	(0.0025)	(0.0287)	(0.0052)	(0.0228)	(0.0041)
Non-Economics 1			-0.0452	-0.0019			1.1393***	0.2120***
			(0.1995)	(0.0086)			(0.1219)	(0.0219)
Observations | Pseudo-R^2	5000	0.1488			5000	0.0758		
Parameters | LL | AIC	14	-1155.7	2339.5		14	-2747.2	5522.4	
Wald Statistic | DF | P-Value	112.5897	5	0.0000		124.7379	5	0.0000	

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Acceptance, Matriculation, and Success AMEs

A4: Simulated Unrestricted Dynamic Estimates (Optimal)

Table 20a: Simulated Unrestricted Dynamic Estimates (Optimal)

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.9027**	-0.4663**	4.8637***	-11.5885***	-2.2558***	-13.1577***	-2.3739***	-9.4807***	-1.3248***
	(0.7564)	(0.1830)	(1.3238)	(1.0232)	(0.1454)	(2.0081)	(0.2886)	(1.3435)	(0.1562)
GRE Quantitative Percentile	0.0208**	0.0051**	0.0008	0.1236***	0.0241***	0.1372***	0.0248***	0.1150***	0.0161***
	(0.0090)	(0.0022)	(0.0150)	(0.0118)	(0.0018)	(0.0223)	(0.0033)	(0.0145)	(0.0016)
Gender	-0.2335*	-0.0572*	-0.0783	0.2266	0.0441	0.1289	0.0233	0.2078	0.0290
	(0.1309)	(0.0319)	(0.2113)	(0.1464)	(0.0283)	(0.2502)	(0.0451)	(0.1766)	(0.0246)
Demographic 1	0.3691**	0.0905**	-0.3430	-0.3014	-0.0587	-0.2640	-0.0476	-0.3380	-0.0472
	(0.1732)	(0.0421)	(0.2885)	(0.1988)	(0.0386)	(0.3744)	(0.0672)	(0.2456)	(0.0342)
Demographic 2	0.0925	0.0227	-0.1806	0.1538	0.0299	0.1708	0.0308	-0.1354	-0.0189
	(0.1911)	(0.0468)	(0.3001)	(0.2215)	(0.0431)	(0.3844)	(0.0693)	(0.2652)	(0.0371)
Advanced Math			0.6870***			0.7737***	0.1396***	0.6827***	0.0954***
			(0.2024)			(0.2535)	(0.0443)	(0.1770)	(0.0237)
Committee Score						0.0783	0.0141	0.0467	0.0065
						(0.0602)	(0.0108)	(0.0672)	(0.0094)
Missing Committee Score								0.2177	0.0304
								(0.1896)	(0.0264)
Year Transition More	0.0664	0.0163	0.0932	0.9840***	0.1915***	-0.9251***	-0.1669***	-0.0466	-0.0065
	(0.1281)	(0.0314)	(0.2069)	(0.1472)	(0.0263)	(0.2513)	(0.0427)	(0.1740)	(0.0243)
Program Rank								-0.0309***	-0.0043***
								(0.0019)	(0.0002)
$\sqrt{\text{MSE}}$			1.9465***						
			(0.0673)						
Obs | Param | LL | AIC	1000	38	-2655.7	5387.5					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

A4: Simulated Unrestricted Dynamic Estimates (Suboptimal)

Table 20b: Simulated Unrestricted Dynamic Estimates (Suboptimal)

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.9027**	-0.4663**	5.2717***	-11.2295***	-2.1801***	-6.7296***	-1.3296***	-9.7587***	-1.3686***
	(0.7564)	(0.1830)	(1.2701)	(1.0144)	(0.1474)	(1.7376)	(0.3202)	(1.3345)	(0.1538)
GRE Quantitative Percentile	0.0208**	0.0051**	-0.0016	0.1226***	0.0238***	0.0550***	0.0109***	0.1175***	0.0165***
	(0.0090)	(0.0022)	(0.0144)	(0.0117)	(0.0018)	(0.0191)	(0.0036)	(0.0145)	(0.0016)
Gender	-0.2335*	-0.0572*	-0.1023	0.2575*	0.0500*	0.0098	0.0019	0.1984	0.0278
	(0.1309)	(0.0319)	(0.2051)	(0.1469)	(0.0283)	(0.2368)	(0.0468)	(0.1766)	(0.0247)
Demographic 1	0.3691**	0.0905**	-0.4893*	-0.8293***	-0.1610***	-0.3340	-0.0660	-0.3897	-0.0547
	(0.1732)	(0.0421)	(0.2701)	(0.1966)	(0.0371)	(0.3394)	(0.0669)	(0.2462)	(0.0343)
Demographic 2	0.0925	0.0227	-0.2744	0.0797	0.0155	0.1764	0.0348	-0.2068	-0.0290
	(0.1911)	(0.0468)	(0.2740)	(0.2135)	(0.0414)	(0.3346)	(0.0659)	(0.2636)	(0.0370)
Advanced Math			0.7219***			0.3580	0.0707	0.6387***	0.0896***
			(0.1966)			(0.2375)	(0.0465)	(0.1763)	(0.0238)
Committee Score						0.2962***	0.0585***	0.0626	0.0088
						(0.0592)	(0.0106)	(0.0657)	(0.0092)
Missing Committee Score								0.2535	0.0356
								(0.1896)	(0.0264)
Year Transition More	0.0664	0.0163	0.0031	1.1108***	0.2156***	-0.7185***	-0.1420***	-0.0473	-0.0066
	(0.1281)	(0.0314)	(0.2010)	(0.1479)	(0.0257)	(0.2397)	(0.0455)	(0.1757)	(0.0247)
Program Rank								-0.0303***	-0.0043***
								(0.0019)	(0.0002)
$\sqrt{\text{MSE}}$			1.9186***						
			(0.0682)						
Obs | Param | LL | AIC	1000	38	-2696.1	5468.2					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

A4: Simulated Restricted Dynamic Estimates (Optimal)

Table 21a: Simulated Restricted Dynamic Estimates (Optimal)

	Advanced Math	Committee Score	Cutoff 1	L(Cutoff)$_1$	Cutoff 2	P(Cutoff)$_2$	Success	Sigma
Intercept	-2.1626***	4.6976***	-0.3078***	0.7351	0.0503	51.26%	-9.5097***	
	(0.7577)	(1.3258)	(0.0655)	(0.6465, 0.8358)	(0.1841)	(42.30%, 60.13%)	(1.2612)	
GRE Quantitative Percentile	0.0240***	0.0028					0.1148***	
	(0.0090)	(0.0150)					(0.0137)	
Gender	-0.2129	-0.0649					0.2357**	
	(0.1303)	(0.2112)					(0.1084)	
Demographic 1	0.3565**	-0.3526					-0.3177**	
	(0.1724)	(0.2903)					(0.1549)	
Demographic 2	0.0948	-0.1805					0.0020	
	(0.1904)	(0.3010)					(0.1594)	
Advanced Math		0.6919***					0.6157***	
		(0.2026)					(0.1367)	
Committee Score							0.0496	
							(0.0383)	
Missing Committee Score							0.2183	
							(0.1819)	
Year Transition More	0.0743	0.0967	-0.0466	0.7017	0.5233**	63.96%	-0.0124	
	(0.1277)	(0.2069)	(0.0675)	(0.6137, 0.8022)	(0.2398)	(54.65%, 72.33%)	(0.1720)	
Program Rank							-0.0310***	
							(0.0019)	
$\sqrt{\text{MSE}}$		1.9468***						
		(0.0673)						
Sigma								0.1390***
								(0.0189)
Obs | Param | LL | AIC	1000	29	-2683.0	5423.9				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

A4: Simulated Restricted Dynamic Estimates (Suboptimal)

Table 21b: Simulated Restricted Dynamic Estimates (Suboptimal)

	Advanced Math	Committee Score	Cutoff 1	L(Cutoff)$_1$	Cutoff 2	P(Cutoff)$_2$	Success	Sigma
Intercept	-2.1764***	4.5383***	-0.2386***	0.7878	0.0226	50.57%	-9.4138***	
	(0.7589)	(1.2591)	(0.0803)	(0.6730, 0.9220)	(0.2017)	(40.79%, 60.30%)	(1.4009)	
GRE Quantitative Percentile	0.0242***	0.0072					0.1066***	
	(0.0090)	(0.0143)					(0.0150)	
Gender	-0.2179*	-0.0641					0.2086*	
	(0.1306)	(0.2029)					(0.1206)	
Demographic 1	0.3420**	-0.5643**					-0.5037***	
	(0.1728)	(0.2716)					(0.1798)	
Demographic 2	0.1011	-0.2640					-0.0204	
	(0.1906)	(0.2738)					(0.1843)	
Advanced Math		0.7292***					0.5164***	
		(0.1970)					(0.1483)	
Committee Score							0.1934***	
							(0.0382)	
Missing Committee Score							0.2727	
							(0.1887)	
Year Transition More	0.0706	0.0108	-0.1333*	0.6895	0.4714*	62.11%	-0.0673	
	(0.1278)	(0.2007)	(0.0712)	(0.5886, 0.8075)	(0.2663)	(51.59%, 71.59%)	(0.1760)	
Program Rank							-0.0305***	
							(0.0019)	
$\sqrt{\text{MSE}}$		1.9197***						
		(0.0684)						
Sigma								0.1776***
								(0.0305)
Obs | Param | LL | AIC	1000	29	-2742.2	5542.4				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

A4: Simulated Unrestricted Static Estimates (Optimal)

Table 22a: Simulated Unrestricted Static Estimates (Optimal)

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-11.5885***	-2.2558***	-13.1577***	-2.3739***	-8.1374***	-1.1678***	-9.4807***	-1.3248***
	(1.0232)	(0.1454)	(2.0081)	(0.2886)	(1.1381)	(0.1344)	(1.3435)	(0.1562)
GRE Quantitative Percentile	0.1236***	0.0241***	0.1372***	0.0248***	0.1079***	0.0155***	0.1150***	0.0161***
	(0.0118)	(0.0018)	(0.0223)	(0.0033)	(0.0133)	(0.0015)	(0.0145)	(0.0016)
Gender	0.2266	0.0441	0.1289	0.0233	0.1399	0.0201	0.2078	0.0290
	(0.1464)	(0.0283)	(0.2502)	(0.0451)	(0.1739)	(0.0249)	(0.1766)	(0.0246)
Demographic 1	-0.3014	-0.0587	-0.2640	-0.0476	-0.2533	-0.0364	-0.3380	-0.0472
	(0.1988)	(0.0386)	(0.3744)	(0.0672)	(0.2359)	(0.0338)	(0.2456)	(0.0342)
Demographic 2	0.1538	0.0299	0.1708	0.0308	-0.1179	-0.0169	-0.1354	-0.0189
	(0.2215)	(0.0431)	(0.3844)	(0.0693)	(0.2581)	(0.0371)	(0.2652)	(0.0371)
Advanced Math			0.7737***	0.1396***			0.6827***	0.0954***
			(0.2535)	(0.0443)			(0.1770)	(0.0237)
Committee Score			0.0783	0.0141			0.0467	0.0065
			(0.0602)	(0.0108)			(0.0672)	(0.0094)
Missing Committee Score							0.2177	0.0304
							(0.1896)	(0.0264)
Year Transition More	0.9840***	0.1915***	-0.9251***	-0.1669***	-0.0587	-0.0084	-0.0466	-0.0065
	(0.1472)	(0.0263)	(0.2513)	(0.0427)	(0.1690)	(0.0242)	(0.1740)	(0.0243)
Program Rank					-0.0310***	-0.0045***	-0.0309***	-0.0043***
					(0.0019)	(0.0002)	(0.0019)	(0.0002)
Obs | Param | LL | AIC	1000	31	-1622.9	3307.8				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

A4: Simulated Unrestricted Static Estimates (Suboptimal)

Table 22b: Simulated Unrestricted Static Estimates (Suboptimal)

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-11.2295***	-2.1801***	-6.7296***	-1.3296***	-8.2825***	-1.1913***	-9.7587***	-1.3686***
	(1.0144)	(0.1474)	(1.7376)	(0.3202)	(1.1392)	(0.1341)	(1.3345)	(0.1538)
GRE Quantitative Percentile	0.1226***	0.0238***	0.0550***	0.0109***	0.1097***	0.0158***	0.1175***	0.0165***
	(0.0117)	(0.0018)	(0.0191)	(0.0036)	(0.0133)	(0.0015)	(0.0145)	(0.0016)
Gender	0.2575*	0.0500*	0.0098	0.0019	0.1355	0.0195	0.1984	0.0278
	(0.1469)	(0.0283)	(0.2368)	(0.0468)	(0.1741)	(0.0250)	(0.1766)	(0.0247)
Demographic 1	-0.8293***	-0.1610***	-0.3340	-0.0660	-0.2776	-0.0399	-0.3897	-0.0547
	(0.1966)	(0.0371)	(0.3394)	(0.0669)	(0.2344)	(0.0337)	(0.2462)	(0.0343)
Demographic 2	0.0797	0.0155	0.1764	0.0348	-0.1809	-0.0260	-0.2068	-0.0290
	(0.2135)	(0.0414)	(0.3346)	(0.0659)	(0.2566)	(0.0369)	(0.2636)	(0.0370)
Advanced Math			0.3580	0.0707			0.6387***	0.0896***
			(0.2375)	(0.0465)			(0.1763)	(0.0238)
Committee Score			0.2962***	0.0585***			0.0626	0.0088
			(0.0592)	(0.0106)			(0.0657)	(0.0092)
Missing Committee Score							0.2535	0.0356
							(0.1896)	(0.0264)
Year Transition More	1.1108***	0.2156***	-0.7185***	-0.1420***	-0.0767	-0.0110	-0.0473	-0.0066
	(0.1479)	(0.0257)	(0.2397)	(0.0455)	(0.1688)	(0.0243)	(0.1757)	(0.0247)
Program Rank					-0.0304***	-0.0044***	-0.0303***	-0.0043***
					(0.0019)	(0.0002)	(0.0019)	(0.0002)
Obs | Param | LL | AIC	1000	31	-1647.5	3357.0				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

▸▸ Conclusion

A4: Simulated Restricted Static Estimates (Optimal)

Table 23a: Simulated Restricted Static Estimates (Optimal)

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	0.6911***	66.62%	0.2080	55.18%	-8.0508***	-9.5892***	
	(0.1817)	(58.30%, 74.02%)	(0.2148)	(44.69%, 65.23%)	(1.1629)	(1.3344)	
GRE Quantitative Percentile					0.1061***	0.1155***	
					(0.0134)	(0.0146)	
Gender					0.1997**	0.1694	
					(0.1017)	(0.1361)	
Demographic 1					-0.2479*	-0.3121	
					(0.1351)	(0.1916)	
Demographic 2					0.0408	-0.0331	
					(0.1509)	(0.1966)	
Advanced Math						0.6628***	
						(0.1426)	
Committee Score						0.0557	
						(0.0426)	
Missing Committee Score						0.2170	
						(0.1838)	
Year Transition More	-0.8478***	46.09%	0.7486**	72.24%	-0.0606	-0.0276	
	(0.2178)	(39.63%, 52.68%)	(0.3087)	(60.18%, 81.76%)	(0.1667)	(0.1730)	
Program Rank					-0.0310***	-0.0310***	
					(0.0019)	(0.0019)	
Sigma							0.1922***
							(0.0246)
Obs | Param | LL | AIC	1000	22	-1623.7	3291.3			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

A4: Simulated Restricted Static Estimates (Suboptimal)

Table 23b: Simulated Restricted Static Estimates (Suboptimal)

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	0.8247***	69.52%	0.2858	57.10%	-8.5784***	-8.3580***	
	(0.2158)	(59.91%, 77.69%)	(0.2418)	(45.31%, 68.13%)	(1.1959)	(1.4863)	
GRE Quantitative Percentile					0.1141***	0.0950***	
					(0.0138)	(0.0158)	
Gender					0.2291**	0.1303	
					(0.1093)	(0.1417)	
Demographic 1					-0.5672***	-0.3661*	
					(0.1465)	(0.2059)	
Demographic 2					-0.0218	-0.0705	
					(0.1599)	(0.2104)	
Advanced Math						0.5399***	
						(0.1477)	
Committee Score						0.1918***	
						(0.0412)	
Missing Committee Score						0.1703	
						(0.1949)	
Year Transition More	-1.0834***	43.57%	0.5451*	69.66%	-0.0713	-0.1035	
	(0.2489)	(37.09%, 50.27%)	(0.3249)	(57.04%, 79.87%)	(0.1664)	(0.1784)	
Program Rank					-0.0306***	-0.0305***	
					(0.0019)	(0.0019)	
Sigma							0.2211***
							(0.0320)
Obs | Param | LL | AIC	1000	22	-1662.3	3368.6			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

A4: Simulated Unrestricted Myopic Estimates (Optimal)

Table 24a: Simulated Unrestricted Myopic Estimates (Optimal)

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	0.0000	0.0000	4.6668***	-11.5885***	-2.2558***	-13.1577***	-2.3739***	-9.4807***	-1.3248***
	(0.0894)	(0.0224)	(0.1940)	(1.0232)	(0.1454)	(2.0081)	(0.2886)	(1.3435)	(0.1562)
GRE Quantitative Percentile				0.1236***	0.0241***	0.1372***	0.0248***	0.1150***	0.0161***
				(0.0118)	(0.0018)	(0.0223)	(0.0033)	(0.0145)	(0.0016)
Gender				0.2266	0.0441	0.1289	0.0233	0.2078	0.0290
				(0.1464)	(0.0283)	(0.2502)	(0.0451)	(0.1766)	(0.0246)
Demographic 1				-0.3014	-0.0587	-0.2640	-0.0476	-0.3380	-0.0472
				(0.1988)	(0.0386)	(0.3744)	(0.0672)	(0.2456)	(0.0342)
Demographic 2				0.1538	0.0299	0.1708	0.0308	-0.1354	-0.0189
				(0.2215)	(0.0431)	(0.3844)	(0.0693)	(0.2652)	(0.0371)
Advanced Math			0.6763***			0.7737***	0.1396***	0.6827***	0.0954***
			(0.2006)			(0.2535)	(0.0443)	(0.1770)	(0.0237)
Committee Score						0.0783	0.0141	0.0467	0.0065
						(0.0602)	(0.0108)	(0.0672)	(0.0094)
Missing Committee Score								0.2177	0.0304
								(0.1896)	(0.0264)
Year Transition More	0.0480	0.0120	0.1023	0.9840***	0.1915***	-0.9251***	-0.1669***	-0.0466	-0.0065
	(0.1265)	(0.0316)	(0.2069)	(0.1472)	(0.0263)	(0.2513)	(0.0427)	(0.1740)	(0.0243)
Program Rank								-0.0309***	-0.0043***
								(0.0019)	(0.0002)
$\sqrt{\text{MSE}}$			1.9512***						
			(0.0674)						
Obs | Param | LL | AIC	1000	30	-2666.4	5392.8					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

A4: Simulated Unrestricted Myopic Estimates (Suboptimal)

Table 24b: Simulated Unrestricted Myopic Estimates (Suboptimal)

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	0.0000	0.0000	4.7588***	-11.2295***	-2.1801***	-6.7296***	-1.3296***	-9.7587***	-1.3686***
	(0.0894)	(0.0224)	(0.1881)	(1.0144)	(0.1474)	(1.7376)	(0.3202)	(1.3345)	(0.1538)
GRE Quantitative Percentile				0.1226***	0.0238***	0.0550***	0.0109***	0.1175***	0.0165***
				(0.0117)	(0.0018)	(0.0191)	(0.0036)	(0.0145)	(0.0016)
Gender				0.2575*	0.0500*	0.0098	0.0019	0.1984	0.0278
				(0.1469)	(0.0283)	(0.2368)	(0.0468)	(0.1766)	(0.0247)
Demographic 1				-0.8293***	-0.1610***	-0.3340	-0.0660	-0.3897	-0.0547
				(0.1966)	(0.0371)	(0.3394)	(0.0669)	(0.2462)	(0.0343)
Demographic 2				0.0797	0.0155	0.1764	0.0348	-0.2068	-0.0290
				(0.2135)	(0.0414)	(0.3346)	(0.0659)	(0.2636)	(0.0370)
Advanced Math			0.7040***			0.3580	0.0707	0.6387***	0.0896***
			(0.1957)			(0.2375)	(0.0465)	(0.1763)	(0.0238)
Committee Score						0.2962***	0.0585***	0.0626	0.0088
						(0.0592)	(0.0106)	(0.0657)	(0.0092)
Missing Committee Score								0.2535	0.0356
								(0.1896)	(0.0264)
Year Transition More	0.0480	0.0120	0.0218	1.1108***	0.2156***	-0.7185***	-0.1420***	-0.0473	-0.0066
	(0.1265)	(0.0316)	(0.2022)	(0.1479)	(0.0257)	(0.2397)	(0.0455)	(0.1757)	(0.0247)
Program Rank								-0.0303***	-0.0043***
								(0.0019)	(0.0002)
$\sqrt{\text{MSE}}$			1.9289***						
			(0.0684)						
Obs | Param | LL | AIC	1000	30	-2708.0	5475.9					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

A4: Simulated Restricted Myopic Estimates (Optimal)

Table 25a: Simulated Restricted Myopic Estimates (Optimal)

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	0.0146	4.6803***	-0.3118***	0.7321	0.0452	51.13%	-9.5732***	
	(0.0889)	(0.1930)	(0.0659)	(0.6433, 0.8331)	(0.1841)	(42.17%, 60.01%)	(1.2633)	
GRE Quantitative Percentile							0.1156***	
							(0.0138)	
Gender							0.2266**	
							(0.1080)	
Demographic 1							-0.3082**	
							(0.1533)	
Demographic 2							0.0055	
							(0.1579)	
Advanced Math		0.6789***					0.5999***	
		(0.2009)					(0.1366)	
Committee Score							0.0496	
							(0.0382)	
Missing Committee Score							0.2197	
							(0.1818)	
Year Transition More	0.0542	0.1004	-0.0430	0.7013	0.5323**	64.05%	-0.0098	
	(0.1261)	(0.2067)	(0.0679)	(0.6129, 0.8024)	(0.2407)	(54.67%, 72.46%)	(0.1720)	
Program Rank							-0.0310***	
							(0.0019)	
$\sqrt{\text{MSE}}$		1.9514***						
		(0.0674)						
Sigma								0.1387***
								(0.0190)
Obs | Param | LL | AIC	1000	21	-2694.3	5430.7				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

A4: Simulated Restricted Myopic Estimates (Suboptimal)

Table 25b: Simulated Restricted Myopic Estimates (Suboptimal)

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	0.0122	4.8050***	-0.2369***	0.7891	0.0217	50.54%	-9.5133***	
	(0.0891)	(0.1878)	(0.0811)	(0.6731, 0.9250)	(0.2029)	(40.71%, 60.33%)	(1.3931)	
GRE Quantitative Percentile							0.1081***	
							(0.0149)	
Gender							0.2007*	
							(0.1204)	
Demographic 1							-0.5140***	
							(0.1790)	
Demographic 2							-0.0254	
							(0.1847)	
Advanced Math		0.7027***					0.5083***	
		(0.1961)					(0.1484)	
Committee Score							0.1894***	
							(0.0385)	
Missing Committee Score							0.2765	
							(0.1884)	
Year Transition More	0.0511	0.0095	-0.1323*	0.6913	0.4816*	62.32%	-0.0638	
	(0.1262)	(0.2016)	(0.0716)	(0.5895, 0.8107)	(0.2684)	(51.71%, 71.88%)	(0.1759)	
Program Rank							-0.0305***	
							(0.0019)	
$\sqrt{\text{MSE}}$		1.9289***						
		(0.0684)						
Sigma								0.1794***
								(0.0308)
Obs | Param | LL | AIC	1000	21	-2754.8	5551.6				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

A4: Simulated Unrestricted Mat-Dynamic Estimates (Optimal)

Table 26a: Simulated Unrestricted Matriculation-Dynamic Estimates (Optimal)

	Advanced Math		Committee Score	Transition		Accept		Success		Matriculate	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.9027**	-0.4663**	4.8637***	-11.5885***	-2.2558***	-13.1577***	-2.3739***	-9.4807***	-1.3248***	-4.6066	-0.6127
	(0.7564)	(0.1830)	(1.3238)	(1.0232)	(0.1454)	(2.0081)	(0.2886)	(1.3435)	(0.1562)	(4.6480)	(0.6187)
GRE Quantitative Percentile	0.0208**	0.0051**	0.0008	0.1236***	0.0241***	0.1372***	0.0248***	0.1150***	0.0161***	0.0593	0.0079
	(0.0090)	(0.0022)	(0.0150)	(0.0118)	(0.0018)	(0.0223)	(0.0033)	(0.0145)	(0.0016)	(0.0459)	(0.0061)
Gender	-0.2335*	-0.0572*	-0.0783	0.2266	0.0441	0.1289	0.0233	0.2078	0.0290	0.4885	0.0650
	(0.1309)	(0.0319)	(0.2113)	(0.1464)	(0.0283)	(0.2502)	(0.0451)	(0.1766)	(0.0246)	(0.5063)	(0.0657)
Demographic 1	0.3691**	0.0905**	-0.3430	-0.3014	-0.0587	-0.2640	-0.0476	-0.3380	-0.0472	0.3587	0.0477
	(0.1732)	(0.0421)	(0.2885)	(0.1988)	(0.0386)	(0.3744)	(0.0672)	(0.2456)	(0.0342)	(0.7459)	(0.1001)
Demographic 2	0.0925	0.0227	-0.1806	0.1538	0.0299	0.1708	0.0308	-0.1354	-0.0189	0.1834	0.0244
	(0.1911)	(0.0468)	(0.3001)	(0.2215)	(0.0431)	(0.3844)	(0.0693)	(0.2652)	(0.0371)	(0.7962)	(0.1069)
Advanced Math			0.6870***			0.7737***	0.1396***	0.6827***	0.0954***	0.8106	0.1078
			(0.2024)			(0.2535)	(0.0443)	(0.1770)	(0.0237)	(0.5773)	(0.0733)
Committee Score						0.0783	0.0141	0.0467	0.0065	-0.7402***	-0.0985***
						(0.0602)	(0.0108)	(0.0672)	(0.0094)	(0.2064)	(0.0235)
Missing Committee Score								0.2177	0.0304		
								(0.1896)	(0.0264)		
Year Transition/Matriculate More	0.0664	0.0163	0.0932	0.9840***	0.1915***	-0.9251***	-0.1669***	-0.0466	-0.0065	3.3280***	0.4427***
	(0.1281)	(0.0314)	(0.2069)	(0.1472)	(0.0263)	(0.2513)	(0.0427)	(0.1740)	(0.0243)	(0.5674)	(0.0331)
Program Rank								-0.0309***	-0.0043***		
								(0.0019)	(0.0002)		
$\sqrt{\text{MSE}}$			1.9465***								
			(0.0673)								
Obs | Param | LL | AIC	1000	46	-2708.1	5508.3							

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

A4: Simulated Unrestricted Mat-Dynamic Estimates (Suboptimal)

Table 26b: Simulated Unrestricted Matriculation-Dynamic Estimates (Suboptimal)

	Advanced Math		Committee Score	Transition		Accept		Success		Matriculate	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.9027**	-0.4663**	5.2717***	-11.2295***	-2.1801***	-6.7296***	-1.3296***	-9.7587***	-1.3686***	-1.0400	-0.1351
	(0.7564)	(0.1830)	(1.2701)	(1.0144)	(0.1474)	(1.7376)	(0.3202)	(1.3345)	(0.1538)	(3.6083)	(0.4721)
GRE Quantitative Percentile	0.0208**	0.0051**	-0.0016	0.1226***	0.0238***	0.0550***	0.0109***	0.1175***	0.0165***	0.0129	0.0017
	(0.0090)	(0.0022)	(0.0144)	(0.0117)	(0.0018)	(0.0191)	(0.0036)	(0.0145)	(0.0016)	(0.0391)	(0.0051)
Gender	-0.2335*	-0.0572*	-0.1023	0.2575*	0.0500*	0.0098	0.0019	0.1984	0.0278	0.1298	0.0169
	(0.1309)	(0.0319)	(0.2051)	(0.1469)	(0.0283)	(0.2368)	(0.0468)	(0.1766)	(0.0247)	(0.4717)	(0.0608)
Demographic 1	0.3691**	0.0905**	-0.4893*	-0.8293***	-0.1610***	-0.3340	-0.0660	-0.3897	-0.0547	0.6327	0.0822
	(0.1732)	(0.0421)	(0.2701)	(0.1966)	(0.0371)	(0.3394)	(0.0669)	(0.2462)	(0.0343)	(0.6281)	(0.0823)
Demographic 2	0.0925	0.0227	-0.2744	0.0797	0.0155	0.1764	0.0348	-0.2068	-0.0290	-0.2222	-0.0289
	(0.1911)	(0.0468)	(0.2740)	(0.2135)	(0.0414)	(0.3346)	(0.0659)	(0.2636)	(0.0370)	(0.6864)	(0.0880)
Advanced Math			0.7219***			0.3580	0.0707	0.6387***	0.0896***	0.5877	0.0763
			(0.1966)			(0.2375)	(0.0465)	(0.1763)	(0.0238)	(0.5321)	(0.0674)
Committee Score						0.2962***	0.0585***	0.0626	0.0088	-0.6092***	-0.0791***
						(0.0592)	(0.0106)	(0.0657)	(0.0092)	(0.1637)	(0.0176)
Missing Committee Score								0.2535	0.0356		
								(0.1896)	(0.0264)		
Year Transition/Matriculate More	0.0664	0.0163	0.0031	1.1108***	0.2156***	-0.7185***	-0.1420***	-0.0473	-0.0066	3.4846***	0.4526***
	(0.1281)	(0.0314)	(0.2010)	(0.1479)	(0.0257)	(0.2397)	(0.0455)	(0.1757)	(0.0247)	(0.6046)	(0.0349)
Program Rank								-0.0303***	-0.0043***		
								(0.0019)	(0.0002)		
$\sqrt{\text{MSE}}$			1.9186***								
			(0.0682)								
Obs | Param | LL | AIC	1000	46	-2749.9	5591.9							

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

A4: Simulated Restricted Mat-Dynamic Estimates (Optimal)

Table 27a: Simulated Restricted Matriculation-Dynamic Estimates (Optimal)

Intercept	-2.1963***	5.4976***	-1.8249***	0.1612	0.0339	50.85%	-9.4675***	-9.8786**	
	(0.7585)	(1.3429)	(0.1759)	(0.1142, 0.2276)	(0.2008)	(41.11%, 60.53%)	(1.2874)	(4.0907)	
GRE Quantitative Percentile	0.0244***	-0.0072					0.1153***	0.1186***	
	(0.0090)	(0.0153)					(0.0140)	(0.0397)	
Gender	-0.2210*	-0.0965					0.1845	0.7496*	
	(0.1307)	(0.2096)					(0.1213)	(0.4548)	
Demographic 1	0.3598**	-0.3130					-0.3019*	0.1162	
	(0.1729)	(0.2878)					(0.1729)	(0.7211)	
Demographic 2	0.0946	-0.1832					-0.0093	0.0811	
	(0.1907)	(0.2982)					(0.1766)	(0.7755)	
Advanced Math		0.6485***					0.6475***	1.0534*	
		(0.2032)					(0.1382)	(0.5829)	
Committee Score							0.0265	-0.6809***	
							(0.0400)	(0.1999)	
Missing Committee Score							0.2380		
							(0.1861)		
Year Transition/Matriculate More	0.0710	0.1591	-1.1147***	0.0529	0.5592**	64.41%	-0.0010	3.0532***	
	(0.1280)	(0.2057)	(0.2799)	(0.0296, 0.0946)	(0.2553)	(54.59%, 73.15%)	(0.1732)	(0.5423)	
Program Rank							-0.0309***		
							(0.0019)		
$\sqrt{\text{MSE}}$		1.9457***							
		(0.0672)							
Sigma									0.1457***
									(0.0234)
Obs | Param | LL | AIC	1000	37	-2755.2	5584.5					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

A4: Simulated Restricted Mat-Dynamic Estimates (Suboptimal)

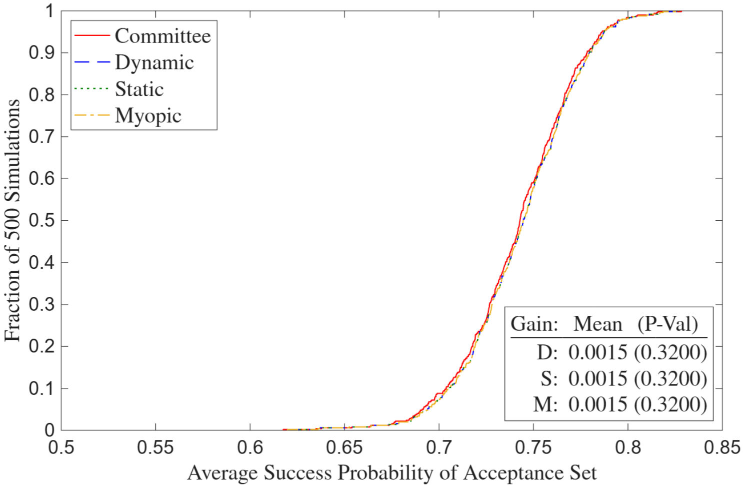
Table 27b: Simulated Restricted Matriculation-Dynamic Estimates (Suboptimal)

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Matriculate	Sigma
Intercept	-2.0816***	5.5356***	-1.4144***	0.2431	0.1778	54.43%	-9.6441***	-6.4840*	
	(0.7582)	(1.2750)	(0.2209)	(0.1576, 0.3748)	(0.2485)	(42.33%, 66.04%)	(1.3938)	(3.3823)	
GRE Quantitative Percentile	0.0230**	-0.0050					0.1098***	0.0774**	
	(0.0090)	(0.0145)					(0.0150)	(0.0363)	
Gender	-0.2276*	-0.1115					0.1637	0.3953	
	(0.1310)	(0.2034)					(0.1438)	(0.4595)	
Demographic 1	0.3530**	-0.4625*					-0.4614**	0.0566	
	(0.1729)	(0.2685)					(0.2103)	(0.6572)	
Demographic 2	0.0922	-0.2781					-0.0923	-0.4297	
	(0.1908)	(0.2722)					(0.2189)	(0.7030)	
Advanced Math		0.7079***					0.5798***	0.7404	
		(0.1960)					(0.1558)	(0.5503)	
Committee Score							0.1840***	-0.5475***	
							(0.0433)	(0.1601)	
Missing Committee Score							0.2500		
							(0.1936)		
Year Transition/Matriculate More	0.0723	0.0357	-1.3730***	0.0616	0.5357	67.12%	-0.0723	3.1502***	
	(0.1280)	(0.2000)	(0.3748)	(0.0274, 0.1385)	(0.3413)	(53.66%, 78.25%)	(0.1795)	(0.5830)	
Program Rank							-0.0305***		
							(0.0019)		
$\sqrt{\text{MSE}}$		1.9147***							
		(0.0677)							
Sigma									0.2346***
									(0.0484)
Obs | Param | LL | AIC	1000	37	-2818.0	5710.0					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

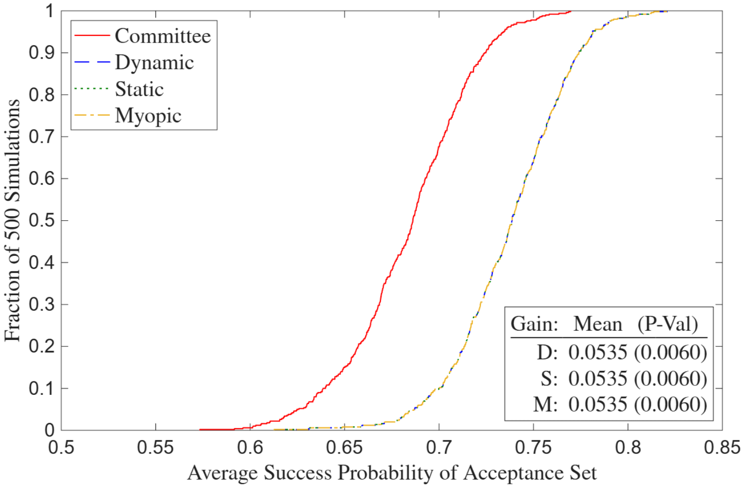
Conclusion

A4: Simulated Deviation Gains (Optimal)



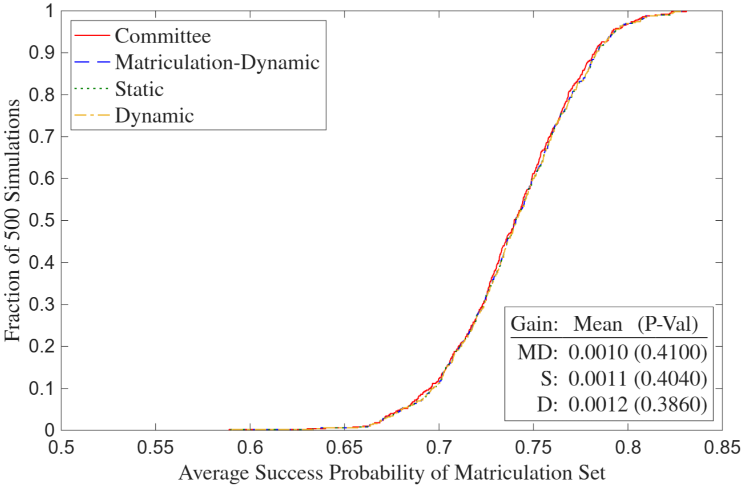
Conclusion

A4: Simulated Deviation Gains (Suboptimal)



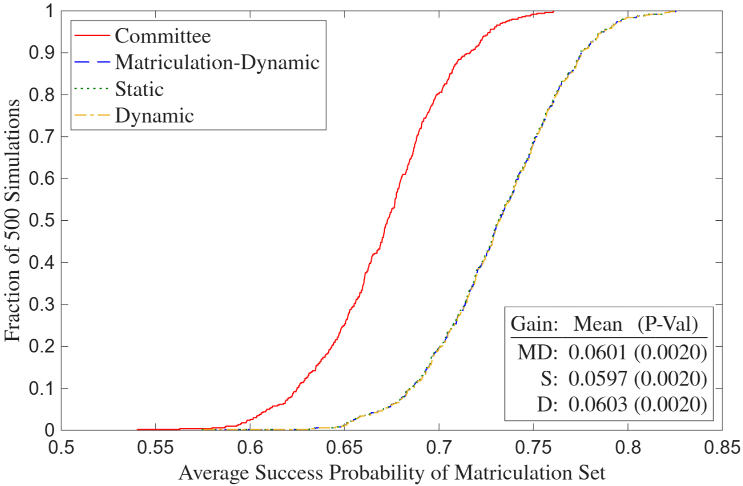
▸▸ Conclusion

A4: Simulated Matriculation Deviation Gains (Optimal)



Conclusion

A4: Simulated Matriculation Deviation Gains (Suboptimal)



Conclusion

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